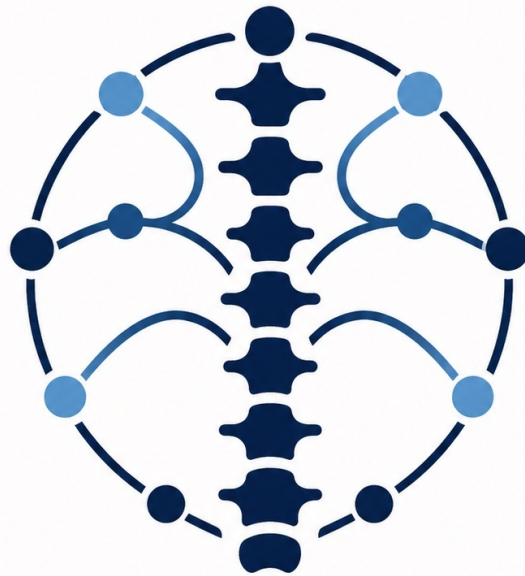




RACHIS

EUROPE'S INTELLIGENCE SPINE

DISCLOSURE WITHOUT SURRENDER

**INTEGRATION AUTHORITY, AND FEDERATED INFORMATION EXCHANGE AS PUBLIC
INFRASTRUCTURE**

James Patrick CITP MBCS

July 2026

TABLE OF CONTENTS

Abstract.....	10
Thesis statement.....	12
Contribution claims.....	13
Reader's guide.....	13
PART I - THE PROBLEM.....	15
Chapter 1 - The coordination failure.....	15
1.1 A question that cannot be answered.....	15
1.2 The instinct to call this a technical problem.....	15
1.3 Institutions cannot agree with themselves.....	16
1.4 What the market actually sells.....	16
1.5 The argument of this document.....	17
1.6 The larger claim: unbundling.....	17
1.7 Structure.....	18
Chapter 2 - What creates the dependency, and what it costs.....	18
2.1 Locating the moat.....	18
2.2 The integration model.....	19
2.3 The economics.....	20
2.4 Why sources decline, and why they are right to.....	20
2.5 The inversion.....	21
2.6 A necessary concession.....	21
2.7 The economics of integration authority.....	22
Definitions.....	22
The prevailing model.....	22
The proposed model.....	23
Result 1 - marginal cost.....	23
Result 2 - the floor.....	23
2.8 Registry coverage.....	23
Result 3.....	24
2.9 Switching cost and the location of knowledge.....	24
Proposition.....	24
Result 4.....	24
Result 5.....	25
2.10 Predictions, and what would falsify them.....	25
Chapter 3 - Why this is not a defence problem.....	26
3.1 The general form.....	26
3.2 Policing.....	26
3.3 Health.....	27
3.4 Regulated industry.....	27
3.5 What generalises, and what does not.....	27

3.6 The promotion of purpose.....	28
3.7 A positioning discipline.....	29
PART II - WHAT HAS BEEN TRIED.....	30
Chapter 4 - The universal model tradition.....	30
4.1 MIP and the JC3IEDM.....	30
4.2 NIEM.....	31
4.3 What the tradition establishes.....	31
Chapter 5 - The federated infrastructure tradition.....	32
5.1 NNEC, and the limits of the programmatic approach.....	32
5.2 AMN and FMN.....	32
5.3 DCGS-A and the DIB.....	32
5.4 X-Road.....	33
Chapter 6 - The sovereignty tradition.....	34
6.1 Gaia-X.....	34
6.2 International Data Spaces.....	34
6.3 NATO's Data Centric Reference Architecture.....	35
6.4 The gap.....	36
Chapter 7 - Ten lessons.....	37
L1 - Federation cannot be procured as a programme.....	37
The claim.....	37
The evidence.....	37
The demand.....	38
L2 - Mapping to a published contract is a proven pattern, not an innovation.....	38
The claim.....	38
The evidence.....	38
The demand.....	38
L3 - The exchange contract grows heavier than its users can bear.....	39
The claim.....	39
The evidence.....	39
The demand.....	39
L4 - Conformance becomes a moat unless it is free, self-serve and end-user reproducible.....	39
The claim.....	39
The evidence.....	40
The demand.....	40
L5 - Enforcement must sit where the incentive sits.....	41
The claim.....	41
The evidence.....	41
The demand.....	41
L6 - Bilateral agreement does not scale; published unilateral contracts do.....	41
The claim.....	41

The evidence.....	42
The demand.....	42
L7 - Central state must be minimised, and each element of it separately justified.....	42
The claim.....	42
The evidence.....	42
The demand.....	43
L8 - The interface decides adoption, whatever the architecture says.....	43
The claim.....	43
The evidence.....	43
The demand.....	43
Honest note.....	43
L9 - Adoption follows mandate or immediate self-interest, never architectural merit	44
The claim.....	44
The evidence.....	44
The demand.....	44
L10 - Custodianship follows running code, and must be capture-resistant from the first day.....	44
The claim.....	44
The evidence.....	44
The demand.....	45
Summary.....	45
PART III - THE MECHANISM.....	47
Chapter 8 - The Expectation.....	48
8.1 Definition.....	48
8.2 Pulled, never pushed.....	48
8.3 Versioned, with compatibility as a governance commitment.....	49
8.4 Testable offline.....	49
8.5 Unilateral adoption.....	50
8.6 The mapping registry and the cost curve.....	51
Mappings are published, and mappings are reusable.....	51
8.7 Keeping the core thin.....	52
Namespaced extensions.....	52
A constitutional constraint on core growth.....	52
A published size discipline.....	52
8.8 What the Expectation does not settle.....	53
Chapter 9 - Mapping and disclosure as separate acts.....	53
9.1 Two artefacts, two authorities.....	53
9.2 The mapping: declarative, not procedural.....	54
9.3 The disclosure policy: four dispositions.....	54
The default is withheld.....	55
9.4 Withheld means absent.....	55

9.5 Derivation constraints.....	56
9.6 The validator.....	57
Schema-derived.....	57
Library-matched.....	57
Statistical.....	58
Two honest limitations.....	58
9.7 The cost, stated honestly.....	58
Chapter 10 - Binding: a JSON and post-quantum profile of ADatP-4778.....	59
10.1 What already exists.....	59
10.2 What the standards do not yet cover.....	59
No JSON.....	59
No post-quantum algorithms.....	60
No selective disclosure.....	60
10.3 The proposed profile.....	60
Canonical identification.....	60
Serialisation.....	60
Label carriage.....	61
Algorithms.....	61
10.4 Selective disclosure.....	61
The leakage that remains, and why it is acceptable.....	62
10.5 The four-part header.....	62
10.6 Relationship to data-centric security maturity.....	63
10.7 A note on conformance.....	63
Chapter 11 - Verification, and why ingress must not classify.....	63
11.1 The checks.....	63
11.2 Ingress is a conformance point, not a component.....	64
11.3 The platform has no classification authority.....	64
11.4 The consequence for cross-domain transfer.....	65
11.5 Why this will be resisted, and for how long.....	65
Chapter 12 - Graduated disclosure and the callback.....	66
12.1 The mechanism.....	66
12.2 Two leakage channels.....	67
Query pattern.....	67
Disclosure by differencing.....	67
12.3 Zero-index deployment.....	68
The index is an optimisation, not an architectural requirement.....	68
12.4 Value follows trust.....	69
PART IV - FOUNDATIONS.....	70
Chapter 13 - Identity is an assertion.....	70
13.1 The conventional treatment.....	70
13.2 The inversion.....	71

13.3 What this makes possible.....	71
13.4 Identity pinning.....	71
Once an entity identifier has been cited, it must remain resolvable indefinitely.....	72
13.5 The cost.....	72
Read-time assembly is expensive.....	72
Candidate generation still requires a matching key.....	72
An empty assertion store is a worse search engine.....	73
Chapter 14 - Time.....	73
14.1 Why this matters more here than elsewhere.....	73
14.2 Six timestamps.....	73
14.3 Supersession is not withdrawal.....	74
14.4 Withdrawal cascades.....	74
14.5 Export.....	74
Chapter 15 - Marking arithmetic and aggregation.....	75
15.1 The arithmetic.....	75
15.2 Every path, without exception.....	75
15.3 Aggregation is created by federation.....	76
15.4 The discipline that has already studied this.....	76
15.5 What else transfers.....	77
Threshold rules.....	77
Feasibility intervals.....	77
The non-perturbative constraint.....	77
A formal result.....	77
15.6 The differencing channel.....	78
Across requesters.....	78
Across time.....	78
Across records.....	78
Shape consistency.....	78
Differencing-aware audit.....	79
Windowed thresholds.....	79
Review queues.....	79
15.7 The honest position.....	79
Chapter 16 - Trust, attestation and key custody.....	80
16.1 Custody.....	80
16.2 Identity.....	80
16.3 Attestation.....	81
16.4 Forward secrecy.....	81
16.5 Key management is the real cost.....	81
16.6 Correlation without an enumerable key.....	82
The resolution.....	82
Why this shape, and not the alternatives.....	83

The online dependency, and its actual extent.....	83
Epochs and rotation.....	84
Costs, stated.....	84
A delivery fallback.....	84
16.7 What cryptography does not achieve.....	84
PART V - SPECIFICATION AND CONFORMANCE.....	86
Chapter 17 - The eight normative contracts.....	86
17.1 What conformance means.....	86
17.2 Why Disclosure is not optional.....	87
17.3 Ingress as a conformance point.....	87
17.4 The non-normative set.....	87
17.5 An open question.....	87
Chapter 18 - Conformance without capture.....	88
18.1 The failure mode.....	88
18.2 The alternative.....	88
18.3 What the suite must exercise.....	89
18.4 Governance separation.....	90
18.5 What this offers back.....	90
PART VI - GOVERNANCE AND ADOPTION.....	91
Governance as constitutional design.....	91
The relevant literature is institutional, not technical.....	91
Ostrom's conditions applied.....	92
The two gaps.....	92
Chapter 19 - The custodian question.....	93
19.1 The contradiction.....	93
19.2 Three futures.....	94
As a product.....	94
As an open standard.....	94
As an institutional programme.....	94
19.3 Sequencing.....	94
19.4 The constitution.....	95
19.5 Building the leaving first.....	96
Chapter 20 - Anti-capture.....	96
20.1 The mechanism to be prevented.....	96
20.2 Conformance without ratification.....	96
20.3 Supermajority and quorum.....	97
20.4 Change of control.....	97
20.5 The limit of governance.....	97
Chapter 21 - Adoption.....	98
21.1 The uncomfortable evidence.....	98
21.2 Levers.....	98

21.3 Pull the lever once.....	99
21.4 The empty store.....	99
21.5 The first source owner.....	99
PART VII - REALISATION.....	101
Chapter 22 - Implementation.....	101
22.1 Language.....	101
22.2 Baseline stack.....	101
22.3 Delivery in spirals.....	102
Spiral 0 - Contracts and walking skeleton. Two weeks.....	103
Spiral 1 - Normative core. Months 1-3.....	103
Spiral 2 - Assurance. Months 3-6.....	103
Spiral 3 - Capability. Months 6-12.....	103
Spiral 4 - Ecosystem. Months 12-18.....	104
22.4 Cross-cutting disciplines.....	104
Chapter 23 - Domain profiles.....	104
23.1 What a profile contains.....	104
23.2 Defence and national security.....	104
23.3 Policing.....	105
23.4 Health.....	105
23.5 Proving neutrality.....	106
PART VIII - ASSESSMENT.....	107
Chapter 24 - What this does not solve.....	107
Aggregation.....	107
Correctness of marking.....	107
Model governance.....	107
Residual correlation risk.....	107
Cross-record derivation at source.....	108
The empty assertion store.....	108
Source-side competence.....	108
Usability.....	108
Fragmented source-product markets.....	108
Graduated sanctions and dispute resolution.....	108
Chapter 25 - Threats to the thesis.....	109
25.1 "The burden simply moves to source owners, who will not carry it".....	109
25.2 "The index is a centralisation that X-Road avoided".....	109
25.3 "The Expectation becomes the new DIB".....	110
25.4 "Usability decides adoption, and this document barely addresses it".....	110
25.5 "A nationally-founded entity cannot credibly custody a multinational specification".....	110
25.6 "The guard argument will not be accepted in time to matter".....	111
25.7 "The unbundling analogy does not hold".....	111

25.8 "Cautious departments will get a poor tool and conclude the model does not work".....	112
Chapter 26 - Conclusion.....	112
Licence.....	114
Scope.....	114
Written work - Creative Commons Attribution 4.0 International (CC BY 4.0).....	114
Software - MIT Licence.....	115
Why permissive.....	115
What this licence is not.....	116
Authorship.....	116
Bibliography.....	117
1. NATO standards — labelling, binding, and reference architecture.....	117
2. Prior federation efforts.....	118
3. Statistical disclosure control.....	118
4. Cryptography.....	119
5. Institutional economics and commons governance.....	120
6. Conformance and standards governance.....	120
7. UK context.....	120
Suggested checking order.....	121
APPENDIX A - A worked example.....	122
A.1 Scope, and why this appendix exists.....	122
A.2 The Expectation.....	122
A.3 The source system.....	124
A.4 The mapping.....	125
A.5 The disclosure policy.....	126
A.6 The validator run.....	128
A.7 The disclosure package.....	129
A.8 Ingress verification.....	130
A.9 A callback, refused.....	131
A.10 The same policy in a policing vocabulary.....	131
A.11 What this appendix broke.....	133
D1 - The connector is stateful, and §10.4 implied otherwise.....	133
D2 - Correlation key management for hash-only disposition is unresolved.....	133
D3 - Two record labels were being conflated.....	133
D4 - The four-part header should be five-part.....	134
D5 - Canonical leaf ordering was underspecified across namespaces.....	134
A refinement rather than a defect.....	134
APPENDIX B - Comparative position.....	135

ABSTRACT

Organisations that hold information under a duty - security classification, statute, professional confidence or contract - routinely cannot share it with those who need it, and the obstruction is organisational rather than technical. The prevailing commercial response supplies imposed coherence: an external party arrives with a method, compels the organisation to write down what it means, and absorbs the political cost of the decisions. It works, and it produces a dependency that is extremely difficult to unwind.

This document locates that dependency precisely. It is not created by the ontology, the index, the graph or the interface, each of which is well preceded and replaceable. It is created by the integration model - vendor engineers embedded inside the customer, learning the source estate and encoding that knowledge in artefacts the customer cannot read, cannot verify and cannot transfer. Cost is linear in sources and the knowledge accrues privately.

An alternative is specified. The federation publishes an *Expectation*: a versioned, offline-testable exchange contract adopted unilaterally rather than negotiated. The source owner maps their own schema to it declaratively, declares field by field what may be released, binds the resulting disclosure cryptographically at source, and ships it. The platform verifies and holds no authority to classify and no key to what was withheld. Withheld fields are absent rather than encrypted; pointer fields resolve only by a callback the source may refuse, bound or revoke.

Four decades of prior attempts - MIP, NIEM, NNEC, DCGS and its integration backbone, FMN, X-Road, Gaia-X and International Data Spaces - are examined and reduced to ten transferable lessons that the proposal is then tested against. Seven contributions are claimed: the Expectation as a unilateral offline-testable contract; a JSON and post-quantum profile of the NATO metadata binding standard supporting field-level selective disclosure; identity as a read-time projection over an append-only assertion store rather than a pipeline stage; the comparative failure analysis itself; the connection of classification aggregation to statistical disclosure control, from which secondary suppression and disclosure by differencing are imported as first-class concerns; a formal model of integration authority yielding the result that switching cost is a linear function of privately-held integration knowledge, and therefore that lock-in under supplier-performed integration deepens with adoption independently of supplier conduct; and the application

of commons design principles to federated exchange, treating the contract, the registries and the conformance regime as the governed commons rather than the data.

The mechanism that changes the economics is mapping reuse. Mappings describe a source *product's* schema and contain no data, so they are publishable and reusable across every organisation running that product, and per-participant integration cost falls as the federation grows rather than remaining constant.

The larger claim is one of unbundling. The prevailing model sells the federation layer and the analytical capability as a single purchase, and it is the bundling that produces the dependency: the second cannot be bought without accepting the first. If the federation layer is instead a published contract with a conformance programme and an independent custodian, suppliers compete *above* the federation rather than *being* it - the same relocation of competition that occurred when networking moved from proprietary vendor stacks to a published protocol suite.

Aggregation, marking correctness and model governance are not solved, and are stated as such.

THESIS STATEMENT

Vendor dependency in information-sharing platforms is not produced by superior software. It is produced by where **integration authority** resides, and switching cost is a linear function of how much integration knowledge a supplier holds privately. Relocating the mapping and disclosure decision to the source owner, binding classification cryptographically at source, and publishing the exchange contract rather than negotiating it, therefore bounds switching cost by construction - producing better security properties at lower marginal cost, with sovereignty as a structural rather than contractual fact, across defence, policing, health and any domain where information is held under a duty.

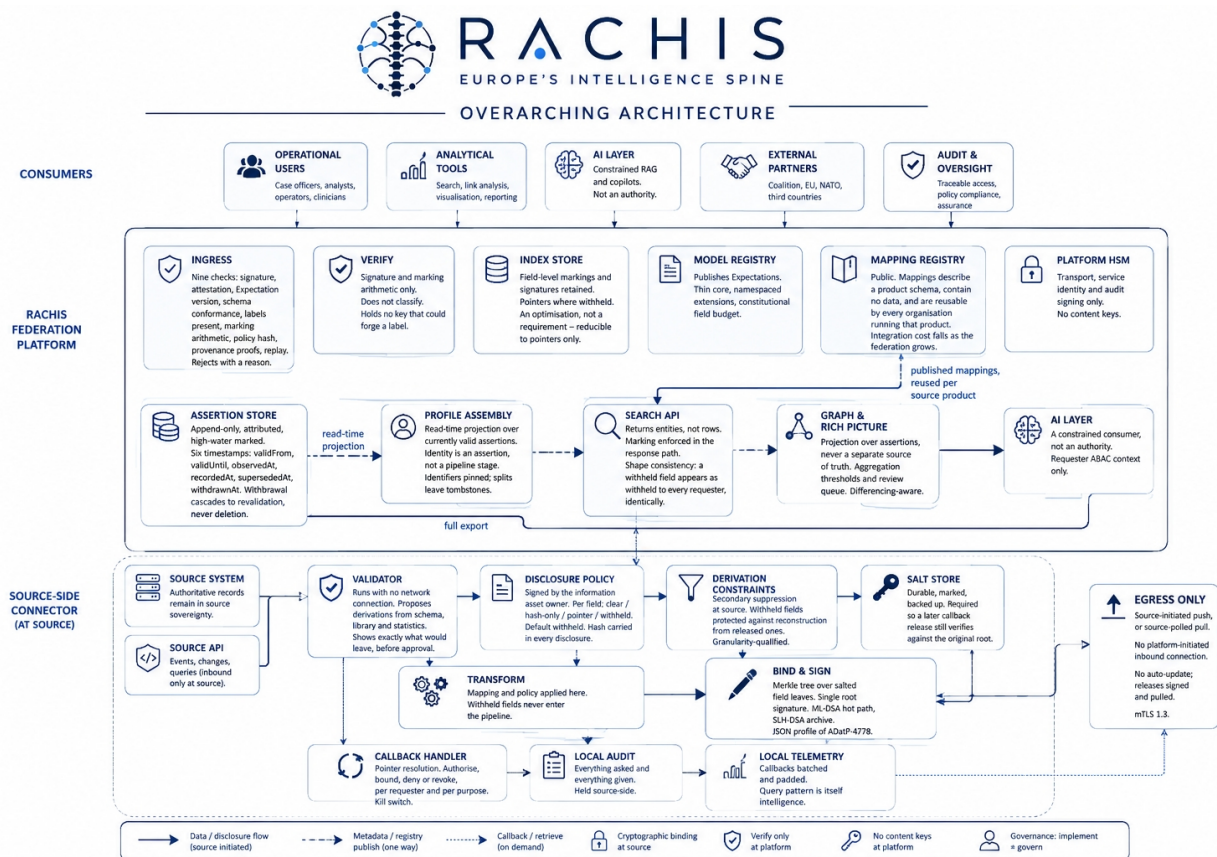


Figure 1: Rachis - High Level Architecture

The consequence is larger than a platform. It is the unbundling of the federation layer from the applications built on it, so that the layer becomes public infrastructure and suppliers compete above it rather than constituting it.

CONTRIBUTION CLAIMS

Seven, deliberately narrow. Everything else in this document is synthesis, and is labelled as such.

1. **The Expectation** as a unilateral, offline-testable exchange contract, distinguished both from negotiated bilateral agreements and from vendor-side integration.
2. **A JSON and post-quantum profile of ADatP-4778** supporting selective disclosure, such that a recipient holding a subset of a record's fields can verify authenticity without learning the existence or extent of the withheld remainder.
3. **Identity as an assertion** - entity resolution as a read-time projection over an append-only attributed store rather than a pipeline stage.
4. **A comparative analysis of federated information-sharing efforts** through the lens of source-owner incentive, yielding ten transferable lessons.
5. **The connection of classification aggregation to statistical disclosure control**, importing secondary suppression and disclosure-by-differencing as first-class concerns in security marking.
6. **A formal model of integration authority** (§§2.7–2.10), establishing that switching cost approximates the stock of privately-held integration knowledge; that under supplier-performed integration this stock grows linearly in the number of connected sources, so that lock-in deepens with adoption independently of supplier conduct or contract terms; and that under source-held integration authority it is bounded and does not grow. The model is stated with its falsification conditions.
7. **The application of commons design principles to federated information exchange** (Part VI), treating the contract, the registries, the conformance regime and the accumulated assertions as the governed commons - the data being never pooled - and using Ostrom's conditions as a diagnostic that surfaces two gaps in the governance proposed here.

READER'S GUIDE

Senior responsible owners will find the argument in Chapters 1–3, 21 and 26, and the exit case in Chapter 19. Risk owners and accreditors should read Chapters 8–12 and 15–16, which describe exactly what leaves a source estate and who decides. Architects and implementers will want Chapters 8–18 and Appendix A. Standards and policy readers are directed to Chapters 7, 10, 18 and Part VI. Readers approaching this from institutional



economics or procurement policy should read §§2.7–2.10 and Part VI's opening, which together contain the argument that the rest of the document exists to serve.

This document uses the Atkinson Hyperlegible font produced by the Braille Institute of America to enhance readability for all (copyright 2020 Braille Institute of America, Inc. The Font Software is licensed under the SIL Open Font License, Version 1.1 . This license is available with a FAQ at: <https://openfontlicense.org>).

PART I - THE PROBLEM

CHAPTER 1 - THE COORDINATION FAILURE

1.1 A QUESTION THAT CANNOT BE ANSWERED

Consider a question of the kind that is asked in operations rooms, custody suites and clinical handovers every day. *Has this vessel been seen before, and by whom? Does this individual appear in any other force's holdings? What else is known about this location?*

The questions are simple. The information required to answer them exists, is held lawfully, and is in most cases held by an organisation that would be willing to share it. The answers are nonetheless routinely unobtainable within the time in which they would be useful, and the reasons have almost nothing to do with the difficulty of the query.

They have to do with the fact that the information sits in thirty systems, owned by different parts of an enterprise or by different nations, governed by different rules, built at different times, and defended by people whose professional exposure is asymmetric - nothing accrues to them for saying yes, and everything accrues to them for saying yes to something that later goes wrong.

1.2 THE INSTINCT TO CALL THIS A TECHNICAL PROBLEM

The instinct of the technical community, when presented with this, is to reach for architecture. Federated query. A common data model. An enterprise service bus, a data lake, a data mesh, a knowledge graph, an ontology. The instinct is understandable and it is not entirely wrong. It is, however, insufficient in a way that four decades of expenditure ought by now to have made obvious.

The technology to query across federated stores has existed since the 1980s. What has not existed is a mechanism by which the owner of a source system can participate without either surrendering control of their holdings or absorbing an unbounded and unquantifiable risk. Absent that mechanism, the architecture is irrelevant, because the feeds never arrive.

The United Kingdom's current programme of work on connecting sensors, deciders and effectors offers an unusually candid illustration, and it is worth citing precisely because the observation comes from an independent institution rather than from an interested vendor. RUSI's assessment of the Digital Targeting Web notes that the organisation charged with

leading its implementation does so in an orchestrating capacity rather than a commanding one, with limited authority to compel. Parliament's own science and technology advisory service has separately noted the constraint that an opaque departmental digital estate places on streamlined process, and the risk that national autonomy is undermined by dependencies on international technology companies.

Neither observation concerns a shortage of architecture. Both concern the ability of an institution to secure the agreement of its own components.

1.3 INSTITUTIONS CANNOT AGREE WITH THEMSELVES

This is the condition underneath the four problems that the intelligence platform market claims to solve, and it is worth stating in its general form.

An organisation of any size holds information in units that were built to serve local purposes, are owned locally, and are governed locally. Any attempt to make that information collectively useful requires those units to agree on what things are called, what they mean, who may see them and under what conditions. That agreement is not a technical act. It is a political act, in which somebody's definition of a person, an asset, an incident or an encounter prevails over somebody else's, and in which somebody accepts a risk they were not previously carrying.

Large institutions are structurally poor at making such decisions about themselves. The authority required is usually distributed, the incentive to move is usually absent at the level where the decision must be taken, and the cost of delay is usually borne by someone other than the person deciding.

1.4 WHAT THE MARKET ACTUALLY SELLS

The prevailing commercial answer to this condition is to sell **imposed coherence**. An external party arrives with a method, a bill, and - decisively - no stake in the internal politics. It compels the organisation to write down what it means. It absorbs the blame for decisions the organisation could not make itself. The software that results is, in an important sense, the artefact of a consulting engagement rather than its purpose.

This works, and it should be said plainly that it works, because a critique which begins by denying the value of the incumbent model will not be read by anyone who has used it. Organisations that buy imposed coherence do get coherence. They get it faster than they

would have achieved it internally, and often they get it at all where internally they would not have.

What they also get is a dependency that is extraordinarily difficult to unwind, because the coherence lives in the vendor's platform rather than in the customer's institution. Chapter 2 examines exactly how that dependency forms, because the mechanism turns out to be more specific - and more avoidable - than is generally assumed.

1.5 THE ARGUMENT OF THIS DOCUMENT

The claim advanced here is not that the analytical capabilities of modern intelligence platforms are unnecessary, nor that they are technically difficult to reproduce. It is that the *integration model* by which those capabilities are connected to source information is a choice; that the choice was made for commercial rather than technical reasons; and that a different choice produces better security properties, lower cost, faster deployment, and sovereign control as a structural fact rather than a contractual assurance.

Chapter 2 makes this precise. Dependency is not a consequence of software quality; it is a linear function of how much integration knowledge a supplier holds privately, and it therefore deepens as a programme succeeds, whatever the supplier's conduct. That result is the reason the remedy has to be architectural rather than contractual.

1.6 THE LARGER CLAIM: UNBUNDLING

There is a further consequence, and understating it would misrepresent the work.

The prevailing model sells two things as one purchase: the **federation layer** - the contract by which information crosses organisational boundaries, the disclosure decision, the marking, the verification, the resolution of identity - and the **applications** built on top of it: search, link analysis, timelines, case management, models. It is the bundling that produces the dependency. A customer cannot buy the second without accepting the first, and having accepted the first they cannot change the second.

This is the condition that obtained in computer networking before the general adoption of a published protocol suite. Vendors supplied the network and the things that ran on it as a single stack, and changing one meant changing both. The transition that followed did not eliminate large suppliers, nor render networking a commodity in any pejorative sense. It relocated the point of competition: a customer could change what ran on the network

without renegotiating the network, and could change the network without renegotiating what ran on it.

The proposition of this document is the same relocation, one layer up. If the federation layer is a published contract with an executable conformance programme and an independent custodian, then **suppliers compete above the federation rather than constituting it**. The layer becomes infrastructure in the strict sense - a shared, generic resource whose value derives from the range of downstream uses it enables rather than from any one of them.

Whether that is achievable is a governance question rather than a technical one, and it is the subject of Part VI. Whether it is desirable is, for any government purchasing shared digital infrastructure, close to self-evident.

1.7 STRUCTURE

Part I sets out the problem, models its economics, and establishes that it is not specific to defence. Part II examines four decades of attempts to solve it and extracts ten lessons that any new proposal must satisfy. Part III specifies the mechanism. Parts IV to VIII address foundations, conformance, governance, realisation and honest assessment respectively.

CHAPTER 2 - WHAT CREATES THE DEPENDENCY, AND WHAT IT COSTS

2.1 LOCATING THE MOAT

It is common to hear that the incumbent platforms are difficult to displace because their capabilities are hard to replicate. This is worth examining, because if it is true then the argument of this document collapses, and if it is false then a great deal of procurement reasoning rests on a mistake.

Take the components in turn.

The ontology - an object, property, link and action model over federated sources, with mappings back to the systems that hold the data. This is a federated conceptual schema. The idea is forty years old, it is taught in undergraduate courses, and a competent team can specify a serviceable one for a given domain in a fortnight. Appendix A does so in a page.

The index - a searchable store supporting keyword, structured and semantic retrieval over indexed content. Multiple mature open-source implementations exist. None of them is where the money is.

The graph and link analysis - relationship traversal, path finding, temporal reconstruction. Well-precedented, widely implemented, and available commercially and open-source.

The interface - search, entity profiles, timelines, cases. Hard to do *well*, as Chapter 7 (L8) records at length, but hard in the way that all product design is hard rather than in a way that constitutes a technical moat.

Entity resolution - the one component that is genuinely difficult, and the one this document treats differently in Chapter 13.

None of these, individually or collectively, explains why an organisation that wishes to stop using a given platform finds that it cannot.

2.2 THE INTEGRATION MODEL

The dependency is created by how source information reaches the platform, and specifically by the following sequence, which is reproduced with minor variation across the sector.

A vendor engineer is embedded inside the customer's environment. Frequently they are cleared. Frequently they remain for years. Their function is to learn the customer's source systems - the schemas, the quirks, the undocumented semantics, the column that meant something different before the 2018 reorganisation, the table that nobody maintains but everybody relies on, the convention by which a null in one field signifies a condition rather than an absence.

That knowledge is then encoded. It is encoded in transformation logic, running on the vendor's infrastructure, expressed in the vendor's tooling, and governed by the vendor's release process. The resulting pipelines become load-bearing: reports depend on them, decisions depend on them, and after a period they are the only description in existence of how the customer's data actually behaves.

Each additional source repeats the cycle. Cost is linear in the number of sources connected. And the knowledge - which is the actual asset - accrues to the vendor rather than to the customer.

2.3 THE ECONOMICS

This produces a cost structure worth stating explicitly, because it is the structure that the alternative must beat.

For n sources, the prevailing model requires n engagements, each involving vendor staff, clearance processes, discovery, negotiation, build and test. Nothing learned in engagement k materially reduces the cost of engagement $k+1$ for the customer, because the learning is embedded in vendor-side artefacts specific to the source in question. The customer's own understanding of their estate does not improve. Their ability to change supplier does not improve. Their marginal cost per source does not fall.

The vendor's position improves markedly with each engagement. They accumulate a description of the customer's data estate that the customer does not possess, cannot easily reconstruct, and cannot transfer.

This is not an accusation of bad faith and should not be presented as one. It is a description of an incentive structure. Any supplier operating this model, however scrupulous, ends up holding something the customer needs and cannot readily obtain elsewhere. The dependency is emergent rather than designed, which is precisely why criticising the suppliers is a category error and criticising the architecture is not.

2.4 WHY SOURCES DECLINE, AND WHY THEY ARE RIGHT TO

The observable consequence is that feeds do not arrive. Programmes with funding, political backing and executive sponsorship find that eighteen months later they have connected four of the thirty systems they intended to, and the remaining twenty-six have reasons.

This is generally attributed to culture, to institutional conservatism, or to the reluctance of data owners to relinquish control. Those explanations are unsatisfying because they are unfalsifiable and because they attribute to irrationality a decision that is, on examination, entirely rational.

Consider what a source system owner is being asked to accept. A supplier's engineering presence inside their environment. A transformation layer over their authoritative record which they cannot fully read and cannot independently verify. A data model whose semantics are determined elsewhere. An arrangement they cannot exit without losing capability that will by then be operationally embedded. And a personal exposure, since it is

their name against the decision if the arrangement produces a disclosure they did not intend.

Against that, the benefit accrues largely to somebody else - the analytical community, the centre, the programme. The risk owner gains little and carries the residual.

Under those terms, declining is the correct decision, and the people declining are doing their jobs properly. Any architecture that requires them to behave otherwise in order to succeed has misidentified its problem.

2.5 THE INVERSION

The change proposed by this document is a single structural one, from which the remainder follows.

The federation publishes what it expects. The source owner decides what to provide, maps it themselves, marks it, signs it, and ships it.

No vendor engineer learns the source system, because the party that already understands it does the work. No transformation logic runs on platform infrastructure, because transformation happens at source under source change control. No opaque agent operates inside the estate, because the connector is open source, small enough to read in an afternoon, deployed by the source owner, and configured by a declarative file they wrote.

The economics change with it. One published contract, plus one declarative mapping per source - and, as Chapter 8 develops at length, mappings for a given source *product* are reusable by every organisation running it, so cost per participant falls as the federation grows rather than remaining constant.

Integration stops being a project and becomes a declaration.

2.6 A NECESSARY CONCESSION

None of the above is novel in the general case. Publishing a contract and having participants conform to it is how financial messaging works, how healthcare interoperability eventually started working, and how the web scaled - nobody integrates with HTTP; they implement it. Chapter 7 (L2) makes the same point about the specifically governmental precedents, and this document would be less credible if it pretended otherwise.

What is novel is the application to information held under a duty, where the *disclosure decision*, the *classification*, and the *cryptographic binding* of the two must all be exercised at source rather than assumed downstream. That is the subject of Part III.

2.7 THE ECONOMICS OF INTEGRATION AUTHORITY

The preceding sections assert a cost advantage. Assertion is not argument, and the claim is precise enough to be modelled. This section states the model; §2.10 states what would falsify it.

DEFINITIONS

- n participating source organisations
- k distinct source *products* among them, $k \leq n$
- D discovery - learning a source system's schema and undocumented semantics
- B build - constructing the transformation
- T test and assurance
- M mapping authorship, under the model proposed here
- m mapping *adaptation*, where the product is already mapped, $m \ll M$
- P disclosure policy - the organisational act of deciding what may be released

THE PREVAILING MODEL

Each source requires a bespoke engagement. Discovery is conducted against a particular deployment rather than against a product, and the resulting artefacts are supplier-held, so nothing learned at source i reduces the customer's cost at source $i+1$:

$$C_{\text{trad}}(n) = n(D + B + T) ; \partial C / \partial n = D + B + T, \text{ invariant in } n$$

The supplier's marginal cost does fall with experience. The customer's does not, and the difference between the two is captured as margin. This is not a criticism; it is the arrangement working as designed.

THE PROPOSED MODEL

Mapping is performed once per product and adapted thereafter; disclosure policy is performed once per organisation and is not reusable, because it is a statement about that organisation's holdings and obligations:

$$C_{\text{rachis}}(n) = kM + (n - k)m + nP$$

average cost = $(kM)/n + (1 - k/n)m + P \rightarrow m + P$ as n grows with k approximately fixed

RESULT 1 - MARGINAL COST

The marginal cost of the n th participant is $(D+B+T)$ in the prevailing model, invariant in n ; and $(m+P)$ in the proposed model for any participant whose product is already in the registry.

RESULT 2 - THE FLOOR

The irreducible cost of participation is P . It cannot be removed by tooling, reuse or automation, because it is a governance act requiring an accountable human decision-maker.

The second result deserves more than a line. **The floor of federation cost is the cost of deciding, which is exactly the cost that ought to be irreducible.** A model that drove participation cost to zero would have done so by removing the disclosure decision - which is the thing this architecture exists to protect. P is not a residual inefficiency in the design. It is the design.

2.8 REGISTRY COVERAGE

Let $\rho(n)$ be the probability that an arriving participant's source product already has a published mapping. Expected marginal cost is then:

$$E[c] = (1 - \rho)M + \rho m + P$$

ρ is governed not by n but by the concentration of the source-product market. Where a small number of products serve most organisations - the observed pattern in police records management, primary care administration, and defence logistics - ρ rises steeply

from the first few mappings. If the three most common products serve a fraction q of the population, then $p \geq q$ after three mappings.

RESULT 3

In a concentrated source-product market, registry coverage is achieved by a number of mappings that is small and *independent of n* .

This is the mechanism behind the claim in §21.3 that the adoption lever need be pulled only once. It is also the point at which the economic argument becomes checkable in advance: source-product concentration in a given domain is observable before any commitment is made, and a domain in which products are highly fragmented is one in which this architecture's economic advantage is materially weaker.

2.9 SWITCHING COST AND THE LOCATION OF KNOWLEDGE

The model above concerns entry. The thesis of this document concerns exit, and a single quantity connects them.

Let A_p be the stock of integration knowledge held by the supplier in a form the customer cannot read, verify or transfer.

PROPOSITION

Switching cost $S \approx A_p$.

To change supplier, a customer must reconstruct whatever the incumbent holds and they do not. Where the mapping, the semantics, the transformation logic and years of accumulated corrections are supplier-side artefacts, reconstruction means repeating discovery - at least nD , plus rebuild and re-assurance.

RESULT 4

Under supplier-performed integration, A_p grows linearly in n . Therefore switching cost grows linearly with adoption: **lock-in deepens as a direct function of the programme's success, independently of the supplier's conduct, contract terms or intentions.**

This is the formal statement of the political-economy claim made informally in §§2.3–2.4, and it explains why criticising suppliers is a category error rather than merely uncharitable. A scrupulous supplier operating this model produces the same S as an unscrupulous one,

because S is a property of where integration authority sits, not of who exercises it. The variable is architecture, not virtue. It also explains why contractual remedies - exit clauses, source-code escrow, data portability warranties - have so poor a record: they act on the symptom while the architecture continues to manufacture the cause.

Under the model proposed here, $A_p \approx 0$ by construction. The mapping is declarative, source-held and publishable. The disclosure policy is source-authored and signed. The assertions are exportable in a format whose interoperability is a conformance requirement (§18.3). The Expectation is public. What remains is operational familiarity - real, but small, and not growing in n .

RESULT 5

Under source-held integration authority, switching cost is bounded and does not grow with adoption.

Results 4 and 5 are the whole thesis, compressed. Everything in Parts III to VII is machinery for holding A_p at zero while still producing a useful federation.

2.10 PREDICTIONS, AND WHAT WOULD FALSIFY THEM

A model that cannot be wrong is not worth stating. Four consequences follow, each observable:

1. In any domain where integration is supplier-performed, incumbent displacement rates fall as the number of connected sources rises - and fall faster than differences in platform capability would explain.
2. Under a published-mapping regime, per-participant integration cost declines with cumulative *registry coverage*, not with elapsed time or supplier experience. The two are separable and distinguishable in data.
3. The advantage is proportional to source-product concentration. Where products are highly fragmented and k approaches n , the two models converge and the economic case in this document largely disappears.
4. Participation cost has a floor at P . Any deployment reporting a materially lower figure has either automated the disclosure decision or is not making one.

Prediction 3 is the important one, because it names the condition under which the argument fails rather than the conditions under which it holds. **A candidate domain should be assessed for source-product concentration before it is assessed for**

anything else in this document. Prediction 4 is the one most likely to be quietly violated in practice, and it is worth auditing for: a federation whose participants are onboarding suspiciously cheaply has probably delegated the disclosure decision to whoever wrote the tooling.

CHAPTER 3 - WHY THIS IS NOT A DEFENCE PROBLEM

3.1 THE GENERAL FORM

The argument so far has used the vocabulary of defence and national security. The structure it describes is not specific to that domain, and establishing this early matters for three reasons: it makes the architecture more likely to be right, since a mechanism that generalises has usually identified the real invariant; it substantially enlarges the population of organisations for whom the design is relevant; and it exposes the design to critique from communities that have thought about parts of the problem for longer.

Stated generally, the condition is this:

An organisation holds information under a duty. A wider community would benefit from being able to ask questions across that information and others like it. No mechanism exists by which the holder can participate without either surrendering control or accepting risk they cannot mitigate.

Every element of that statement is domain-neutral. The duty may arise from security classification, from statute, from a common-law obligation of confidence, from professional ethics, or from contract. The mechanism required is the same.

3.2 POLICING

Police forces hold intelligence under handling conditions that govern dissemination, source protection and retention. Forces are separate legal entities with separate information asset owners. Sharing between them is lawful, frequently necessary, and routinely difficult.

Three features distinguish the policing case usefully. Source protection is often the dominant sensitivity rather than the content itself, which makes the derivation problem of §9.5 acute - narrative detail can identify a reporting source even where no field names them. Purpose limitation is a *legal* constraint rather than good practice, and an access request without a stated lawful basis is not merely poor discipline but unlawful processing. And collateral intrusion - the acquisition of information about people who are not the

subject of interest - is a recognised category of harm with its own review requirements, which is the policing name for a problem this document meets again in Chapter 15 as aggregation.

3.3 HEALTH

Health systems hold records under consent, statutory basis, and a duty of confidence, distributed across primary care, secondary care, community and social care providers who are separate data controllers. The clinical case for sharing is direct and the sensitivity is high.

The health case contributes two things to the general architecture. Its sensitivity is *conditional* in a way defence classification is not: the same record may be freely shareable for direct care, restricted for planning, and prohibited for research, which makes purpose a first-class attribute of every access decision rather than a discretionary field. And re-identification risk is a mature discipline with decades of literature, established methods and working tooling - the same problem this document treats as aggregation, studied by a community that has never had cause to talk to the security marking community. Chapter 15 attempts to introduce them.

3.4 REGULATED INDUSTRY

Financial institutions, utilities and critical infrastructure operators hold information whose sharing is simultaneously required by regulation and constrained by commercial confidence and competition law. The pattern is the same: a holder under a duty, a community that would benefit, and no participation mechanism that preserves control.

The domain is included here for completeness rather than developed, since defence, policing and health are sufficient to demonstrate the invariant.

3.5 WHAT GENERALISES, AND WHAT DOES NOT

The design rule that emerges is stated once and applied throughout:

The marking vocabulary is pluggable. The marking arithmetic is not.

The *arithmetic* - that a derived artefact inherits the strictest marking among its inputs, that caveats union rather than intersect, that relaxation is a deliberate attributed act rather than an emergent property of a transformation - is domain-independent. It is the same

operation whether the labels are classifications, handling codes, consent states or commercial sensitivities.

The *vocabulary* differs entirely, and must. This is not an innovation of the present work: it is the pattern already established by the NATO labelling standards, where a security policy information file supplies the classification scheme against which labels are interpreted, and where the standard's own worked examples use a commercial classification policy rather than a military one. That the labelling standard demonstrates itself on a corporate scheme is direct evidence within the standards family that the mechanism was never defence-specific.

	Defence	Policing	Health	Regulated industry
Vocabulary	Classification, caveats, releasability	Handling codes, source protection	Consent state, lawful basis, opt-out	Commercial confidence, regulatory scope
Duty arises from	Security policy	Statute and common law	Consent, statute, confidence	Regulation and contract
Aggregation appears as	Classification uplift	Collateral intrusion	Re-identification	Sensitive inference
Purpose is	Good practice	A legal test	A legal test	Contractual
Adoption lever	Programme mandate, accreditation	Statutory duty, inspection	Statutory framework, funding	Regulatory requirement

Appendix A.10 demonstrates the point concretely rather than by assertion: the same disclosure policy structure, the same dispositions, the same derivation semantics, expressed in a policing vocabulary, differing in two substantive respects and no structural ones.

3.6 THE PROMOTION OF PURPOSE

One consequence of taking the generalisation seriously deserves stating in advance, because it changes an architectural priority.

In defence practice, recording why someone accessed a record is good discipline. In policing and health it is a legal test, and an access without a lawful basis and a declared purpose is a breach rather than an oversight. Purpose-based access control therefore ceases to be a refinement of role-based access and becomes a primary control, with

purpose as a mandatory, enforced, audited attribute of every request - including every callback in Chapter 12, where it appears among the required fields rather than the optional ones.

This makes the defence case stronger rather than diluting it. A defence deployment inherits a control designed to satisfy a legal standard, which is a better position than a legal deployment inheriting a control designed to satisfy a policy one.

3.7 A POSITIONING DISCIPLINE

A caution, since the temptation that follows from this chapter is obvious and would be damaging.

Claiming universality badly weakens every particular case. An architecture presented as applicable to defence, policing, health, finance and local government will be received in each of those communities as designed for none of them, and the reception will be correct. The discipline adopted here is therefore: lead with one domain for any given audience; prove neutrality by *shipping* a second profile rather than by asserting neutrality; and deploy the general case only in answer to the question of whether the design is parochial.

This document leads with defence and national security, demonstrates policing and health, and mentions regulated industry once.

PART II - WHAT HAS BEEN TRIED

Almost every idea in Part III has been attempted before by people with more money and better access than the present author. Part II examines those attempts in three traditions - the universal model, the federated infrastructure, and the sovereignty initiative - and Chapter 7 distils them into ten lessons that are used as tests throughout the remainder of the document.

The purpose is not scholarly completeness. It is to establish which failure modes are structural and therefore predictable, so that a new proposal can be assessed against them rather than against its own enthusiasm.

CHAPTER 4 - THE UNIVERSAL MODEL TRADITION

4.1 MIP AND THE JC3IEDM

The Multilateral Interoperability Programme has pursued command and control interoperability since 1998, with the aim of achieving international interoperability of C2 information systems at all levels in support of multinational operations. Its central artefact, the Joint Consultation, Command and Control Information Exchange Data Model, was ratified as a NATO standardisation agreement and supported by 27 nations.

The mechanism is precisely the one this document proposes. Each participating system maps its own information - which may be described by a proprietary data model - onto the common model, and back again. The common model is published; the mapping is local; participation does not require anyone to replace their system.

Two features of MIP deserve particular attention because they anticipate arguments made later here. It is explicit that the programme is not empowered to direct how nations develop their own systems, which is a governance posture rather than a technical one and is the correct posture. And it maintained a test reference system against which implementations could be exercised, which identified and resolved specification problems at an early stage and materially increased test coverage - a conformance capability of the kind Chapter 18 argues is essential and which, notably, the more recent NATO binding standards do not yet have.

The instructive failure is one of accretion rather than of design. The JC3IEDM conglomerates more than twenty years of modelling experience and merges a large number of separate information exchange requirements into a single model. The result is comprehensive and heavy. Its successor work has pursued a leaner information model, which is an admission worth noting.

4.2 NIEM

The National Information Exchange Model applied a comparable pattern to United States civil government: a published core model, domain extensions, documented exchange packages, conformance rules, and a public clearinghouse in which exchange packages are registered and made available for reuse.

Three observations transfer directly.

The clearinghouse instinct was correct and is adopted here in stronger form. Registering exchange artefacts so that others may reuse them is the mechanism by which per-participant cost falls, and Chapter 8 extends it from exchange packages to mappings.

Artefact complexity throttled it. NIEM's own guidance concedes that exchange package design, development and assembly may be difficult and time-consuming, particularly when undertaken manually, and that developers will generally prefer tool support. A published contract that requires a specialist to satisfy has reintroduced the specialist, and Chapter 9's insistence that mappings be declarative rather than procedural is a direct response.

Adoption was driven by mandate. Recipients of federal grants for projects implementing XML-based information exchange were required to use NIEM and to register the resulting artefacts. This is uncomfortable for anyone who would prefer good architecture to be self-recommending, and it is the central evidence for Chapter 7 (L9).

4.3 WHAT THE TRADITION ESTABLISHES

Mapping to a published contract works, at coalition scale, across sovereign boundaries, over decades. It is not a hypothesis in need of demonstration.

It establishes equally that published models accrete until participation becomes expensive, that the complexity of the exchange artefact governs adoption more than the elegance of the model, and that even a demonstrably workable pattern requires an external lever to achieve scale.

CHAPTER 5 - THE FEDERATED INFRASTRUCTURE TRADITION

5.1 NNEC, AND THE LIMITS OF THE PROGRAMMATIC APPROACH

NATO Network Enabled Capability set out to connect alliance and national systems into a coherent whole. It was ambitious, extensively documented, and did not deliver on its promise as experienced by forces deployed early in the Afghan area of operations.

The diagnosis matters more than the outcome. NNEC's structure assumed that federation could be specified centrally and then delivered, and that the resulting capability would arrive as a whole. It did not, and the reasons - scope, duration, the divergence between what was specified and what deployed forces needed, the impossibility of holding nearly thirty nations to a single delivery timetable - recur wherever the same structure is attempted.

5.2 AMN AND FMN

What worked instead was assembled under pressure. The Afghan Mission Network was constructed from national and NATO networks already present in theatre, connected because coalition partners could not otherwise share what they knew and were taking casualties as a result.

NATO institutionalised the lesson as Federated Mission Networking, and did so with unusual candour about its predecessor. The founding logic states that the capability cannot be developed as one large acquisition programme as NNEC envisaged, and must instead evolve through spiral development, with the requirements for each spiral established by military need.

Three principles from FMN are adopted directly in this document. Participants retain control of their own contributions rather than surrendering them to a central authority. Governance is maintained as a distinct concern from technology, with its own bodies and its own cadence. And delivery proceeds in increments that are individually useful, so that value does not depend on the completion of a distant end state.

5.3 DCGS-A AND THE DIB

The Distributed Common Ground System - Army is the most instructive failure in the record, and the most uncomfortable for the argument of this document, because its architecture was substantially the one proposed here.

The programme consumed more than three billion dollars over roughly a decade. An internal assessment concluded that it was not an enterprise system capable of replacing the multiple intelligence ground processing systems it was intended to supersede. Its lessons-learned reporting recorded soldiers working for more than ten hours on a network analysis product only to lose it to a system failure, and thereafter conducting link analysis in PowerPoint.

Its integration layer, the DCGS Integration Backbone, was by its own description a cohesive set of modular, community-governed, standards-based data services focused on enterprise information sharing, providing a common framework for exposing and transforming data and enabling users to discover and access information across a wide range of distributed sources. It included a metadata catalogue permitting search across distributed libraries in the manner of a web search engine.

Published contract. Community governance. Federated discovery. Search across sources that retain their holdings. Any reader who has reached this point will recognise the description.

It lost. It lost to a commercial platform with a superior interface and a forward-deployed integration model - the very model this document identifies as the source of dependency. And in the procurement dispute that followed, compatibility with the DIB was among the capabilities the Army asserted was unavailable from commercial products, in defending its decision not to solicit commercially. A standard created to open the estate was deployed to close it.

Two lessons follow, and they pull in different directions. Conformance becomes a moat when demonstrating it is expensive and the gate is held by an interested party - Chapter 7 (L4), discharged in Chapter 18. And architectural correctness confers no protection whatever against rejection at the workstation - Chapter 7 (L8), which this document discharges least well and says so.

5.4 X-ROAD

The counter-example, and the most durable system examined here.

Estonia's X-Road has operated for over two decades and now runs in more than twenty-five countries. Rather than storing information in a central repository, it links existing databases through a standardised protocol, so that each provider retains ownership and control of its own holdings. Custodianship rests with the Nordic Institute for Interoperability Solutions, a

non-profit established in 2017 by two governments and funded at around one million euros annually, with the software developed in the open.

Three findings transfer. Central state is a liability, and X-Road's longevity is not incidental to holding almost none of it - Chapter 7 (L7), which tells directly against the index this document proposes and is answered in §12.3. Custodianship can be modest, multi-governmental, and durable, which is the template for Chapter 19. And its own institutional assessment holds that the primary obstacles to interoperability are organisational, political and legal rather than technological - a conclusion reached from twenty years of operating a working system, which is a better basis than most.

CHAPTER 6 - THE SOVEREIGNTY TRADITION

6.1 GAIA-X

Gaia-X was launched to establish a European framework for data infrastructure sovereignty, with a substantial founding membership and an elaborate governance structure. Working prototype services were anticipated by the summer of 2021. Concrete implementations beyond limited proofs of concept did not materialise, and observers have characterised the intervening period in terms of consensus failure and analysis paralysis, with the initiative producing governance apparatus and presentations rather than software.

It also attracted a criticism that any successor must take seriously. Once major non-European cloud providers were admitted to membership, the initiative was widely held to have lost the purpose for which it had been created - the incumbents whose dominance motivated it acquired a seat at the table that defined the response to their dominance.

Two lessons, both discharged in Part VI. Governance established before there is running code produces governance without product - Chapter 7 (L10). And an open initiative without explicit anti-capture provisions will be captured, because the parties with the most resource to commit to its governance are precisely those with the most interest in its outcome.

6.2 INTERNATIONAL DATA SPACES

The International Data Spaces initiative and the Eclipse Dataspace Components that implement its protocol represent the most technically proximate prior art to the present work, and the comparison repays attention.

The IDS connector runs in the data provider's environment and mediates exchange. Usage policies are attached to data, describing permitted and prohibited uses. A metadata broker supports discovery. Participation is governed by identity and certification. Read at that level of abstraction, the resemblance to the architecture of Part III is close.

Two structural differences separate them, and both are decisive for the classified case.

Enforcement location. IDS attaches usage restrictions to transferred data and relies on the consumer's connector to honour them. The provider releases, and thereafter trusts. This is coherent where contractual recourse exists and commercial reputation is at stake. It is not a proposition a classification authority can accept, because it requires disclosure to precede control. The architecture here withholds at source, so that there is nothing downstream requiring enforcement - Chapter 7 (L5).

Agreement structure. IDS is built on contract negotiation between provider and consumer, with a defined negotiation protocol preceding transfer. Every additional participant multiplies the negotiations, and coordination cost grows quadratically. A published contract adopted unilaterally grows linearly - Chapter 7 (L6). For a federation of forty forces or two hundred providers, this is the difference between feasible and not.

A third observation from the IDS community bears directly on Chapter 9. Practitioners identify the practical description and enforcement of context-dependent usage policies, particularly by domain experts without software engineering backgrounds, as a major open challenge. That is precisely the problem the declarative disclosure policy and the validator are intended to address, and the fact that a mature adjacent community regards it as unsolved should be read as a caution rather than an opportunity.

6.3 NATO'S DATA CENTRIC REFERENCE ARCHITECTURE

The most significant finding of this survey is that the architectural direction argued for in this document has already been endorsed at alliance level.

The Data Centric Reference Architecture for the Alliance, approved in May 2025 and issued as a NATO Unclassified reference architecture, describes an approach built on data-as-a-product principles, standardised interfaces, and federations of data meshes and data spaces. It states that a federated model supports national sovereignty by permitting decentralised governance of data assets while adhering to common alliance frameworks and standards.

That is the thesis of this document, ratified thirteen months before it was written.

The consequence is a substantial reframing and a considerable strengthening. This work does not propose an alternative to the endorsed direction; it proposes the concrete disclosure mechanism that the endorsed direction requires and does not specify. A reference architecture describes what should be true. It does not say how a source owner decides what to release, how that decision is bound to the released data, how a recipient verifies the binding, or how a withheld field is later obtained. Those are the questions Part III answers.

The related labelling standards are similarly established rather than aspirational. Confidentiality label syntax and metadata binding are both ratified, with binding profiles published for a range of formats and protocols, and alliance work on data-centric security maturity is presently converging on the level at which cryptographic binding and encryption enter scope. Chapter 10 positions its contribution as a profile within that framework rather than as a competitor to it.

6.4 THE GAP

Taken together, the three traditions leave a specific and narrow gap.

The universal model tradition established that mapping to a published contract works, and that the contract must be kept light and the artefacts reusable. The federated infrastructure tradition established that participants must retain control, that delivery must be incremental, that central state is a liability, and that conformance must not become a gate. The sovereignty tradition established that governance must follow running code and resist capture, that enforcement must sit with the party bearing the risk, and - in the alliance's own reference architecture - that federated, sovereignty-preserving data sharing is the endorsed direction.

What none of them supplies is a mechanism by which a source owner declares, in a form their own governance can approve and their own staff can audit, exactly what may leave their estate; binds that declaration cryptographically to the data at the point of release; and retains the ability to refuse, revoke and disconnect thereafter.

That mechanism is the subject of Part III.

CHAPTER 7 - TEN LESSONS

The preceding three chapters surveyed four decades of attempts to make information cross organisational boundaries: the universal-model tradition of MIP and NIEM, the federated-infrastructure tradition of NNEC, DCGS, FMN and X-Road, and the sovereignty tradition of Gaia-X, International Data Spaces and NATO's Data Centric Reference Architecture.

That survey is not decoration. Almost every idea in the remainder of this document has been tried before, in some form, by people with more money and better access than the present author. The purpose of this chapter is to extract from those attempts the constraints that any new proposal must satisfy, and to state them precisely enough that they can be used as tests rather than sentiments.

Ten lessons follow. Each is stated as a claim, evidenced from the record, and translated into a demand on the architecture. Every subsequent design decision in this document cites the lesson it answers, or is explicitly marked as unforced. Where a lesson is imperfectly discharged, that is said here rather than left for a reader to discover.

The plan for this work anticipated eight. The evidence supported ten, and merging them further would have concealed distinctions that do real work later.

L1 - FEDERATION CANNOT BE PROCURED AS A PROGRAMME

THE CLAIM

Large single-acquisition efforts to deliver cross-organisational information sharing fail with striking regularity, and the ones that succeed are assembled incrementally under operational pressure.

THE EVIDENCE

NATO's Network Enabled Capability was ambitious and did not deliver on its promise, as experienced by forces deployed early in the Afghan area of operations. What worked was the Afghan Mission Network, assembled from national and NATO networks already in the field. When NATO institutionalised the lesson as Federated Mission Networking, it did so with the explicit finding that the capability could not be developed in one large acquisition programme as NNEC had envisaged, and would instead evolve through spiral development with each spiral's requirements set by military need. On the national side, DCGS-A

consumed more than three billion dollars and a decade before an internal assessment concluded it was not an enterprise system capable of replacing the multiple intelligence ground processing systems it was meant to supersede.

THE DEMAND

Delivery in spirals, each of which is operationally useful standing alone and none of which depends on the completion of a later one. No phase of the roadmap may be justified solely as a precondition for a subsequent phase. Discharged in Chapter 22.

L2 - MAPPING TO A PUBLISHED CONTRACT IS A PROVEN PATTERN, NOT AN INNOVATION

THE CLAIM

The core mechanism proposed in this document - a published common model against which each source maps its own schema - is thirty years old and has worked at coalition and national scale.

THE EVIDENCE

The Multilateral Interoperability Programme has operated on precisely this basis since 1998: each command and control information system maps its information, which may be described by a proprietary data model, onto the JC3IEDM and back again. The result was ratified as a NATO standardisation agreement and supported by 27 nations. NIEM applied the same pattern at national scale across justice, homeland security and human services, with a published core model, documented exchange packages and a public clearinghouse. X-Road has linked government databases through a standardised protocol for over two decades and now operates in more than 25 countries.

THE DEMAND

Intellectual honesty first: the is not novel and must not be presented as such, because doing so invites dismissal by anyone who has heard of MIP. What is new is what is added to the pattern - the disclosure decision, the cryptographic binding, and the unilateral rather than negotiated character of adoption. Discharged in Chapter 8, and reflected in the contribution claims above.

L3 - THE EXCHANGE CONTRACT GROWS HEAVIER THAN ITS USERS CAN BEAR

THE CLAIM

Published common models accrete. Left ungoverned, they become so large, and their exchange artefacts so laborious to construct, that the cost of participation rises to meet the cost of the bespoke integration they were meant to replace.

THE EVIDENCE

JC3IEDM conglomerates more than twenty years of command and control modelling experience and merges a large number of separate information exchange requirements into a single model - an achievement, and a warning. NIEM's own specification concedes that exchange package design, development and assembly may be difficult and time-consuming, especially if done manually, and that developers will generally prefer tool support to reduce the complexity of designing, constructing, changing and managing them. A published contract that requires specialist tooling and specialist knowledge to satisfy has reintroduced the specialist.

THE DEMAND

Three things. A thin normative core, with domain-specific requirements carried in namespaced extensions that never migrate inward without demonstrated multi-party need - a constraint that belongs in the constitution rather than in engineering judgement. A mapping artefact that is declarative rather than procedural, readable by a non-programmer, and diffable in a change board paper. And tooling that generates a candidate mapping by introspecting the source schema, so that the source owner edits rather than authors. Discharged in Chapters 8, 9 and 19.

L4 - CONFORMANCE BECOMES A MOAT UNLESS IT IS FREE, SELF-SERVE AND END-USER REPRODUCIBLE

THE CLAIM

A published integration standard confers advantage on whoever controls certification. Where conformance is expensive to demonstrate and the gate is held by the incumbent programme, the standard becomes a barrier to entry rather than a lowering of it - the precise inversion of its purpose.

THE EVIDENCE

The DCGS Integration Backbone was, by its own description, a cohesive set of modular, community-governed, standards-based data services for enterprise information sharing, with a metadata catalogue supporting search across distributed sources. It is structurally the architecture proposed here. In the litigation that followed the DCGS-A2 solicitation, compatibility with the DIB was among the capabilities the Army asserted was unavailable from commercial items, in defending its decision not to solicit a commercial product. A standard intended to open the estate was deployed to close it.

The pattern recurs in the standards this document builds upon. ADatP-4778's conformance model requires a Protocol Implementation Conformance Statement, extra information for testing, approved test cases and scripts, and testing by an authorised laboratory. The standard-related document acknowledges that no formal reference facility for testing external implementations has been established and that the testing capability remains under investigation. That was published in December 2020.

The counter-example is instructive. The Certified Kubernetes programme invites any vendor to run an open test suite and submit the results for review; participation is free to foundation members and non-profits; certification is versioned to the current release and the two preceding it; and the programme states as an explicit objective that the self-tests should be verifiable by end users of each qualifying offering. Ninety-plus certified offerings exist. The programme works because running it is cheap, submitting is public, and the customer can re-run the vendor's tests themselves.

THE DEMAND

Self-certification with published evidence rather than a gate. A conformance suite that is free, containerised, runnable offline and against any implementation including the reference one. An audit right rather than an approval right. And the end-user reproducibility objective adopted verbatim, because it is the specific mechanism that prevents the certifier becoming the bottleneck. Discharged in Chapter 18.

L5 - ENFORCEMENT MUST SIT WHERE THE INCENTIVE SITS

THE CLAIM

Control mechanisms located away from the party who bears the risk are honoured unevenly and cannot be verified by that party. The only durable enforcement point is the one the risk owner operates.

THE EVIDENCE

The International Data Spaces architecture attaches usage restrictions to data and enforces them at the consumer side - the data provider transmits, and the recipient's connector is trusted to honour the attached policy. This is a coherent design for commercial data ecosystems with contractual recourse. It is not one a classification authority can accept, because it requires the source to release the information first and rely on the recipient's software to constrain what happens next.

The prevailing commercial intelligence model exhibits the same displacement in a different direction. Transformation logic is authored and operated by vendor engineers on vendor infrastructure. The source owner cannot fully read it, cannot independently verify it, and cannot reconstruct it if the relationship ends. The incentive to preserve that arrangement sits with the vendor; the risk of it sits with the source.

THE DEMAND

Withholding happens at source, before egress, so that there is nothing downstream to enforce. Fields the policy does not release do not enter the transform pipeline, are not encrypted-and-retained, and are not reconstructible by the platform. The platform's inability to read them is a property of what it holds rather than of how it behaves. Discharged in Chapters 9 and 11.

L6 - BILATERAL AGREEMENT DOES NOT SCALE; PUBLISHED UNILATERAL CONTRACTS DO

THE CLAIM

Federations that require each pair of participants to negotiate terms grow their coordination cost quadratically and stall. Federations that publish a contract once and let participants adopt it independently grow linearly and keep growing.

THE EVIDENCE

IDS builds on contract negotiation between provider and consumer, with a policy handshake and negotiation protocol. Every new participant multiplies the negotiations. By contrast MIP, NIEM and X-Road each publish a technical contract once and are adopted many times over; X-Road still requires federation and data-exchange agreements for cross-ecosystem trust, but the *technical* terms of participation are settled in advance and identically for everyone.

THE DEMAND

The *Expectation* is published, versioned and adopted unilaterally. A source owner may connect without negotiating with any other participant, and the disclosure decision is likewise unilateral - no consumer has standing to negotiate what a source releases. Bilateral instruments remain necessary for legal and trust purposes, but they are not preconditions for technical participation. Discharged in Chapter 8.

L7 - CENTRAL STATE MUST BE MINIMISED, AND EACH ELEMENT OF IT SEPARATELY JUSTIFIED

THE CLAIM

The durability of a federation is inversely related to how much it holds. Every item of centrally-held state is a governance liability, a security liability, and a reason for a cautious source to decline.

THE EVIDENCE

X-Road's longevity is not incidental to its design: rather than storing data in a central repository, it links existing databases through a standardised protocol, so that each provider retains ownership and control. It has run for over two decades and spread to dozens of countries while holding, itself, essentially nothing. Efforts proposing substantial central holdings have fared markedly worse.

This lesson bears directly against the architecture proposed here, which does hold an index and an assertion store. That is an asymmetry, and it must be answered rather than finessed.

THE DEMAND

Every element of central state is justified individually or removed. The index is an optimisation rather than an architectural requirement: a federation electing to disclose nothing but pointers must still function, degrading in latency rather than in capability, which reduces the system to something X-Road-shaped when a participant wishes it to. The assertion store - the one genuinely novel central artefact - is exportable in full and hostable on the customer's own infrastructure. Discharged in Chapters 12 and 14.

L8 - THE INTERFACE DECIDES ADOPTION, WHATEVER THE ARCHITECTURE SAYS

THE CLAIM

Correctness at the data layer confers no protection against rejection at the workstation. Users abandon systems that obstruct them, and they abandon them for whatever is to hand.

THE EVIDENCE

The Army's own lessons-learned reporting on DCGS-A recorded soldiers working for over ten hours on a network analysis product only to have it disappear after a system failure or reset, and thereafter conducting link analysis in PowerPoint. A three-billion-dollar programme was displaced, in the field, by presentation software - not because its architecture was wrong in any respect a reader of this document would recognise, but because it did not survive contact with the people who had to use it.

THE DEMAND

The reference interface is a product with usability as an explicit objective, not a demonstration harness. Adoption is measured by analyst task completion, not feature coverage. Offline and degraded-mode behaviour is a first-class requirement rather than a later increment.

HONEST NOTE

This is the lesson least well discharged by the present work. The architecture is specified in detail; the interface is not, and the roadmap allocates it a single phase. That imbalance is stated in Chapter 25 among the threats to the thesis, and it should be read as an admission rather than an oversight.

L9 - ADOPTION FOLLOWS MANDATE OR IMMEDIATE SELF-INTEREST, NEVER ARCHITECTURAL MERIT

THE CLAIM

No significant information-sharing federation in the record scaled because participants were persuaded by its design. Each scaled because a legal instrument, a funding condition, or an operational necessity made participation the path of least resistance.

THE EVIDENCE

NIEM's reach owes much to the requirement that recipients of grants for projects implementing XML-based information exchange use it, and register the resulting artefacts publicly. X-Road grew inside a legal framework built alongside it, with a funded institutional custodian. The Afghan Mission Network was assembled because coalition partners were taking casualties and could not share what they knew. Federated Mission Networking followed with command direction behind it. Gaia-X, which had neither a mandate nor an immediate operational necessity, delivered no concrete implementations beyond limited proofs of concept against a timeline that had promised working prototype services by 2021.

THE DEMAND

Identify the specific lever in each domain - statutory duty, inspection regime, funding condition, accreditation route, operational imperative - and treat acquisition of the first source as a political problem with a technical answer rather than the reverse. Then make the second and fortieth sources close to free, through mapping reuse, so that the lever need only be pulled once. Discharged in Chapters 21 and 8 respectively.

L10 - CUSTODIANSHIP FOLLOWS RUNNING CODE, AND MUST BE CAPTURE-RESISTANT FROM THE FIRST DAY

THE CLAIM

A standards custodian established before there is anything to steward produces governance without product. A custodian established without defences against its own membership produces product without independence.

THE EVIDENCE

Gaia-X was launched with a European governance structure and a substantial membership, and did not deliver concrete implementations beyond limited proofs of concept; observers

describe lack of consensus and analysis paralysis while the intended beneficiaries lost patience. It also attracted a specific criticism worth taking seriously: that once the major non-European cloud providers were admitted, the initiative lost the purpose for which it was created. Kubernetes took the opposite sequence - unilateral development until adoption made donation meaningful, and only then transfer to a foundation, with a certification programme whose change control, test mechanics and process governance sit with three separate bodies. X-Road followed a comparable path, with a modest non-profit institute founded by two governments and funded at around one million euros a year, established after the technology was already in national production.

THE DEMAND

Announce custodian intent immediately, because it costs nothing and answers the lock-in objection at first contact. Write the constitution now, alongside the draft contracts, because the question of what may never be proprietary is a question about specific interfaces. Donate after independent implementations exist, because a specification with a single implementation is a product carrying governance overhead. And write anti-capture provisions from the outset: implementers may conform without acquiring the right to ratify, and the balance of ratification rests with sovereign customers rather than suppliers. Discharged in Chapters 19 and 20.

SUMMARY

#	Lesson	Principal evidence	Discharged in
L1	Federation cannot be procured as a programme	NNEC, DCGS-A, FMN	Ch 22
L2	Mapping to a published contract is proven, not novel	MIP/JC3IEDM, NIEM, X-Road	Ch 8
L3	The exchange contract grows heavier than its users can bear	JC3IEDM, NIEM IEPDs	Ch 8, 9, 19
L4	Conformance becomes a moat unless free and reproducible	DIB, ADatP-4778, Kubernetes	Ch 18
L5	Enforcement must sit where the incentive sits	IDS, the forward-deployed model	Ch 9, 11
L6	Bilateral agreement does not scale	IDS vs MIP, NIEM, X-Road	Ch 8
L7	Central state must be minimised and justified	X-Road	Ch 12, 14

#	Lesson	Principal evidence	Discharged in
L8	The interface decides adoption	DCGS-A	Ch 22, 25
L9	Adoption follows mandate or self-interest	NIEM, X-Road, AMN, Gaia-X	Ch 21, 8
L10	Custodianship follows running code, and must resist capture	Gaia-X, Kubernetes, X-Road/NIIS	Ch 19, 20

Two of these are uncomfortable for the present proposal. L7 tells against holding a central index at all, and L8 identifies the dimension on which this work is weakest. They are retained in their strong form for that reason. A set of lessons chosen to flatter the design that follows would be worth nothing, and would be recognised as such by exactly the readers whose agreement matters.

PART III - THE MECHANISM

The three chapters of Part II established that the pattern proposed here - sources mapping their holdings onto a published contract - is well precedented and has succeeded at coalition and national scale. What has not been done is to combine that pattern with a disclosure decision exercised at source, a cryptographic binding that makes classification a matter of verification rather than administration, and a contract adopted unilaterally rather than negotiated.

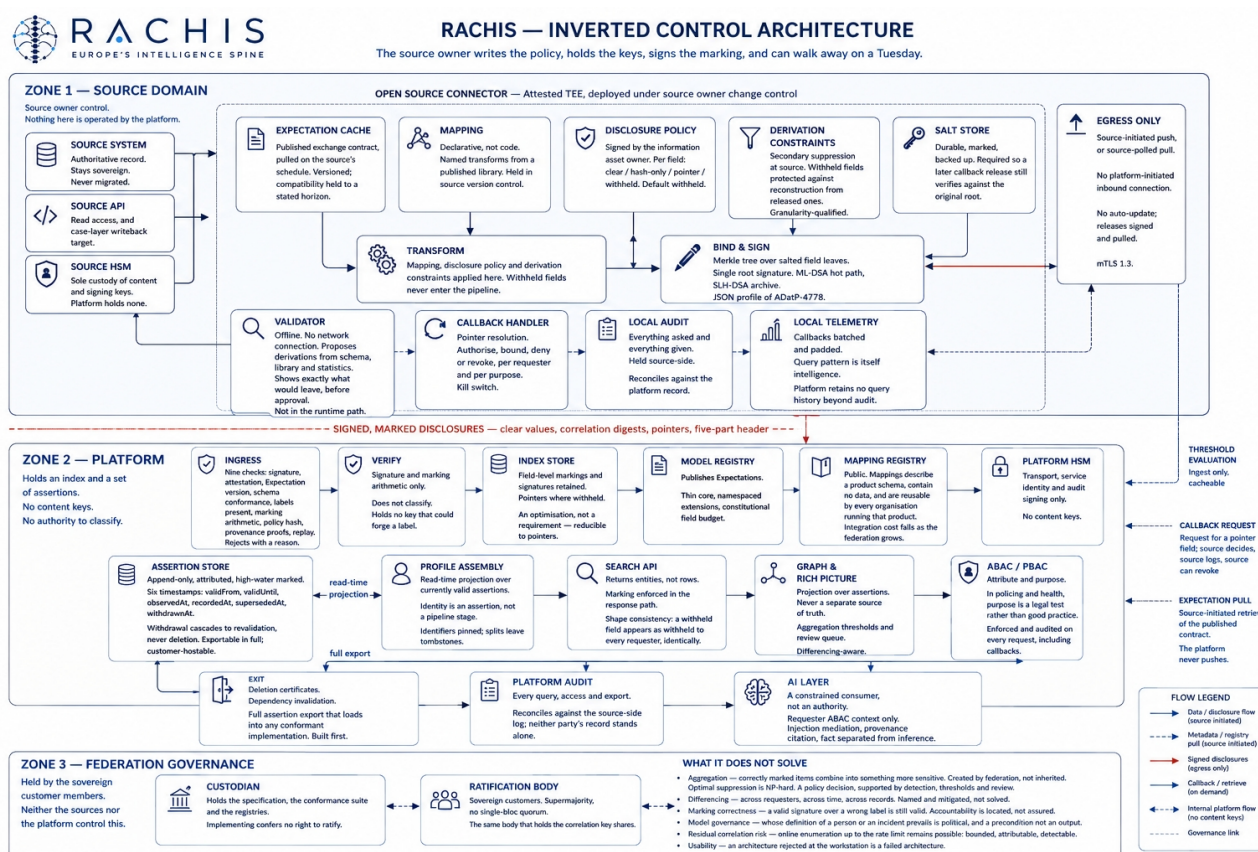


Figure 2: RACHIS Inverted Control Architecture Lower Level Design

Part III specifies that combination. Chapter 8 defines the Expectation. Chapter 9 separates the technical act of mapping from the governance act of disclosure, and introduces derivation constraints as the source-side answer to secondary suppression. Chapter 10 sets out the binding as a profile of an existing NATO standard rather than a new one. Chapter 11 describes verification at the boundary and argues that the entry point must

have no classification authority whatsoever. Chapter 12 deals with graduated disclosure, the callback path, and the leakage channels that both create.

CHAPTER 8 - THE EXPECTATION

8.1 DEFINITION

An **Expectation** is a published, versioned, machine-readable statement of what a federation wishes to receive. It is authored by the primary customer through the model registry - not by the supplier of any implementation, and not by the platform operator.

For each entity type in scope, an Expectation specifies:

- **Fields**, with names, types, cardinality and format constraints
- **Obligation**, distinguishing fields required for a record to be valid at all from those that are optional
- **Enumerations**, where controlled vocabularies exist, with the source's own values mapped rather than replaced
- **Marking requirements**: which fields must carry a confidentiality label, which policy identifiers are acceptable, and which handling caveats the federation will honour
- **Validation rules**, published in advance, so that the checks applied at ingest are knowable before anything is sent
- **Temporal obligations**: which of the bitemporal timestamps of Chapter 14 must be supplied and which the platform derives
- **Provenance obligations**: what must accompany a record for it to be attributable

None of this is new, and Chapter 7 (L2) says so plainly. The JC3IEDM has done substantially this since 1998, and NIEM's exchange packages perform an equivalent function at national scale. The Expectation earns its place in this document not through its contents but through four properties of how it is published, adopted and tested.

8.2 PULLED, NEVER PUSHED

A source owner's connector retrieves the current Expectation on a schedule the source owner sets, over a connection the source owner initiates. The platform has no route into the source domain and acquires none. Schema evolution reaches a source only when that source goes looking for it.

This is a small design decision with a disproportionate consequence. It means the entire architecture can be stated to a risk owner in a single negative: *nothing we operate can reach into your estate*. No inbound firewall rule, no persistent agent answering to a supplier, no update pushed on someone else's timetable. Chapter 7 (L5) requires that enforcement sit where the incentive sits; a pulled contract is the same principle applied to configuration.

It follows that there is no auto-update. Connector releases are signed and pulled by the source owner under their own change control. A connector that phones home for new versions has reconstructed the intrusive agent with better manners, and the claim of inverted control would be cosmetic.

8.3 VERSIONED, WITH COMPATIBILITY AS A GOVERNANCE COMMITMENT

An Expectation carries a canonical identifier and a version identifier, following the pattern already established in the NATO binding profiles: a stable canonical URN identifying the contract, and a versioned URN identifying the particular edition. Subsequent versions retain the canonical identifier and define new version identifiers, with deprecation stated explicitly.

A source conformant to version n remains conformant when version $n+1$ publishes, until a declared deprecation horizon. Sources are never broken by a change they did not choose to adopt.

This is a governance commitment before it is a technical one, and it belongs in the constitution rather than in an engineering standard. A custodian that can silently invalidate existing mappings holds the same power over participants that a vendor holds over a customer whose pipelines it operates. The commitment must therefore bind the custodian, be stated in the licence, and carry a defined minimum horizon.

8.4 TESTABLE OFFLINE

A source owner can validate a candidate mapping against an Expectation with no network connection to the platform whatsoever, and can see precisely what would be disclosed, field by field, before anything is disclosed.

This is the single most important property in the design, and it does three separate jobs.

First, it is the trust property. Every other assurance in this document - the signing, the attestation, the audit reconciliation - asks a risk owner to believe something about a

system they do not operate. Offline validation asks them to believe nothing. They run a tool inside their own estate, on their own data, and read the output. The claim "we cannot see what you do not send" becomes checkable by the person who bears the consequence of it being false.

Second, it is the answer to Chapter 7 (L4). Conformance becomes a moat when demonstrating it is expensive and the gate is held by an interested party. If any source owner can determine conformance locally, at zero cost, without submitting anything to anyone, there is no gate to hold. The certification programme of Chapter 18 rests on the same mechanism: the tests run anywhere, against anything, including the reference implementation.

Third, it decouples participation from procurement. A source owner can evaluate the entire proposition - what would leave, in what form, under what marking - before signing anything, requesting funding, or admitting a supplier to a conversation. The failure mode identified in Chapter 7 (L9), where the cost of the first step exceeds any individual's authority to approve it, is substantially mitigated by making the first step free and internal.

8.5 UNILATERAL ADOPTION

A source owner adopts an Expectation without negotiating with any other participant. There is no contract handshake, no policy negotiation protocol, no bilateral agreement as a precondition of technical participation.

Chapter 7 (L6) records why this matters. Architectures built on pairwise negotiation grow their coordination cost quadratically; those built on published contracts grow linearly. The International Data Spaces model, which requires provider and consumer to negotiate terms before transfer, is coherent for commercial ecosystems with contractual recourse and impractical for a federation of forty police forces or two hundred health providers.

Legal instruments remain necessary - a federation agreement, a data-sharing agreement, an information-sharing protocol under the relevant statutory regime. The claim is narrower and precise: those instruments are not preconditions for *technical* participation, they do not vary the technical contract, and they need not be renegotiated when a new participant joins.

Note the symmetry. Adoption is unilateral, and so is disclosure. No consumer has standing to negotiate what a source releases. A requester may ask, through the callback path of Chapter 12, and the source may refuse without explanation.

8.6 THE MAPPING REGISTRY AND THE COST CURVE

The preceding sections describe a system whose per-source integration cost is low. The following describes the mechanism by which it becomes lower as the federation grows, which is the strongest economic argument available to this thesis.

MAPPINGS ARE PUBLISHED, AND MAPPINGS ARE REUSABLE.

Where many organisations operate the same underlying product - a records management system used by forty police forces, a patient administration system deployed across a health economy, a logistics package fielded by several nations - the mapping from that product's schema to a given Expectation is substantially the same for every deployment. The first organisation to write it does the work; the thirty-ninth adopts it and edits the local variations.

This is safe to publish, and the reason is worth stating carefully because it will be challenged. A mapping describes the relationship between a *product's* schema and a *published* contract. It contains no data. It reveals nothing about what any particular organisation holds, how much of it, or about whom. It does not contain the disclosure policy, which is where organisational sensitivity actually lives and which is never published. What a mapping discloses is that a given product's `PERSON_DTL.SURNAME_TXT` corresponds to the Expectation's `person.familyName` - information already available to anyone with the product's documentation.

The economic consequence is a different curve, not a better position on the same one. The forward-deployed model's cost is linear in sources, borne separately for each engagement, with the resulting knowledge accruing privately to the supplier. The registry model's cost is sub-linear, borne once per product rather than once per deployment, with the knowledge accruing publicly to the community of participants.

NIEM had the right instinct here and was throttled by artefact complexity - its own guidance concedes that exchange package design and assembly is difficult and time-consuming when done manually. Chapter 7 (L3) requires that the artefact be light enough for reuse to be genuinely cheaper than authorship. That constraint governs the mapping format specified in Chapter 9 and is the reason the format is declarative.

The registry also supplies the answer to Chapter 7 (L9). If adoption requires a mandate or an operational imperative, and such levers are scarce and expensive to pull, then the design objective is to require the lever exactly once. Mapping reuse means the second,

tenth and fortieth participants face a cost small enough to fall below the threshold at which anyone needs to be compelled.

8.7 KEEPING THE CORE THIN

Chapter 7 (L3) is the lesson most likely to be violated by success. Every participant who joins arrives with a legitimate requirement the core model does not meet, and each accommodation is individually reasonable. Twenty years later the model conglomerates two decades of accumulated requirements and the cost of participation has risen to meet the cost of the bespoke integration it replaced.

Three provisions, of which only the first is technical:

NAMESPACED EXTENSIONS

Any participant may extend an Expectation within a namespace they control. Extensions are carried, indexed and marked exactly as core fields are, and a record bearing extensions remains conformant. The technical cost of extension is therefore near zero, which removes the pressure to expand the core.

A CONSTITUTIONAL CONSTRAINT ON CORE GROWTH

Migration of a field from an extension namespace into the core requires demonstrated use by multiple independent participants, and is ratified by the custodian under the rules of Chapter 19 - not by an implementer, and not by whichever participant is currently the largest. The constraint belongs in the constitution because engineering judgement will not hold against sustained institutional pressure.

A PUBLISHED SIZE DISCIPLINE

The core Expectation for a domain has a stated field budget, and exceeding it requires the same ratification as any other constitutional change. This is crude, and it is crude deliberately: sophisticated governance mechanisms have repeatedly failed to prevent model bloat, and a hard number that forces an explicit decision has better odds than a principle that invites interpretation.

8.8 WHAT THE EXPECTATION DOES NOT SETTLE

The Expectation determines the *form* in which information is exchanged. It does not determine whose definition of a person, an asset, an incident or an encounter prevails. That is a political question, it is the question that the prevailing commercial model is genuinely paid to absorb, and this architecture does not answer it.

The consequence must be stated rather than glossed. A federation adopting RACHIS requires a body with authority to ratify its Expectations, and standing up that body is a precondition rather than a consequence of deployment. Where such a body cannot be constituted - because the participants are peers with no common authority, or because the question of whose definition wins is precisely what is in dispute - this architecture will not help, and no amount of namespaced extension will substitute for the missing decision.

Chapter 24 returns to this as the most significant of the problems the thesis does not solve.

CHAPTER 9 - MAPPING AND DISCLOSURE AS SEPARATE ACTS

9.1 TWO ARTEFACTS, TWO AUTHORITIES

Having retrieved an Expectation, a source owner produces two artefacts. They are distinct in content, in author, and in approval path, and conflating them is a design error that recurs in comparable systems.

The **mapping** states how local fields correspond to Expectation fields. It is a technical act, performed by someone who understands the source schema, and its correctness is a factual question.

The **disclosure policy** states what may cross the boundary. It is a governance act, performed by someone with authority to release information, and its correctness is a matter of judgement.

In most organisations these are different people, answering to different governance, on different timescales. A schema change requires a technical review; a disclosure change requires an information asset owner, and in several domains a formal decision under a statutory regime. Holding them in one artefact forces every technical correction through a governance process it does not need, and - more dangerously - allows a disclosure change to be made under cover of a technical one.

They are therefore separate documents, separately signed, separately versioned, and separately auditable. Only the disclosure policy's hash accompanies disclosures, because only the disclosure policy determines what left.

9.2 THE MAPPING: DECLARATIVE, NOT PROCEDURAL

The mapping handles the ordinary awkwardness of real data: field renaming, type coercion, unit conversion, date normalisation, enumeration translation, concatenation and splitting, and derivation of a stable local identifier. Where the source holds something the Expectation does not anticipate, a namespaced extension carries it. Where the Expectation requires something the source does not hold, the field is absent and the record is either valid without it or rejected locally with a stated reason.

The governing constraint is absolute and bears repetition: **the mapping is configuration, not code.**

It must be readable by someone who is not a programmer, reviewable by someone not cleared to the platform, and diffable in a change board paper. This is not an aesthetic preference. The entire inspectability claim of this architecture rests on the source-side components being small enough and plain enough for the source owner's own staff to audit. The moment a source must write arbitrary code to participate, three things happen: the audit becomes a software review rather than a document review; the specialist reappears; and mapping reuse (§8.6) collapses, because code is far harder to adopt from a registry than declarations.

Where a transformation genuinely cannot be expressed declaratively - an unusual encoding, a proprietary binary format, a legacy date convention nobody documented - the correct response is a *named, published, reusable transform* added to the standard library and available to every participant, not an escape hatch permitting arbitrary logic. This constrains what is expressible, which is the point.

9.3 THE DISCLOSURE POLICY: FOUR DISPOSITIONS

For each mapped field, the source owner declares one of four dispositions.

Disposition	Behaviour	Typical application
Clear	The value is released and indexed	Attributes the source is content to publish within the federation

Disposition	Behaviour	Typical application
Hash-only	A keyed digest is released, permitting exact-match correlation but not disclosure	Identifiers and names where linkage is wanted and the value is not to be shared
Pointer-only	An opaque reference is released; the value remains at source, retrievable only by callback	Sensitive detail the source will release case by case, to named requesters, for stated purposes
Withheld	The field does not leave. It is not indexed, not pointed to, and its absence is not distinguishable from non-existence	Everything the source has not decided to share

These are not arbitrary. They correspond precisely to the **non-perturbative** methods of statistical disclosure control - suppression, generalisation and pseudonymisation - which is a more useful observation than it first appears and is developed in Chapter 15.

The correspondence also settles a design question that would otherwise be open. Perturbative methods, which protect by altering values, are excluded. The disclosure control literature already establishes why: perturbed data may be unacceptable or may invalidate analytical conclusions in domains such as medicine and official statistics. The same objection applies with greater force here. An intelligence record, a custody record or a patient record cannot carry injected noise, because the downstream act is a decision about a specific individual rather than an inference about a population. RACHIS is confined to non-perturbative protection, and this is a principled confinement rather than an omission.

THE DEFAULT IS WITHHELD

A field with no declared disposition does not cross. Participation is opt-in at field granularity, and silence is refusal.

9.4 WITHHELD MEANS ABSENT

A withheld field does not enter the transform pipeline. There is no encrypted copy held centrally, no redacted placeholder, no key escrowed elsewhere that could reveal it. The platform's inability to read it is a property of what the platform holds, not of how the platform behaves.

This is a deliberate rejection of the obvious alternative. It would be architecturally tidier to disclose everything under encryption and enforce access cryptographically at the centre, and a substantial literature exists on searchable encryption schemes that would permit it.

That literature also establishes why it should not be done. Practical searchable encryption schemes leak access patterns, search patterns and volume, and these leakages are not theoretical curiosities. Islam, Kuzu and Kantarcioglu demonstrated in 2012 that a semi-honest server with some statistics about a database and knowledge of a small fraction of queries could recover the contents of all queries with better than ninety per cent accuracy. Subsequent work has weakened the adversary's requirements considerably: recent query-recovery attacks achieve around eighty-five per cent success without exact knowledge of the stored documents, beginning from roughly ten known queries and using only a distributionally similar dataset that was never indexed. File-injection attacks allow an active adversary to recover query contents by introducing a small number of crafted documents. Schemes that obfuscate access patterns remain vulnerable through search-pattern leakage.

A source owner is being asked to accept a specific proposition: *the platform holds your data in a form it cannot read*. The disclosure control literature says that proposition is conditional on assumptions about the platform's auxiliary knowledge that no source owner can verify and no accreditor should accept. Selective disclosure makes a stronger and simpler claim - *the platform does not hold it* - and that claim requires no cryptographic assumption at all.

9.5 DERIVATION CONSTRAINTS

A per-field disposition table is insufficient, and the reason is a well-established result in a field that security marking practice has largely ignored.

In statistical disclosure control, suppressing a sensitive cell is only the first step. **Primary suppression** removes the sensitive value; **secondary** or **complementary suppression** removes further values that are not themselves sensitive but from which the primary value could be reconstructed - typically through marginal totals. Omitting the second step is not a subtle failure. A systematic review of web-based public health query systems found that in nine of the systems examined, suppressed cells were recoverable, either by examining marginal totals for rows and columns containing suppressions or by differencing published results.

The equivalent failure in a disclosure policy is immediate and obvious once stated. Withholding a date of birth achieves nothing if age and a reference date are released in clear. Withholding a precise location achieves nothing if a postcode and a facility identifier

are released. Withholding an identifier achieves nothing if the released hash is computed over a value with insufficient entropy to resist enumeration.

The disclosure policy therefore admits **derivation constraints** alongside dispositions. A constraint declares that a field is derivable from a set of other fields:

```
person.dateOfBirth is derivable from { person.age,  
record.observedAt }
```

The enforcement rule is that a derivable field inherits the *most permissive* disposition among the sets from which it can be derived, and if that is more permissive than its declared disposition, the policy does not validate. The source owner resolves the conflict by demoting one member of the deriving set - the secondary suppression decision - or by accepting the derived disclosure explicitly.

The connector enforces constraints at transform time, before egress, in keeping with Chapter 7 (L5). Constraint evaluation is part of policy application, not a downstream check, and a policy that does not validate does not run.

9.6 THE VALIDATOR

Derivation constraints impose analytical work on the source owner, and Chapter 7 (L3) warns that analytical work is what kills adoption. The mitigation is tooling that proposes constraints rather than requiring the source owner to discover them.

The validator operates in three modes, in increasing order of confidence:

SCHEMA-DERIVED

Functional dependencies declared or inferable from the source schema - primary and foreign keys, computed columns, generated fields, check constraints - yield derivations directly. These are proposed with high confidence and are usually correct.

LIBRARY-MATCHED

A published library of common derivations, maintained alongside the Expectation registry and reusable in the same way mappings are: date of birth from age and reference date; coordinates from full postcode; identity from name and date of birth in combination; institutional location from an internal reference code. This library is a community artefact

and grows with use, and it is the mechanism by which one organisation's discovered derivation protects every subsequent participant.

STATISTICAL

Where the source owner permits the validator to run against real data locally - never leaving the estate - it can detect empirical near-dependencies that the schema does not declare, using standard measures of association. This is the weakest mode and the most likely to produce false positives, and it is optional for that reason.

The validator proposes; the source owner disposes. Every proposed constraint is accepted, rejected or modified by a human with authority, and the record of that decision forms part of the signed policy. This division matters: the tool must not become the classification authority, which would reintroduce at the source exactly the displacement of judgement that Chapter 11 forbids at the centre.

TWO HONEST LIMITATIONS

The validator cannot detect derivations that depend on knowledge held outside the source system, and a determined analyst with external data will always be able to derive things no local tool could anticipate. And within-record constraints do not address derivation *across* records or across sources - releasing a single postcode is different from releasing ten thousand of them, and neither the source nor its connector can see the aggregate. That second problem is genuinely central, it cannot be solved at source, and it is the subject of Chapter 15.

9.7 THE COST, STATED HONESTLY

Section 8.6 claimed that source-side effort is small. Derivation constraints make that claim less comfortable, and it should be restated accurately rather than defended.

A disclosure policy is no longer a flat table of field dispositions. It is a table plus a set of constraints, some proposed automatically and some requiring judgement, each of which must be resolved by someone with authority to release information. For a schema of a hundred fields, expect the validator to propose perhaps ten to twenty candidate derivations, of which most are dismissed quickly and a handful require genuine thought.

That is hours of work for someone who knows the data, against months for a vendor engineer who does not - but it is hours, not minutes, and it requires the involvement of a

governance authority rather than only a technician. Any claim that participation is effortless would collapse on contact with a real schema, and would deserve to.

CHAPTER 10 - BINDING: A JSON AND POST-QUANTUM PROFILE OF ADATP-4778

10.1 WHAT ALREADY EXISTS

NATO maintains three related standards governing the labelling of information. ADatP-5636, the NATO Core Metadata Specification, defines the core metadata elements supporting interoperable exchange. ADatP-4774 defines the confidentiality metadata label syntax - classification, policy identifier, caveats, creation time - as a structured XML element. ADatP-4778 defines the binding mechanism: how a label of any sort is bound to a finite data object such that the association can be verified.

The binding produces a **Binding Data Object**, a structured artefact recording that a given data object and a given label belong together. ADatP-4778.2 profiles the application of that mechanism to specific formats and protocols - electronic mail, instant messaging, office documents, web services, arbitrary files - and specifies two cryptographic profiles for strong binding, using XML Signature with either digital signatures or keyed-hash message authentication codes.

Three binding approaches are defined: **embedded**, where the binding object sits inside the data object; **encapsulated**, where the binding object contains the data; and **detached**, where the two reside separately and reference one another.

This work builds on all of it. The contribution claimed here is a profile, not a replacement, and the distinction is both accurate and strategically important. An architecture that proposes to supersede a ratified standardisation agreement invites a multi-year conversation with a standardisation body. One that proposes a subprofile within an existing framework invites a technical review.

10.2 WHAT THE STANDARDS DO NOT YET COVER

Three gaps, of which two are explicitly anticipated by the standards themselves.

NO JSON

The cryptographic profiles specify XML Signature and Cryptographic Message Syntax. ADatP-4778.2 states directly that further revisions may be required to add subprofiles for

other cryptographic protocols such as JSON Web Signature. The gap is named in the document and has not been filled.

NO POST-QUANTUM ALGORITHMS

The signature algorithm tables permit RSA, DSA and ECDSA over the SHA-2 family; SHA-384 is mandatory for digest under the XML Signature profile. The document anticipates revision to introduce new algorithms or deprecate existing ones. FIPS 204 and FIPS 205 postdate the profile, and NATO and CCEB work on data-centric security maturity is presently converging on the level at which encryption enters scope.

NO SELECTIVE DISCLOSURE

This is the substantive gap. ADatP-4778 supports binding labels to *parts* of a data object - the profile's own examples bind distinct labels to individual paragraphs within a document by fragment reference. What it does not support is a recipient holding *some* of the bound parts and verifying their authenticity without learning about the parts they do not hold. The XML Signature Manifest structure digests each reference separately in a flat list, so a recipient receives the identifiers and digests of everything bound, whether or not they receive the corresponding data.

For document labelling that is unproblematic. For field-level disclosure across a sovereignty boundary it defeats the purpose: a source that withholds a field would nonetheless disclose the field's identity, its existence, and a digest that is vulnerable to enumeration for low-entropy values.

10.3 THE PROPOSED PROFILE

CANONICAL IDENTIFICATION

The profile follows the established naming pattern, with a stable canonical identifier and a versioned identifier that is superseded rather than mutated, and the deprecation of prior versions stated explicitly.

SERIALISATION

JSON, with a specified canonicalisation, signed under JSON Web Signature - the subprofile the parent document anticipates. Binding is detached: the binding object is a separate

structure referencing the disclosed fields, which suits a record-oriented rather than document-oriented exchange and keeps the disclosed payload free of binding syntax.

LABEL CARRIAGE

The 4774 label structure is carried unchanged in a JSON representation with a documented lossless mapping to and from the XML form, so that a record crossing between an XML-native and a JSON-native environment retains an identical label. Preserving 4774 semantics exactly is a constraint on this work rather than an option.

ALGORITHMS

ML-DSA (FIPS 204) on the hot path, where verification volume is high. SLH-DSA (FIPS 205) where archival longevity is required and no lattice assumption should be load-bearing across a thirty-year retention horizon. Hybrid classical-and-post-quantum operation during transition. SHA-384 as the mandatory digest, aligning with the existing profile rather than diverging from it.

10.4 SELECTIVE DISCLOSURE

The construction is straightforward, which is a virtue in something that must be implemented correctly by several independent parties.

For a record conforming to an Expectation, each field yields a leaf:

```
leaf_i = H( canonical_field_name || canonical_value ||  
            field_label || salt_i )
```

Leaves are ordered canonically by field name, and a Merkle tree is constructed over them. The record-level label and the four-part header (§10.5) are bound as additional leaves. The **root is signed once**, in the source HSM, under the profile's algorithms.

A disclosure package contains the signed root, the header, and for each released field the leaf value, its salt, and the inclusion proof - the sibling hashes along the path to the root. Withheld fields contribute their leaves to the tree and nothing else.

The properties this delivers:

- **Authenticity of the part.** A recipient verifies each released field against the signed root without possessing the whole record.

- **One signature per record.** Signing cost does not scale with field count, which matters at ingest volumes.
- **Native redaction.** Withholding requires no re-signing and no source-side action beyond omission. The same signed record supports different disclosure outcomes to different requesters - necessary for the callback path of Chapter 12, where a field withheld at ingest is later released to a named requester and must still verify against the original signature.
- **Per-leaf salting.** Salts defeat enumeration of low-entropy values, which a bare digest would not.

THE LEAKAGE THAT REMAINS, AND WHY IT IS ACCEPTABLE

Tree shape reveals the number of bound fields. This is not a disclosure, because the Expectation is public: the field set is known to anyone who reads the contract. What a recipient learns from tree shape is how many of a known set were present, which is materially different from learning what they contained, and in most cases the record's conformance already implies the presence of required fields. Where even field presence is sensitive - a policing context in which the existence of a marker is itself protected - the profile permits padding to a fixed arity at the cost of a modestly larger tree, and the Expectation may require it.

10.5 THE FOUR-PART HEADER

Every disclosure package carries a header binding four assertions, each independently verifiable at ingest:

Element	Answers
Expectation identifier and version	Which contract does this record claim to satisfy?
Disclosure policy hash	Which release rules produced it?
Connector measurement	Which code, in which configuration, in which attested environment?
Source signing identity	Which organisation asserts the content and the label?

The header is bound into the Merkle tree and therefore covered by the same signature.

This extends what 4778 conventionally binds. The standard binds *metadata* to *data*; the header additionally binds the *provenance of the disclosure decision* - the contract, the

policy, the software and the authority. The value is not immediate but eventual. Presented with a record years later, an inquiry can establish which schema it conformed to, which release rules were in force at the time, which software produced it, and who stood behind it. None of that depends on the honesty of the platform, because the platform holds no key that could forge any of it.

10.6 RELATIONSHIP TO DATA-CENTRIC SECURITY MATURITY

NATO's data-centric security work describes maturity in levels, with basic labelling at the lower end and cryptographic binding and encryption above it. The profile specified here sits at the level where binding becomes cryptographic and encryption enters scope, which is the level at which alliance and CCEB work is currently converging.

That timing is favourable and should be exploited rather than assumed. A profile contributed while the relevant standardisation work is live has a materially different reception from one arriving after positions have settled, and Chapter 19's recommendation to publish early is grounded partly in this.

10.7 A NOTE ON CONFORMANCE

The conformance model in ADatP-4778.2 requires an implementation conformance statement, extra information for testing, approved test cases and scripts, and evaluation by an authorised laboratory. The document acknowledges that no formal reference facility for testing external implementations has been established and that the testing capability remains under investigation. It was published in December 2020.

This is not a criticism of the standard, which is sound. It is Chapter 7 (L4) observed within the standards family this work adopts: a heavyweight, gated conformance model produces, after some years, no operational means for an implementer to demonstrate conformance. The programme described in Chapter 18 takes the opposite approach and would - if it worked - supply executable conformance tests for a profile of 4778 that the parent standard does not presently have. That is offered as a contribution rather than a reproach.

CHAPTER 11 - VERIFICATION, AND WHY INGRESS MUST NOT CLASSIFY

11.1 THE CHECKS

A disclosure package is admitted only if every one of the following holds:

1. The signature verifies against a trust anchor the federation recognises for the asserting source.
2. The attestation evidence corresponds to a published, currently permitted connector measurement.
3. The declared Expectation version exists and is not beyond its deprecation horizon.
4. The record conforms to that Expectation: required fields present, types correct, enumerations valid, format constraints satisfied.
5. Every field carries a label, and every label is expressed against a policy identifier the federation accepts.
6. Marking arithmetic is satisfied: no field's label exceeds the record's, no released field's label contradicts the Expectation's marking requirements.
7. The disclosure policy hash is well-formed and recorded.
8. The provenance manifest is complete and internally consistent, and inclusion proofs verify against the signed root.
9. The package is not a replay, evaluated against the binding timestamp and a windowed record of accepted bindings.

Failures are rejected or quarantined with a reason returned to the source. Nothing is coerced into shape, and nothing is admitted with a defect noted for later attention, because a defect noted for later attention is a defect admitted.

11.2 INGRESS IS A CONFORMANCE POINT, NOT A COMPONENT

It is tempting to draw ingress as a box. It is more accurately a *place where several contracts are jointly enforced* - Trust supplies signature verification, Provenance supplies proof checking, Model supplies conformance, Policy supplies marking arithmetic.

The specification should therefore state the checks, their ordering, and their failure semantics, and leave the realisation open. Implementers will reasonably build it as a middleware chain, a set of filters, or a service, and the choice has no bearing on conformance. Specifying it as a component invites people to build a component they did not need - a small instance of Chapter 7 (L3).

11.3 THE PLATFORM HAS NO CLASSIFICATION AUTHORITY

Ingress verifies. It does not classify. It has no authority to assign, upgrade, downgrade or infer a label, and - more importantly - it holds no key that would allow it to.

This distinction is worth labouring, because it is the difference between two entirely different accreditation propositions. A component that exercises classification judgement is a trusted process: it must be evaluated for correctness of judgement, its logic must be assured, and every new data type it encounters potentially reopens that evaluation. A component that verifies a signature over a label asserted by an accredited authority elsewhere is a verifier: its correctness is a cryptographic property, its evaluation surface is small, and new data types do not disturb it.

Marking, in this architecture, is arithmetic performed over assertions made elsewhere. The platform cannot relabel because it cannot forge.

11.4 THE CONSEQUENCE FOR CROSS-DOMAIN TRANSFER

The strongest and least obvious claim available to this thesis follows from the preceding section, and it inverts a widely-held assumption about guards.

Cross-domain solutions are expensive to accredit primarily because they perform content inspection. A guard examines what passes and forms a judgement about whether it may - semantic evaluation executed inside a box, requiring assurance against every data type it may encounter and re-evaluation whenever a new one appears. The guard becomes the bottleneck on adding sources, and the cost of that bottleneck is why federation stalls at the domain boundary.

Where content arrives already bound to a label asserted by an accredited authority at source, the guard's necessary function reduces substantially: verify the signature, evaluate the marking arithmetic against the transfer direction, enforce the result. The trusted computing base shrinks. The evaluation surface shrinks with it. New sources and new data types do not require the guard to be re-evaluated, because the guard is not reasoning about content.

11.5 WHY THIS WILL BE RESISTED, AND FOR HOW LONG

The argument above is correct and it will not be accepted quickly.

It asks an accreditation community to substitute *verification of an assertion for inspection of content*, and to accept that the assurance formerly provided by inspection is now provided by an accredited authority at the far end of a signature chain. That is a genuine transfer of trust, not merely a relocation of function, and the reasonable question - what if the source asserts a wrong label - has an honest answer that does not fully satisfy: the

signature is valid, the assertion is wrong, and accountability lies unambiguously with the identified asserting authority.

That is an improvement on the alternative, in which a wrong label diffuses into a pipeline and accountability cannot be located at all. It is not equivalent to inspection, and pretending otherwise would be the fastest way to lose the argument.

The recommendation is therefore procedural rather than technical. The claim should be published as a technical paper and debated on its merits before it appears in any proposal, deployment or business case. Novelty is expensive in the accreditation community for sound reasons, and the timescale for a claim of this kind is measured in years. Planning otherwise will produce a programme whose critical path runs through a conversation nobody has started.

CHAPTER 12 - GRADUATED DISCLOSURE AND THE CALLBACK

12.1 THE MECHANISM

Pointer-only disclosure permits a source to participate meaningfully while releasing almost nothing. It should be understood as a primary capability rather than a fallback, because it is what allows adoption to begin below the threshold at which a cautious risk owner must make an irreversible decision.

When a requester needs a field released as a pointer, the disclosure service issues a request to the source's callback handler. The request carries the requester's identity, organisation, nationality, role, clearance, stated purpose, device posture, mission or operational context, and requested duration.

The source evaluates it against their own policy, logs it locally whatever the outcome, and may release the value in clear; release it encrypted directly to the requesting user or workload under a short-lived signed token; release it for a bounded period after which access lapses without further action; deny it; or revoke a release previously granted, terminating future access.

Released values verify against the record's original signed root through their inclusion proofs (§10.4). No re-signing is required, and a value released months after ingest carries the same provenance as one released at the outset.

Forward secrecy applies to the callback channel. Ephemeral session keys ensure that a later compromise does not retrospectively expose past disclosures - a precondition for any risk owner authorising standing retrieval rather than case-by-case release.

12.2 TWO LEAKAGE CHANNELS

Graduated disclosure creates two channels that the architecture must address explicitly rather than discover in operation.

QUERY PATTERN

The pattern of callback requests is itself informative. Which sources are being asked about, how often, in what sequence, and in temporal correlation with external events, constitutes intelligence about the requesting organisation's interests - and in a multinational federation, the source is not necessarily a party that should hold it.

This is the same problem the searchable encryption literature has studied at length, and its findings apply here with the roles reversed: it is not the platform observing the source, but the source observing the platform's users. The mitigations are correspondingly familiar. Callbacks are batched where latency permits, padded with cover requests where it does not, and the platform retains no query history beyond what audit requires. None of this is complete protection, and the residual is a matter for the federation's threat model rather than the architecture.

DISCLOSURE BY DIFFERENCING

This is the more serious channel and it is not addressed by any current design in this class.

Statistical disclosure control identifies differencing - comparing two releases at differing levels of detail - as a principal risk, alongside small counts and attribute disclosure. In a federated index with attribute-based access control, differencing arises structurally: two requesters with different attribute sets receive different views of the same underlying record, and comparing those views reveals the shape of what was withheld from one of them. The same effect arises temporally, when a single requester compares a result before and after a callback grant, and cumulatively, when a requester with access to many records infers a withheld field from the pattern of what is released across them.

Chapter 9's derivation constraints do not reach this. They operate within a record, at source, where the source can see the fields in question. Differencing operates across records, across requesters and across time, at the centre, where no source can see it.

The mitigations available are partial and belong to Chapter 15, which treats aggregation properly: query auditing that detects differencing patterns rather than merely recording queries; consistency of response across attribute contexts, so that a field withheld from one requester is presented identically rather than absently; and thresholding on the number of records a requester may difference within a window. None of these is a solution. Naming the channel, and designing the audit to detect its exploitation, is the honest position.

12.3 ZERO-INDEX DEPLOYMENT

Chapter 7 (L7) argues against holding central state, and observes that X-Road's longevity owes much to holding essentially none. The architecture proposed here holds an index and an assertion store, which is a real asymmetry. This section answers it.

THE INDEX IS AN OPTIMISATION, NOT AN ARCHITECTURAL REQUIREMENT.

A federation may elect to disclose nothing in clear and nothing as hash - every field pointer-only. The platform then holds an index of pointers, provenance references and record-level labels, and nothing whatever of the content. Search under those conditions resolves through callbacks: the platform identifies candidate records by their pointers and requests resolution from each source, which authorises or denies per requester.

This works. It is slower - every result costs a round trip to a source that may be disconnected, and result ranking is severely constrained without content to rank on. It is also, functionally, X-Road with an entity model: a broker that routes requests to sovereign holders and holds nothing itself.

Two consequences follow, and both are strategically useful. First, a cautious participant can begin at the point of maximum caution and increase disclosure as confidence accumulates, without any architectural change and without renegotiating anything. Second, the answer to a risk owner asking what happens if they trust the platform with nothing is not a concession but a specification: here is the deployment, here is its performance, here is what it costs you in latency and ranking quality.

The assertion store cannot be reduced in the same way, being the one genuinely novel central artefact. Its answer is different and is given in Chapter 14: exportable in full, in a documented format, and hostable on the customer's own infrastructure.

12.4 VALUE FOLLOWS TRUST

The design consequence of graduated disclosure is that the system's usefulness to a participant is proportional to what that participant has released. A department disclosing nothing but pointers obtains a slow federated lookup. A department disclosing widely obtains rich correlation, entity assembly and pattern analysis.

This should be stated first, and stated by its proponents, because it is the obvious criticism and owning it is what makes the remainder credible. It is not a defect that has been engineered around. It is the intended property: **value follows trust rather than preceding it**, and a federation in which participants can obtain full value while contributing nothing would be one in which the disclosure decision had no meaning.

PART IV - FOUNDATIONS

Part III specified how information crosses the boundary. Part IV addresses what happens to it afterwards: how identity is established without a black box, how time is represented so that an inquiry can be answered, how markings behave under combination, and what the trust and key architecture must provide.

Three of these four chapters make claims that are conventional in adjacent disciplines and unusual in this one. That is the intended effect.

CHAPTER 13 - IDENTITY IS AN ASSERTION

13.1 THE CONVENTIONAL TREATMENT

Entity resolution is normally a pipeline stage. Records enter, a resolution engine applies deterministic, probabilistic and increasingly embedding-based matching, and resolved entities emerge. The engine's confidence scores are internal to it. Its decisions are explicable in aggregate and difficult to explain individually. The resulting identity is a piece of platform state that must thereafter be reconciled against everything else the platform holds.

This is defensible engineering and it produces three problems that are not engineering problems.

It cannot answer an inquiry. Asked why two records were treated as the same person, the honest response describes a model, a training regime and a threshold. That is not the answer the question wanted, and in a context where the merge contributed to a consequential decision it is not an adequate one.

It is difficult to reverse. A merge that is later found wrong requires the resolution state to be corrected, downstream artefacts derived from it to be identified, and the correction to be propagated - a data repair operation rather than an ordinary transaction.

And it places judgement in the wrong hands. The engine decides; the analyst who knows the case is presented with a result. Where the analyst disagrees, their disagreement is an override recorded outside the mechanism rather than an input to it.

13.2 THE INVERSION

In this architecture, **the profile an analyst sees is a read-time projection over all currently valid assertions.**

A merge is an assertion. So is a split, an alias, a disputed link, a confirmed identification, a confidence downgrade. Each is an attributed, evidenced, bitemporal, reversible statement held in the same append-only store as every other assertion, subject to the same challenge, supersession and withdrawal semantics described in Chapter 14.

There is no separate resolution state. There is nothing to reconcile, because there is only one store.

13.3 WHAT THIS MAKES POSSIBLE

Every merge retains the evidence that justified it and the identity of the person or process that made it. An inquiry asking why two records were treated as one receives a name, a timestamp, an evidence set and a reasoning note.

Reversal becomes ordinary. A merge is withdrawn by asserting its withdrawal; the projection recomputes; dependent assertions are flagged for revalidation under §14.4. No repair operation is required and no state is corrected, because nothing was overwritten.

Automated matching becomes a *proposal*. A matching process is a first-class asserting party, distinguished from a human by its identity rather than by its mechanism, and its assertions carry a lower default confidence and appear in an adjudication queue. Where an analyst confirms one, the confirmation is a new assertion attributable to them. Where they reject it, the rejection is equally recorded, and the record of rejections is exactly the training signal a matching process needs.

Human adjudication ceases to be a corrective layered over an engine and becomes the resolution mechanism, with automation as an accelerant. This is the arrangement the accountability requires, and it is also - not incidentally - the arrangement that produces better labelled data.

13.4 IDENTITY PINNING

The choice imposes one hard requirement, and it is not optional.

ONCE AN ENTITY IDENTIFIER HAS BEEN CITED, IT MUST REMAIN RESOLVABLE INDEFINITELY.

An analyst who references an entity in a disseminated product, a case file, a warrant application or a targeting recommendation has created an external dependency on that identifier. A projection that reshuffles between sessions - assigning a different identifier to the same cluster of records because an assertion changed - is not merely inconvenient. It breaks the citation, and it breaks it silently.

The mechanics required are modest but must be specified rather than assumed. An entity identifier, once assigned, is never reassigned. Where a merge is withdrawn and an entity splits, the original identifier becomes a tombstone that resolves to its successors together with the assertion that caused the split, so that a citation of the original resolves to a comprehensible account rather than to nothing. Where two entities merge, both prior identifiers resolve to the merged entity and remain valid citations.

Stability of provisional identifiers is therefore a first-class requirement of the Assertions contract, not an optimisation, and it is among the properties the conformance suite must test.

13.5 THE COST

Three costs, stated rather than minimised.

READ-TIME ASSEMBLY IS EXPENSIVE

Projecting a profile over a large assertion set on every request will not perform at scale without materialisation, and materialisation reintroduces derived state that must be invalidated correctly. The derived state is a cache rather than a record - reconstructible from the assertion log and never authoritative - but caches that must be invalidated on a bitemporal condition are not trivial.

CANDIDATE GENERATION STILL REQUIRES A MATCHING KEY

Projection resolves what to do with candidates; it does not produce them. Across sources with no common identifier, candidate generation depends on the hash-only disposition of §9.3 and therefore on the correlation scheme of §16.6. Where a federation elects the reduced delivery option described there, or operates at zero disclosure, cross-source candidate generation degrades to pointer resolution and the identity model is correspondingly weaker.

AN EMPTY ASSERTION STORE IS A WORSE SEARCH ENGINE

The model's advantages accrue as assertions accumulate. In the first months of a deployment, there are none, and the system offers federated lookup and nothing else. This is a real adoption obstacle and is addressed in §21.4 rather than dismissed here.

CHAPTER 14 - TIME

14.1 WHY THIS MATTERS MORE HERE THAN ELSEWHERE

The distinction between when something was true and when it became known is the substance of assessment. A system that cannot represent the distinction cannot answer the question that every inquiry eventually asks: *when did you know, and what had you already done?*

Most operational systems record one timestamp, sometimes two, and reconstruct the rest from audit logs of varying quality. This is adequate until it is not, and the moment at which it becomes inadequate is invariably the moment at which the answer matters most.

14.2 SIX TIMESTAMPS

Every assertion and every disclosed fact carries, at minimum:

Timestamp	Meaning	Supplied by
validFrom	When the fact became true in the world	Source
validUntil	When it ceased to be true, where known	Source
observedAt	When it was observed by whatever collected it	Source
recordedAt	When the federation learned of it	Platform
supersededAt	When a later assertion replaced it	Platform
withdrawnAt	When it was retracted as incorrect	Platform

The first four are conventional bitemporality with an intermediate observation time, which the intelligence case requires because the gap between observation and recording is frequently where the operationally significant delay lives.

14.3 SUPERSESSION IS NOT WITHDRAWAL

The final two are the ones systems most often conflate, and the conflation is consequential.

An assertion that has been **superseded** was correct and has been replaced by something more current. A vessel's position at 04:12 is superseded by its position at 09:30; the earlier statement remains true of the earlier time.

An assertion that has been **withdrawn** was wrong. It was not true of any time, the conclusion drawn from it was unfounded, and everything derived from it is now suspect.

A store that records only "no longer current" cannot distinguish these, and therefore cannot answer whether a decision taken on the basis of a superseded fact was reasonable at the time - which is the entire question in a retrospective examination. The distinction costs one field and is not negotiable.

14.4 WITHDRAWAL CASCADES

An assertion derived from a withdrawn fact is **flagged for revalidation**, not deleted.

The distinction is deliberate. The derived conclusion may remain correct on other evidence, and deleting it destroys the record of the reasoning - which is precisely what an inquiry is trying to examine. What must happen is that the dependency is visible, the analyst who made the derived assertion is notified, and the assertion carries a visible flag until it is either re-evidenced or withdrawn in turn.

This requires that dependency links be recorded when a derived assertion is made, which is a Provenance responsibility and one of the harder contracts to implement correctly. An assertion that cites its supporting evidence is cheap; one that cites it in a form that supports automated cascade on withdrawal is not.

14.5 EXPORT

The assertion store is the one genuinely novel item of central state, and Chapter 7 (L7) requires that every such item be separately justified or removed. Its justification is that it is the accumulated institutional knowledge of the federation and cannot exist anywhere else. Its answer to the lesson is different from the index's: not reduction, but **portability**.

Assertions export in full, in a documented format, with their bitemporal fields, their attribution, their evidence links, their markings and their dependency graph intact. A

customer may host the store on their own infrastructure. A customer withdrawing from the federation takes their assertions with them, in a form another implementation can ingest, and Chapter 18's conformance suite tests that a conformant implementation can load an export produced by any other.

This is what makes the fifth principle demonstrable rather than promised, and it is the mechanism behind the claim in Chapter 19 that the leaving was built first.

CHAPTER 15 - MARKING ARITHMETIC AND AGGREGATION

15.1 THE ARITHMETIC

Markings propagate by high-water mark. An artefact derived from several inputs inherits the strictest classification among them and the union of their handling caveats. Relaxation - declassification, caveat removal, downgrade - is a deliberate act performed by an authority competent to perform it, attributed to a named party, and recorded as an assertion like any other. It is never an emergent property of a transformation.

This is conventional and uncontroversial. What is neither is the requirement that it hold everywhere.

15.2 EVERY PATH, WITHOUT EXCEPTION

Propagation must be correct on the path from index to search result, from search result to profile projection, from projection to assertion, from assertion to graph traversal, from graph to any model-generated output, and from any of these to any export, print, screen capture or downstream system.

A single path that fails to propagate correctly ends the project. Not degrades it: ends it. The disclosure of one item of information to one party not entitled to it, traced to a defect in the platform's marking logic, is a programme-terminating event in every domain considered in this document.

Two consequences follow. The marking engine must be **small** - small enough that a competent reviewer can hold its entire behaviour in mind, which argues for a purpose-built implementation over a general-purpose policy engine whose evaluation semantics are themselves a subject of study. And its correctness must be established by **property-based testing** rather than by example: generated label lattices, generated combination sequences, and invariants asserted over them - that output classification is never below the

maximum of inputs, that caveats never diminish, that no sequence of operations produces relaxation without an attributed relaxation assertion. This is precisely the domain that property-based testing exists for, and it is among the reasons Chapter 22 specifies the tooling it does.

15.3 AGGREGATION IS CREATED BY FEDERATION

Correctly marked items may combine to constitute something more sensitive than any of them. Fifty releasable facts may jointly disclose a capability, a method, a source or an individual.

The important observation, and the one usually left implicit, is that **this problem does not exist while the data sits in thirty separate systems**. It is created by the act of federation. The architecture proposed here is not inheriting a difficulty from its sources; it is manufacturing one, and it owes its participants an account of how it will be handled.

No guard catches it, because the judgement is semantic rather than syntactic. No signature prevents it, because every constituent item is correctly signed. And the platform is precisely the instrument performing the combination.

15.4 THE DISCIPLINE THAT HAS ALREADY STUDIED THIS

Official statistics has addressed the identical structural problem for forty years under the name **statistical disclosure control**, and the two communities do not cite one another.

The correspondence is not loose. SDC concerns the release of information in a form that permits legitimate use while preventing the reconstruction of protected values from what is released. It distinguishes **identity disclosure**, in which a subject is identified, from **attribute disclosure**, in which a protected characteristic of a subject is determined - a distinction that maps exactly onto "who is this" and "what is true of them". It has developed a taxonomy of risk that includes small counts, attribute disclosure and **disclosure by differencing**. And it has a mature practice, an established literature, a working toolset and, notably, a documented record of failure when its methods are applied incompletely.

Chapter 9 has already imported the first and most immediately useful concept. **Primary suppression** removes a sensitive value; **secondary** or **complementary suppression** removes further values, not themselves sensitive, from which the primary value could be reconstructed. Omitting the second step is a common and well-documented failure: a systematic review of web-based public health query systems found that in nine of the

systems examined, suppressed cells were recoverable, either from marginal totals or by differencing published results. Derivation constraints (§9.5) are secondary suppression exercised at source.

15.5 WHAT ELSE TRANSFERS

Disclosure by differencing is treated below in §15.6.

THRESHOLD RULES

SDC practice defines sensitivity rules - the (n,k) rule, the p% rule, minimum cell counts - that determine when an aggregate is too revealing. A federation may adopt analogous thresholds governing when a result set, a graph expansion or a model output requires review before release. These are crude and they are usable, which is more than can be said for most alternatives.

FEASIBILITY INTERVALS

Where a value is suppressed, SDC computes the interval of possible values consistent with what was published. The equivalent question - given what a requester has been shown, how tightly is a withheld field constrained - is directly computable for structured fields and is the natural quantitative basis for a review threshold.

THE NON-PERTURBATIVE CONSTRAINT

As established in §9.3, methods that protect by altering values are excluded here for the same reason SDC excludes them in medicine and official statistics: perturbation invalidates the analytical conclusion where the conclusion concerns a specific individual rather than a population. This confines the available toolkit and the confinement is principled.

A FORMAL RESULT

The cell suppression problem - selecting a minimum-cost set of suppressions such that no sensitive value is reconstructible - is strongly NP-hard. This is the rigorous grounding for the claim that aggregation cannot be automated. It is not that the engineering is hard or the semantics are unclear; it is that optimal suppression is computationally intractable in general, and any system claiming to solve it is either operating on a restricted case or is wrong.

15.6 THE DIFFERENCING CHANNEL

This is the risk that Chapter 12 identified and deferred, and it is the most serious unsolved problem in the architecture.

SDC identifies differencing - comparing releases at differing levels of detail - as a principal risk class. In a federated index with attribute-based access control it arises structurally and in three forms.

ACROSS REQUESTERS

Two analysts with different attributes receive different views of the same record. Comparing the views reveals the shape of what was withheld from one of them.

ACROSS TIME

One analyst compares a result before and after a callback grant, or before and after a change in their own attributes. The delta is the disclosure.

ACROSS RECORDS

An analyst with access to many records infers a withheld field from the pattern of what is released across the population, which is the classical SDC case and the one its literature addresses best.

Derivation constraints do not reach any of these. They operate within a record, at source, where the source can see the fields concerned. Differencing operates across records, requesters and time, at the centre, where no source can see it and where the platform is the only party positioned to detect it.

Four mitigations are available, none complete.

SHAPE CONSISTENCY

A field withheld from a requester is presented *as withheld*, identically to every requester, rather than being absent for some and present for others. Two views then differ in values, never in structure, which removes the most trivial differencing channel entirely. This is a hard design requirement and it follows directly from the SDC literature; it also costs something real, since acknowledging that a field exists is itself a disclosure in some

policing contexts, which is why §10.4 provides for padding where field presence is protected.

DIFFERENCING-AWARE AUDIT

The audit must be able to detect the *pattern* of differencing, not merely record queries. Repeated near-identical queries under varying contexts, systematic enumeration of a record population, and query sequences that bracket a suppressed value are all detectable, and detecting them is a different capability from logging.

WINDOWED THRESHOLDS

Limits on the number of records a requester may retrieve within a window, on the number of distinct attribute contexts under which a given record may be examined, and on the rate of callback requests against a single source.

REVIEW QUEUES

Results that cross a threshold are held for human release rather than returned. This is the mechanism of last resort and its cost is latency in exactly the cases where latency is least welcome.

15.7 THE HONEST POSITION

Aggregation is a policy decision made by an accountable human, supported by detection, thresholds and review. It is not solved here, it is not solved elsewhere, and it is provably not solvable in the general case.

What this architecture contributes is threefold: it names the problem as one it creates rather than inherits; it imports a mature body of method from a discipline that has studied it for four decades; and it specifies the detection and review machinery as a first-class concern rather than a deferred one.

Any supplier claiming to have automated aggregation control should be asked which restricted case they have solved. There will be one, and it will be narrower than the claim.

CHAPTER 16 - TRUST, ATTESTATION AND KEY CUSTODY

16.1 CUSTODY

Content keys and signing keys reside in the source's hardware security module and do not leave it. The platform's keys cover transport, service identity, audit signing and platform-originated assertions. There exists no key held by the platform capable of producing a valid source signature or of decrypting content that was not disclosed to it.

This is stated as an architectural invariant rather than a configuration option, because a design in which the platform *could* hold content keys but is *configured* not to is a different proposition entirely, and a risk owner will correctly treat it as such.

The connector additionally holds durable state that was not anticipated in the original design (Appendix A.11, D1): the per-leaf salts for withheld and pointer fields, retained so that a later callback release produces a leaf verifying against the original signed root. That salt store is sensitive, must be backed up, must be protected at the classification of the fields it relates to, and must be addressed by the Exit contract when a source withdraws.

16.2 IDENTITY

Four categories of identity are verified, and conflating them is a common source of weakness.

- **Organisations** hold signing identities anchored in the federation's trust hierarchy, and it is an organisation, not a person or a machine, that asserts a classification.
- **Workloads** - connectors, services, model endpoints - hold attested identities bound to a measured software state.
- **Users** hold identities carrying the attributes on which policy operates: role, clearance, vetting level, organisation, nationality, and the purpose declared for a given session.
- **Devices** hold posture attestations, since a legitimate user on an unmanaged endpoint is a different risk from the same user on a managed one, and several of the callback policies in Appendix A depend on the distinction.

16.3 ATTESTATION

The connector executes in a trusted execution environment producing evidence - a measurement of the loaded software, verifiable by both the source owner and the platform against a published reference value.

This answers a question that would otherwise have no good answer: *how do I know the connector is running the disclosure policy I wrote, and not a modified one?* The source owner verifies because they wish to know their policy is being applied. The platform verifies because it will not accept disclosures from an unmeasured or unrecognised connector. Neither party is required to trust the other, and neither is required to trust the supplier.

The measurement is published, which means a third party can determine whether a given deployment is running the software it claims. This is the same property as the offline testability of §8.4, applied to execution rather than to configuration.

16.4 FORWARD SECRECY

The callback channel uses ephemeral session keys, so that compromise of long-term key material does not retrospectively expose past disclosures.

This is a precondition rather than a refinement. A source owner authorising *standing* retrieval - as distinct from case-by-case release - is accepting that requests will be served without individual review for some period. That acceptance is conditional on a later compromise not unravelling everything released in the interim.

16.5 KEY MANAGEMENT IS THE REAL COST

Per-source signing keys. Rotation schedules that do not invalidate historical signatures. Revocation that propagates across enclaves including disconnected ones. Trust anchors spanning sovereign estates with different accreditation authorities. Compromise recovery procedures that can be executed under operational pressure. Archive verification across a thirty-year horizon during which algorithms will be deprecated.

None of this is intellectually interesting and all of it is necessary. It will consume more effort than the cryptography itself, it is the most likely source of schedule slippage in Chapter 22's roadmap, and any plan that treats it as a subsidiary workstream is wrong.

16.6 CORRELATION WITHOUT AN ENUMERABLE KEY

Hash-only disposition (§9.3) exists so that two sources computing a digest over the same value produce the same output, permitting the platform to propose a correlation without either source disclosing the value. This is the mechanism on which cross-source candidate generation depends (§13.5), and it is therefore load-bearing for the identity model rather than incidental to it.

The naive construction - a keyed hash under a key shared across the federation - fails, and fails in a way that a competent adversary will find immediately. The values for which correlation is wanted are names, organisations, places and structured identifiers, whose value spaces are small enough to enumerate exhaustively. Any participant holding the key can build a lookup table over the plausible space offline and thereafter invert every hash-only field emitted by every other participant. In a multinational federation, a scheme whose privacy depends on every participant declining to do something they can do silently, offline and undetectably is not a scheme.

THE RESOLUTION

The mechanism is selected per field, according to the entropy of the field's value space, and the Expectation declares which applies.

1. **Tier A - federation key.** For fields whose values carry sufficient entropy that exhaustive enumeration is infeasible - internal case references, cryptographic identifiers, opaque system keys - a keyed hash under a federation key held in source hardware modules is sufficient. Enumeration is not merely expensive but infeasible, so the shared key discloses nothing. This tier has no online dependency, works disconnected, and is simple enough to audit by inspection.
2. **Tier B - threshold oblivious pseudorandom function.** For fields whose values are enumerable - which is most of the fields for which correlation is actually wanted - the digest is computed by an oblivious pseudorandom function whose key is split across the sovereign customer members under a threshold scheme. The source blinds its input, obtains evaluations from a threshold of members, and unblinds locally. No party learns the input. No party holds the key. Enumeration requires online queries that are authenticated, rate-limited, attributable and audited, and acquiring the key at all requires collusion among a threshold of sovereign participants.

An Expectation that declares no tier for a field does not permit hash-only for that field. Consistent with §9.3, silence is refusal.

WHY THIS SHAPE, AND NOT THE ALTERNATIVES

Distributing the key across the sovereign customer members rather than vesting it in a service operator has a consequence beyond cryptographic hygiene. It makes capture of the correlation function require the same threshold of the same body that Chapter 20 requires for capture of the specification. The governance provision and the cryptographic provision defend the same boundary, which is a better arrangement than either alone.

A single-operator oblivious function would have been simpler and would have created precisely the chokepoint this architecture exists to avoid: a party whose cooperation is required for the federation to function, and whose leverage grows with adoption. Pairwise keys were rejected on architectural rather than cryptographic grounds, since they reintroduce the quadratic coordination cost that Chapter 7 (L6) identifies as fatal.

THE ONLINE DEPENDENCY, AND ITS ACTUAL EXTENT

Tier B requires connectivity at the moment a digest is computed. Three observations bound the cost.

The dependency exists **at ingest and at policy change, not at query**. A federation operating entirely on previously emitted records requires no access to the correlation function at all. Search, projection, graph traversal and callback are unaffected.

Evaluations are cacheable, because the function is deterministic. A connector operating in a disconnected or intermittently connected environment requests evaluations for the values it holds while connected and caches the outputs alongside its salt store (§16.1). Values first encountered while disconnected are emitted as pointer-only and upgraded to hash-only on reconnection. This is the same graceful degradation the architecture applies elsewhere, and it means disconnected edge operation is constrained rather than prevented.

Zero-index deployment is unaffected. A federation disclosing nothing but pointers does not use hash-only at all, and therefore has no dependency on the correlation function whatsoever. The tension between this mechanism and Chapter 7 (L7) that an earlier formulation of this work identified turns out not to exist: the dependency is a consequence of choosing to correlate, and a participant who declines to correlate does not acquire it.

EPOCHS AND ROTATION

Rotation invalidates existing correlations under any scheme, since a new key produces new digests. Correlation keys are therefore versioned as **epochs**, identified in the disclosure package by the correlation key identifier the format already carries (Appendix A.7). Epochs overlap: sources re-emit on their own schedule during the overlap window, and the platform maintains an epoch bridge from records emitted under both. No source is compelled to re-emit on anyone else's timetable, which is the same principle as the Expectation compatibility commitment of §8.3.

COSTS, STATED

Threshold oblivious evaluation is real cryptography and a materially larger implementation than a keyed hash. The available libraries are less mature than the primitive's age would suggest. A threshold of members must be reachable at ingest, which is an availability requirement on a set of organisations that have not previously had one. And the tier declaration requires the model authority to make an entropy judgement per field - a judgement they may get wrong, and one for which the validator can advise but not decide.

A DELIVERY FALLBACK

If threshold evaluation proves too large for the first delivery spiral, a coherent reduced version exists: ship Tier A only, restrict hash-only to high-entropy fields in the Expectation, and route all low-entropy correlation through pointer and callback, where the source decides case by case. Candidate generation across sources is then substantially weaker and entity resolution correspondingly poorer, but the architecture remains consistent, nothing is disclosed that should not be, and the package format already accommodates the upgrade without a breaking change.

This is recorded because a cryptographic decision that blocks delivery is not a resolution.

16.7 WHAT CRYPTOGRAPHY DOES NOT ACHIEVE

An authorised officer at a source who applies the wrong classification produces a perfectly valid signature over a wrong label. Verification confirms that the assertion was made by the identified authority and has not been altered. It does not confirm that the assertion is true.

This must be stated before an accreditor finds it, and it should be stated as what it is: a relocation of accountability rather than an achievement of correctness. The gain is real - in

the alternative, a wrong marking diffuses into a pipeline and accountability cannot be located at all - but it is a gain in attributability, not in assurance, and describing it otherwise would forfeit the credibility that the rest of the argument depends on.

PART V - SPECIFICATION AND CONFORMANCE

CHAPTER 17 - THE EIGHT NORMATIVE CONTRACTS

17.1 WHAT CONFORMANCE MEANS

An implementation is conformant if it implements eight contracts. Everything else - the index, the interface, the graph engine, the storage layer, the model integration - is implementation choice, and an implementer may substitute freely.

The contracts are specified in full elsewhere; Part III describes the mechanism each serves, and this chapter states the division and defends it.

#	Contract	Governs
1	Connect	Connector lifecycle, Expectation retrieval, mapping format and standard transform library, policy application, derivation constraint evaluation, salt custody, signing, callback handling, source-side audit, update mechanism, failure behaviour
2	Model	Expectation structure, versioning, compatibility semantics, extension namespaces, codelists, deprecation, core growth constraint
3	Policy	Disposition language, derivation constraints with granularity, evaluation model, attribute sources, purpose semantics, conflict resolution, decision audit format
4	Trust	Key hierarchies, trust anchors, certificate lifecycle, cryptographic profiles, attestation evidence, correlation key handling, compromise recovery
5	Provenance	Provenance manifest, Merkle construction and canonical ordering, signature formats, the five-part header, dependency links, selective disclosure proofs
6	Ingress	Validation sequence, failure semantics, quarantine, replay protection, error reporting
7	Assertions	Assertion types, attribution, confidence, challenge, supersession, withdrawal, cascade, bitemporal semantics, identity pinning, export format
8	Disclosure	Callback protocol, disclosure object, encryption flow, token format, continuous authorisation, revocation, notification, shape consistency

17.2 WHY DISCLOSURE IS NOT OPTIONAL

An earlier drafting formulation of this work listed seven contracts, omitting Disclosure on the grounds that callback is a capability rather than a primitive. That omission would have been fatal.

If the callback path is optional, an implementation may be certified conformant while requiring sources to surrender content in full - which is the prevailing commercial model with a conformance badge attached. Every claim in this document about inverted control rests on withheld fields being resolvable by request, with the source deciding. It is the difference between the architecture and its imitation, and it must therefore be tested for.

17.3 INGRESS AS A CONFORMANCE POINT

Ingress is specified as a sequence of checks with defined ordering and failure semantics rather than as a component. Implementers will realise it as middleware, a filter chain or a service, and the choice does not affect conformance. Specifying it as a box would cause people to build a box they did not need, which is Chapter 7 (L3) operating at small scale.

17.4 THE NON-NORMATIVE SET

Core, Index, Entity, Search, Graph, Audit, Exit, AI, Studio and SDK are present in the reference implementation and will be present in most serious deployments. None is normative. An implementer may use any search technology, any graph projection, any storage engine, any interface and any model provider.

17.5 AN OPEN QUESTION

Audit and Exit sit uncomfortably outside the normative set.

Dual-sided audit reconciliation is what makes the fourth principle real: without it, the platform's record of what was accessed stands alone and is unverifiable by the source. Exit is what makes the fifth principle demonstrable rather than promised. Both are arguably as load-bearing as any of the eight.

They are non-normative in the current formulation for a sequencing reason rather than a principled one: the normative set should be small enough that a second implementation is achievable, and expanding it before any second implementation exists optimises for the wrong thing. The expectation is that both are promoted after the first external

implementation, and this is flagged here as an unsettled question rather than a settled division, so that a reader who notices the tension finds it already acknowledged.

CHAPTER 18 - CONFORMANCE WITHOUT CAPTURE

18.1 THE FAILURE MODE

Chapter 7 (L4) established that a published standard becomes a barrier to entry when demonstrating conformance is expensive and the gate is held by an interested party. The DCGS Integration Backbone is the demonstrative case: an open, community-governed integration standard whose conformance requirement was subsequently cited in defence of a decision not to solicit commercially.

The pattern is not confined to failed programmes. The NATO binding profiles this work builds upon specify conformance via implementation conformance statements, extra information for testing, approved test cases and scripts, and evaluation by an authorised laboratory. The standards-related document acknowledges that no formal reference facility for testing external implementations has been established and that the testing capability remains under investigation. It was promulgated in December 2020.

This is not a criticism of the standard, which is technically sound. It is an observation that a heavyweight, gated conformance model produces, after some years, no operational route by which an implementer can demonstrate conformance - and that the absence of such a route is itself a barrier, whether or not anyone intended it as one.

18.2 THE ALTERNATIVE

The Certified Kubernetes programme provides a working model that transfers almost without modification.

Any vendor may run an open test suite and submit the results for review. Participation is free to foundation members and to non-profit organisations. Certification is versioned, covering the current release and the two preceding it. Optional functionality is handled through profiles carrying their own test sets and their own marks. A participant whose offering is modified must resubmit within a defined period. Ninety-plus certified offerings exist.

The programme works because running the tests is cheap, submitting is public, and the result is verifiable. Its governing terms state as an explicit objective that the self-tests should be verifiable by end users of each qualifying offering.

That objective is adopted here verbatim, and it is the single most important provision in this chapter. A customer evaluating a supplier's conformance claim does not depend on the certifier's diligence, or on the supplier's honesty, or on a laboratory's availability. They run the tests themselves. There is no gate, because there is nothing to gate.

18.3 WHAT THE SUITE MUST EXERCISE

Executable tests, containerised, runnable offline, against any implementation including the reference one:

- **Signature and binding:** construction and verification under both algorithm profiles, canonical leaf ordering across core and extension namespaces, inclusion proof verification, five-part header binding, replay detection.
- **Marking arithmetic:** property-based, over generated label lattices and generated operation sequences, asserting the invariants of §15.2.
- **Dispositions:** all four, including that a withheld field's absence is indistinguishable from non-existence, and that shape consistency (§15.6) holds across differing requester contexts.
- **Derivation constraints:** evaluation including granularity semantics, conflict detection, and refusal to validate a policy with an unresolved conflict.
- **Callback:** authorisation, denial, bounded duration, expiry, revocation, and verification that a value released post-ingest still verifies against the original root.
- **Bitemporal semantics:** supersession distinct from withdrawal, cascade flagging, as-of queries.
- **Identity pinning:** identifier stability across assertion changes, tombstone resolution after split, prior-identifier resolution after merge.
- **Export and import:** an export from any conformant implementation loads into any other with assertions, attribution, markings, temporal fields and dependency links intact.
- **Negative tests:** malformed packages, replayed packages, unlabelled fields, invalid or deprecated Expectation versions, forged signatures, revoked keys, unrecognised connector measurements.

A published reference corpus of synthetic data with known-correct expected outputs accompanies the suite, so that a failure is diagnosable rather than merely reported.

18.4 GOVERNANCE SEPARATION

The Kubernetes programme separates three functions across three bodies: change control over what conformance means, the mechanics of the test suite, and the process and policy of the certification programme. The separation is not ceremonial. It prevents the party that writes the tests from also deciding what they must test and who has passed.

The same separation is specified in Chapter 19's constitutional arrangements.

18.5 WHAT THIS OFFERS BACK

If the programme works, it produces executable conformance tests for a profile of ADatP-4778 - a capability the parent standard does not presently have and has been seeking since 2020.

That is offered as a contribution to the standards family this work builds upon, and it is a more useful thing to bring to a standardisation conversation than a proposal.

PART VI - GOVERNANCE AND ADOPTION

GOVERNANCE AS CONSTITUTIONAL DESIGN

The governance question in this work is usually posed as administrative: who owns the specification, who runs certification, who pays for the custodian. That framing is too small, and it is the framing under which most standards bodies are captured.

The question is better understood as **constitutional design** - the specification of powers, of their limits, and of the mechanisms by which those limits are enforced against parties who would benefit from exceeding them. The distinguishing feature of a constitutional provision is that it binds the rule-maker rather than only the rule-follower, and several of the provisions proposed here have exactly that character. The compatibility commitment of §8.3 binds the custodian, not the participants. The field budget of §8.7 binds the model authority against its own future preferences. The change-of-control clause of §20.4 binds the founding company against its own future owners. These are pre-commitments, and pre-commitment against one's own later incentives is the defining constitutional act.

THE RELEVANT LITERATURE IS INSTITUTIONAL, NOT TECHNICAL

Ostrom's work on common-pool resources established that shared resources are frequently governed successfully by their own users, without either privatisation or central authority, where certain design conditions hold - and identified eight such conditions inductively from a large body of empirical cases. The knowledge commons literature that followed adapted the framework to informational resources, on the important observation that a commons is not a resource, a community, a place or a thing, but the *institutional arrangement* of those elements.

That observation applies directly. RACHIS is an institutional arrangement, and the resource it governs is **not the data**, which is never pooled and never leaves its owner's control. The commons here consists of the Expectation and its registry, the mapping registry, the conformance regime, and the accumulated assertions. Those are jointly produced, jointly consumed, non-rival in use, and subject to precisely the failure modes the commons literature documents: free-riding, capture by the best-resourced participants, degradation through unmanaged growth, and erosion of the rules by parties with more capacity for rule-making than for rule-following.

Reading the architecture against Ostrom's conditions is therefore a diagnostic rather than a decoration, and it is used as one below. It surfaces two gaps that the preceding chapters did not identify.

OSTROM'S CONDITIONS APPLIED

#	Condition	Provision here	Where
1	Clearly defined boundaries - of the resource and of who may use it	Conformance defines membership; sovereign customer members are distinguished from implementers; the commons is enumerated rather than assumed	§17.1, §20.2
2	Rules congruent with local conditions	Namespaced extensions; domain profiles; the disclosure policy is local by construction and is never published	§8.7, Ch 9, Ch 23
3	Collective-choice arrangements - those affected can modify the rules	Ratification rests with sovereign customers; implementers are heard in working groups and do not vote	§20.2, §20.3
4	Monitoring, by monitors accountable to users	Self-certification with published evidence and end-user reproducibility ; dual-sided audit reconciliation	Ch 18, §16.1
5	Graduated sanctions	Absent. The document specifies only a binary: conformant or not	<i>gap - see below</i>
6	Rapid, low-cost conflict resolution	Named but unspecified. §19.4 requires the constitution to settle dispute resolution; no mechanism is proposed	<i>gap - see below</i>
7	Minimal recognition of the right to organise by external authorities	The custodian must be recognised by participating states; this is the third-country participation and IP-control problem	§19.2
8	Nested enterprises - polycentric governance for larger systems	Federation of instances, national deployments, one-way replication, disconnected edge; the correlation key is polycentric by construction	§16.6, §22.3

THE TWO GAPS

Graduated sanctions. The conformance regime as specified admits only two states. An implementation either passes the suite or it does not, and a participant either honours the marking rules or is excluded. Ostrom's finding is that regimes offering only the maximum sanction end up applying none, because the cost of applying it exceeds the cost of the infraction in every ordinary case.

The remedy is a ladder rather than a cliff: a published defect notice against a named implementation; conditional certification with a remediation window; suspension of a specific conformance profile while others stand; suspension of registry write access; and delisting only as the terminal step. Each is applied by the certification function under §18.4, each is public, and each is appealable under the mechanism the next paragraph requires. This should be written into the constitution and is not currently in it.

Conflict resolution. §19.4 lists dispute resolution among the questions the constitution must settle, and the present work does not settle it. Ostrom's condition is not merely that a mechanism exists but that it is *rapid and low-cost* - because a resolution process more expensive than the dispute is functionally an absence of one, and it advantages whichever party has the deeper reserves. That is a capture vector, and it is the vector Gaia-X's critics describe.

A sketch, offered as a starting point rather than a specification: technical disputes over whether an implementation conforms are resolved by re-running the suite, publicly, with the result binding - a mechanism that costs the price of compute and admits no advocacy. Disputes over what the specification *means* escalate to the change-control body under §18.4 with a fixed clock and a default in favour of the existing text. Disputes between a participant and the custodian go to an arbitral panel whose composition excludes implementers. The design goal throughout is that the cheap path and the correct path are the same path.

Both gaps were found by applying the literature rather than by internal review, which is a reasonable argument for having applied it.

CHAPTER 19 - THE CUSTODIAN QUESTION

19.1 THE CONTRADICTION

The fifth principle holds that no customer is trapped by the platform or its supplier. If a single company authors the specification, operates certification and sells the reference implementation, that principle is not satisfied. It is lock-in with better manners, and the first competent commercial reader will say so in a room where it matters.

The contradiction must be resolved deliberately, visibly, and before it is raised by someone else.

19.2 THREE FUTURES

AS A PRODUCT

The supplier owns specification, implementation and certification. Simplest, least defensible, and the criticism above stands unanswered. Not recommended.

AS AN OPEN STANDARD

An independent foundation holds the specification. Any party may implement. Certification operates independently. The founding company builds and sells the flagship implementation and competes on execution. This is the recommended option.

AS AN INSTITUTIONAL PROGRAMME

The specification belongs to an alliance, union or consortium body. Multiple suppliers compete on implementation. The founding company is original architect and principal implementer.

The third is the larger commercial opportunity, because governments fund ecosystems more generously than they buy products. It carries costs that enthusiasm tends to skip: third-country participation and intellectual property provisions that a nationally-founded entity must navigate deliberately rather than discover; and standardisation timelines measured in years, over which an architecture ratified late will have been superseded twice. A route via a sympathetic national delegation is generally faster than the front door and preserves more design authority.

The options are not exclusive. A foundation-held specification can subsequently be adopted by an institutional programme; the reverse sequence is considerably harder.

19.3 SEQUENCING

The argument for the second option often invokes the internet, and the invocation deserves examination. Internet standards followed running code and multiple independent implementations. This work presently has neither. A specification with one implementation is a product carrying governance overhead, and a foundation established before there is anything to steward is a foundation with a logo and a mailing list.

Gaia-X is the cautionary instance and Kubernetes the instructive one: unilateral development until adoption made donation meaningful, then transfer, with the donation received as significant precisely because there was something to receive. X-Road followed a comparable path, its custodian institute established after the technology was in national production.

The sequence recommended here:

1. **Announce custodian intent immediately.** It costs nothing, it answers the lock-in objection at first contact, and it usefully constrains subsequent decisions.
2. **Write the constitution now**, alongside the draft contracts, because the question of what may never be proprietary is a question about specific interfaces.
3. **Build the reference implementation and the conformance suite.**
4. **Secure two or three independent implementations** of the eight contracts.
5. **Donate**, at the point where donation transfers something real.

19.4 THE CONSTITUTION

Ten pages. Not about software. It must settle:

- Ownership of the specification, the licence, and the legal form of the custodian
- Who may propose a change, and who may ratify one
- How compatibility commitments (§8.3) bind the custodian, and the minimum deprecation horizon
- The constraint on core Expectation growth (§8.7), including the field budget and the ratification required to exceed it
- Who certifies, who certifies the certifier, and the separation of functions from §18.4
- Dispute resolution between implementers and custodian
- What is permanently open: the eight contracts, wire formats, marking arithmetic, conformance suite, connector, and the mapping and Expectation registries
- What remains commercial: reference implementation, support, deployment, enablement, interface
- **What becomes of the specification if the founding company fails, is acquired, or undergoes a change of foreign ownership control**

That final clause is the one a risk owner will actually read, and it should be drafted with that reader in mind rather than with a lawyer's.

19.5 BUILDING THE LEAVING FIRST

The commercial structure - specification, reference implementation, certification - produces an unusual and defensible position, and it should be stated plainly rather than buried in a governance annex.

Every enterprise platform has an onboarding story. Almost none has an exit story. Shipping the Exit capability - deletion certificates, dependency invalidation, attested evidence of completion, and a tested export that loads into a competitor's implementation - as an early rather than late deliverable is the most credible demonstration available that the fifth principle is meant.

Most software companies spend years making it easier to join. This one spent its first year making it easy to leave.

That sentence is worth more in a procurement conversation than any architectural claim in this document, and it is only true if Exit precedes the product it exits.

CHAPTER 20 - ANTI-CAPTURE

20.1 THE MECHANISM TO BE PREVENTED

Gaia-X's difficulties are usually attributed to complexity and slow delivery. The more instructive criticism is structural: once the major non-European cloud providers were admitted to membership, the initiative was widely held to have lost the purpose for which it existed. The parties whose dominance motivated the response acquired standing in defining it.

This is not a failure of vigilance. It is the predictable outcome of an open governance body funded by participation, in which the parties with the most resource to devote to governance are precisely those with the most at stake in its outcome. Any successor that does not design against it will experience it.

20.2 CONFORMANCE WITHOUT RATIFICATION

The central provision: **implementing the specification confers no governance rights.**

Any party may implement, certify, sell and profit. No party acquires, by implementing, the ability to propose or ratify changes to what conformance means. Ratification rests with a

body whose membership is drawn from sovereign customers - the organisations that hold the information and bear the duty - rather than from suppliers.

Suppliers are heard. They sit in technical working groups, they contribute to the test suite, they raise defects and propose extensions. They do not vote on what conformance is, because the value of conformance to a customer is precisely that it is not defined by the people selling it.

20.3 SUPERMAJORITY AND QUORUM

Changes to the normative contracts require a supermajority of sovereign customer members, with a quorum that cannot be met by any single national bloc. Extensions within namespaces require nothing, which is the pressure release valve that makes the constraint sustainable (§8.7).

20.4 CHANGE OF CONTROL

The constitution must specify what happens on acquisition, insolvency or change of foreign ownership control of the founding company or any principal implementer: automatic transfer of specification rights to the custodian if not already transferred; continuation of the conformance programme independent of any implementer; and a defined period during which customers may exercise Exit with contractual support.

These provisions are unusual to write and unpleasant to negotiate. They are also the ones that make the fifth principle credible to a reader who has watched a supplier be acquired.

20.5 THE LIMIT OF GOVERNANCE

None of this is proof against determined capture over a long enough period. Governance provisions slow capture, make it visible, and raise its cost. They do not prevent it, and a document claiming otherwise would be making a promise that no constitution in any domain has ever kept.

The strongest available protection is not constitutional but technical: a specification simple enough to fork, a conformance suite anyone can run, and an export format that works. If the custodian is captured, the participants leave, and the architecture has been built so that leaving is possible.

CHAPTER 21 - ADOPTION

21.1 THE UNCOMFORTABLE EVIDENCE

No significant information-sharing federation in the record scaled because its participants were persuaded by its design.

NIEM's reach owes much to the requirement that recipients of federal grants for projects implementing XML-based information exchange use it and register the resulting artefacts. X-Road grew inside a legal framework constructed alongside it, supported by a funded institutional custodian. The Afghan Mission Network was assembled because coalition partners were taking casualties and could not share what they knew; Federated Mission Networking followed with command direction behind it. Gaia-X, which had neither a mandate nor an immediate operational necessity, delivered no concrete implementations beyond limited proofs of concept against a timeline that had promised working prototypes years earlier.

The pattern is consistent and it is unwelcome to anyone who would prefer good architecture to recommend itself. Architecture determines whether a federation *can* work. Something else entirely determines whether it *is adopted*.

21.2 LEVERS

The design consequence is that a deployment plan must identify its lever explicitly, and the lever differs by domain.

Domain	Available levers
Defence and national security	Programme mandate; accreditation route conditional on conformance; alliance standard adoption; operational necessity in a live coalition
Policing	Statutory duty to share; inspection regime findings; conditions attached to national capability funding
Health	Statutory framework; conditions on funding or commissioning; regulatory requirement
Regulated industry	Regulatory reporting obligation; supervisory expectation

Two of these - operational necessity and inspection findings - cannot be manufactured and can only be waited for. The remainder can be pursued, and pursuing them is a different activity from selling software, requiring different people and a longer horizon.

21.3 PULL THE LEVER ONCE

If levers are scarce and expensive, the design objective follows directly: **require the lever exactly once**.

This is the strategic function of the mapping registry (§8.6). The first participant in a domain is expensive - a mandate, a funded pilot, a political conversation, a mapping written from scratch. The fortieth participant running the same source product adopts a published mapping, writes a disclosure policy, and connects. If the marginal cost of the fortieth participant is low enough, no lever is required for them at all: participation falls below the threshold at which anyone needs to be compelled.

This is the only mechanism in the architecture that produces a genuinely different cost curve rather than a better position on the existing one, and it is the argument to lead with in any commercial conversation.

21.4 THE EMPTY STORE

A second adoption obstacle is internal to the design and is created by Chapter 13's identity model.

The assertion store is worth little until analysts contribute to it. In the first months of a deployment there are no assertions, so there are no confirmed identities, no adjudicated merges and no accumulated judgement. The system offers federated search and nothing more, and federated search alone is a weaker proposition than the one that justified the deployment.

Three mitigations. **Import** existing structured judgements where they exist - confirmed identity links, adjudicated matches, curated watch lists - as assertions attributed to their original authors, so that the store starts populated rather than empty. **Sequence the value proposition** honestly, so that the first phase is sold as federated discovery and the assertion layer as the compounding benefit, rather than promising the compounded state at launch. And **reduce assertion friction** to near zero in the interface, because an assertion that costs an analyst two minutes will not be made.

21.5 THE FIRST SOURCE OWNER

Everything in this document is downstream of one organisation agreeing to connect one system.

The characteristics of a good first source are specific and worth stating, because the temptation is to take whoever volunteers. It should run a **widely deployed source product**, so that the mapping written for it is immediately reusable and §21.3 begins operating from the second participant. It should have **at least one competent person** willing to own the mapping, since the model depends on this and no amount of tooling substitutes for it. It should face a **lever** already, rather than requiring one to be created. And it should have an **existing relationship** with whoever is proposing the work, because the first participant is buying a proposition that has no reference customers.

Absent a named answer to this, the present work is a well-founded architecture in search of a customer, and it should be described as such rather than as a product.

PART VII - REALISATION

CHAPTER 22 - IMPLEMENTATION

22.1 LANGUAGE

Python, and the justification is not convenience.

The central claim of this architecture is that a source owner can inspect the software that applies their disclosure policy. That claim is worth exactly as much as the population of people capable of performing the inspection. A risk owner, an information asset owner, a departmental security officer or an internal audit function reading Python will in most cases understand what they are reading. The same readers presented with Rust or C++ will in most cases not, and will fall back on assurance by reputation - which is the thing this design exists to avoid.

Choosing the language that the customer community can audit is therefore a security decision rather than an engineering preference. It happens also to be well supplied with the cryptographic, schema, validation and web tooling the work requires.

The caveat, stated rather than discovered. The index and the ingest hot path will eventually require something faster. The mitigation is to keep every contract language-neutral from the outset - JSON Schema and OpenAPI documents generated from the models rather than hand-written - so that a component may be replaced without touching the specification. This is the same discipline that makes independent implementation possible, so it costs nothing beyond the intention to maintain it.

22.2 BASELINE STACK

Concern	Selection	Rationale
Runtime	Python 3.12+, uv	Reproducible, air-gap installable
Contracts	Pydantic v2	Single source of truth; emit JSON Schema and OpenAPI
Services	FastAPI	Generated interface documents per contract
Transport	httpx, mTLS 1.3, certificate pinning source-side	Source-initiated only (§8.2)

Concern	Selection	Rationale
Post-quantum signature	liboqs-python / equivalent	ML-DSA hot path, SLH-DSA archive; abstracted behind a signer interface, binding maturity assumed to change
Classical crypto, HSM	cryptography, PKCS#11	Hybrid operation during transition
Merkle construction	stdlib hashing, own tree, SHA-384	Small, testable, specification-exact; aligns with the existing binding profile
Salt store	Encrypted local store at source	Durable connector state (Appendix A.11, D1); backed up, marked, covered by Exit
Correlation	Federation-key HMAC (tier A); threshold OPRF (tier B)	Per-field by value-space entropy (§16.6); evaluations cached for disconnected operation
Policy evaluation	Declarative format, purpose-built evaluator	§15.2: the marking path must be small enough to reason about exhaustively
Assertions	PostgreSQL, SQLAlchemy 2.0	Append-only, bitemporal ranges, no in-place update
Index	OpenSearch	Permissive licence; pgvector for small deployments
Graph	Projection over assertions	Never a separate source of truth (§13.2)
Attestation	SEV-SNP / TDX evidence verification	Measurement published (§16.3)
Testing	pytest, property-based generation	Marking arithmetic and Merkle proofs are exactly this domain
Quality	Strict linting and typing	Non-negotiable in code whose value is that it can be read
Packaging	OCI images, vendored dependencies	Air-gap installation is a requirement

22.3 DELIVERY IN SPIRALS

Chapter 7 (L1) forbids the single-programme structure. Delivery therefore proceeds in spirals, each of which is operationally useful standing alone. No spiral is justified solely as a precondition for a later one, and each is capable of being the last.

SPIRAL 0 - CONTRACTS AND WALKING SKELETON. TWO WEEKS

Models for all eight contracts. An end-to-end demonstration on synthetic data: the federation publishes an Expectation; a connector retrieves it, applies a short mapping and a small disclosure policy with one derivation constraint, signs, ships; ingress verifies and indexes; search returns entities; an analyst makes an assertion; the projection changes; a pointer field is requested and the source refuses. Unaccredited and ugly, and considerably more persuasive than a document. Delivered alongside the constitution's first draft, because the contracts and the constitution constrain each other.

SPIRAL 1 - NORMATIVE CORE. MONTHS 1-3

Connect with two real plugins. Model registry with versioning and compatibility semantics. Policy engine with four dispositions and derivation constraints including granularity. Trust with software keys, then hardware. Provenance with Merkle manifests, inclusion proofs and the five-part header. Ingress with full validation and quarantine. Assertions with bitemporality, supersession, withdrawal and identity pinning. Disclosure with the callback protocol.

Conformance tests written alongside each contract, never after. This is the single scheduling decision most likely to determine whether the certification programme of Chapter 18 ever exists.

SPIRAL 2 - ASSURANCE. MONTHS 3-6

Dual-sided audit with reconciliation. Exit with deletion certificates, dependency invalidation and a tested export. Attestation end to end. Hardware key custody and the salt store. Entity projection with adjudication. Search with marking enforcement in the response path and shape consistency. Conformance suite as a containerised, runnable product.

SPIRAL 3 - CAPABILITY. MONTHS 6-12

Graph and timeline projection with aggregation thresholds and review queue. The reference interface, treated as a product rather than a harness (L8). Model integration as a constrained consumer. Scale work where measurement rather than instinct demands it. SDK with generated clients and a local harness.

SPIRAL 4 - ECOSYSTEM. MONTHS 12-18

Certification programme operational. Second and third independent implementations. Custodian established and specification donated. Federation across instances: one-way replication, and disconnected edge operation with signed, order-independent assertion merge on reconnection.

22.4 CROSS-CUTTING DISCIPLINES

- Every marking decision property-tested. One leak ends the project (§15.2).
- Every contract change versioned with an explicit compatibility statement (§8.3).
- Conformance tests written with the contract they test.
- Threat model maintained as a living document from spiral 0.
- **The connector remains small enough to read in an afternoon.** If it ceases to be, the central claim of the architecture has been lost regardless of what else works.

CHAPTER 23 - DOMAIN PROFILES

23.1 WHAT A PROFILE CONTAINS

The generalisation argued in Chapter 3 is realised as a **domain profile**: a bounded set of artefacts that adapt the architecture to a domain without altering any of the eight contracts.

A profile comprises a label vocabulary and its policy information file; the codelists the domain requires; Expectations for the domain's principal entities; the attributes a callback request must carry; auto-deny rules; aggregation thresholds and their review triggers; and a statement of the adoption lever available in that domain.

Everything else is shared. The dispositions, the derivation semantics, the binding construction, the header, the verification sequence, the bitemporal model, the identity projection and the conformance suite are identical across every profile. This is the practical content of the claim that the vocabulary is pluggable and the arithmetic is not.

23.2 DEFENCE AND NATIONAL SECURITY

Labels follow the confidentiality metadata syntax with a national or alliance policy identifier: classification, caveats, releasability. Expectations cover vessels, aircraft, persons, locations, organisations and events. Callback requests carry clearance and nationality, and

releasability caveats are evaluated at source, as demonstrated in Appendix A.9. Aggregation thresholds are framed as classification uplift, with review by an authority competent to make the uplift determination.

The adoption lever is programme mandate or an accreditation route conditional on conformance.

23.3 POLICING

Labels express handling conditions and source protection rather than classification. The dominant sensitivity is frequently the identity of a reporting source rather than the content of a report, which makes the derivation constraint of §9.5 operationally central: narrative detail can identify a source where no field names one, and Appendix A.10 demonstrates exactly this case.

Callback requests carry lawful basis and declared purpose as mandatory attributes, and a request lacking either is auto-denied - not as a matter of policy discipline but because processing without them would be unlawful. Aggregation is framed as collateral intrusion, with review against proportionality.

The adoption lever is statutory duty, inspection regime finding, or a condition attached to national capability funding.

23.4 HEALTH

Labels express consent state, lawful basis and opt-out status. Purpose classes - direct care, planning, research - are the primary determinant of access, and the same record may be freely available under one and prohibited under another. This is the domain in which the promotion of purpose to a primary control (§3.6) is most obviously correct.

Aggregation is framed as re-identification risk, which is fortunate, because it is the domain where the statistical disclosure control methods imported in Chapter 15 originate and where established practice, thresholds and tooling already exist.

The adoption lever is statutory framework or a condition on commissioning and funding.

23.5 PROVING NEUTRALITY

Section 3.7 warned that claiming universality weakens every particular case. The discipline follows: **neutrality is demonstrated by shipping a second profile, not by asserting that a second profile is possible.**

A single-profile implementation that describes itself as domain-neutral has made a claim nobody can check. A two-profile implementation with a shared conformance suite has made one that anybody can. The roadmap of §22.3 should therefore be read as producing a defence profile in spirals 1 and 2 and a second profile in spiral 3, rather than a general capability throughout.

PART VIII - ASSESSMENT

CHAPTER 24 - WHAT THIS DOES NOT SOLVE

Eight problems, stated before a reader finds them.

AGGREGATION

Correctly marked items combine into something more sensitive. This is created by federation rather than inherited from sources, optimal suppression is strongly NP-hard, and the decision remains a human policy determination supported by detection, thresholds and review (§15.7).

CORRECTNESS OF MARKING

A valid signature over a wrong label is valid. What is achieved is the location of accountability with an identified authority, which is an improvement on diffusion into a pipeline and is not the same as assurance (§16.7).

MODEL GOVERNANCE

The Expectation determines the form of exchange, not whose definition of a person or an incident prevails. A federation requires a body with authority to ratify Expectations, and constituting that body is a precondition of deployment rather than a consequence of it. Where the question of whose definition wins is itself the dispute, this architecture will not help (§8.8).

RESIDUAL CORRELATION RISK

The two-tier correlation scheme of §16.6 removes offline enumeration but does not remove enumeration entirely: a participant may query the oblivious function up to its rate limit. The residual is bounded, attributable and detectable rather than silent and offline, which is the achievable position rather than an ideal one. The entropy judgement determining which tier applies to a field is also fallible, and a field wrongly classed as high-entropy is enumerable by any key holder.

CROSS-RECORD DERIVATION AT SOURCE

Derivation constraints operate within a record, where the source can see the fields concerned. A source cannot see what its releases combine into across a population, and no source-side mechanism can be constructed that would let it (§9.6).

THE EMPTY ASSERTION STORE

The identity and knowledge layers are worth little until analysts contribute, and the first months of any deployment offer federated search alone (§21.4).

SOURCE-SIDE COMPETENCE

The model requires at least one capable person per participating organisation willing to own a mapping. Where that person does not exist, an enablement offer is needed - documentation, examples, a validator and a few days of support - and the distinction between helping someone write a mapping they own and writing a pipeline they cannot read must be held absolutely, because it is the distinction on which the whole argument rests.

USABILITY

Specified in one chapter and one spiral, against an architecture specified in eighteen. See Chapter 25.

FRAGMENTED SOURCE-PRODUCT MARKETS

The economic argument of §2.8 depends on a small number of products serving many organisations. Where products are highly fragmented, registry coverage never arrives, per-participant cost does not fall, and the cost models converge. This is a stated falsification condition rather than a risk to be managed, and it should be tested against a candidate domain before anything else in this document is.

GRADUATED SANCTIONS AND DISPUTE RESOLUTION

Identified by applying the commons-governance literature in Part VI and not resolved here: the conformance regime offers only a binary, and no rapid low-cost conflict mechanism is

specified. Both belong in the constitution and both are capture vectors while they remain open.

CHAPTER 25 - THREATS TO THE THESIS

The strongest available arguments against this work, stated as their proponents would state them, with the best response available and an honest indication of where the response is weak.

25.1 "THE BURDEN SIMPLY MOVES TO SOURCE OWNERS, WHO WILL NOT CARRY IT"

The argument. The forward-deployed model exists because source owners lack the capacity, the funding and the inclination to do integration work. Relocating that work to them does not make it happen; it makes it not happen, more cheaply.

The response. The work is genuinely smaller - a declarative mapping against a published contract, written by someone who already knows the data, against months of discovery by someone who does not. It requires no clearances, no supplier on site and no procurement, so it can begin without a funding cycle. And it produces an artefact the source owner keeps. The mapping registry means that for most participants the work is adoption and editing rather than authorship.

Where this is weak. It still requires one competent, willing person per organisation. Where that person does not exist, this architecture has no answer that does not begin to look like the model it criticises.

25.2 "THE INDEX IS A CENTRALISATION THAT X-ROAD AVOIDED"

The argument. The most durable federation in the record holds nothing. This one holds an index and an assertion store, and every item of central state is a governance liability and a reason to decline.

The response. The index is specified as an optimisation rather than a requirement, and zero-disclosure deployment is a supported pattern that reduces the system to a broker holding pointers (§12.3). The assertion store cannot be reduced but is exportable in full and hostable customer-side (§14.5).

Where this is weak. Zero-index deployment is specified and, at the time of writing, unbuilt. Until a working instance exists, this is an argument rather than a demonstration. It is also,

by construction, the deployment in which cross-source correlation is weakest, since a federation disclosing only pointers has nothing to correlate on.

25.3 "THE EXPECTATION BECOMES THE NEW DIB"

The argument. A published contract with a conformance requirement is precisely what was weaponised against commercial alternatives in the case this document itself cites. Nothing prevents the same outcome here.

The response. Offline testability, free and containerised conformance tests, self-certification with published evidence, end-user reproducibility, separated governance functions, and a constitutional constraint on core growth. These are specifically designed against the failure mode, and Chapter 18 exists for this argument alone.

Where this is weak. Every one of those provisions is a promise about future governance made by a party that does not yet hold any. They become credible when the custodian exists and not before.

25.4 "USABILITY DECIDES ADOPTION, AND THIS DOCUMENT BARELY ADDRESSES IT"

The argument. A three-billion-dollar programme was displaced by presentation software. This work devotes eighteen chapters to architecture and one to the interface.

The response. The imbalance is acknowledged in Chapter 7 (L8) and again here. The architecture is what this document contributes; the interface is a product problem with its own discipline, and specifying it badly would be worse than specifying it briefly.

Where this is weak. This is a reasonable answer to a reviewer and not an answer to the risk. If the interface is poor, everything else in this document is irrelevant, and the response above does not change that.

25.5 "A NATIONALLY-FOUNDED ENTITY CANNOT CREDIBLY CUSTODY A MULTINATIONAL SPECIFICATION"

The argument. The proposition's appeal rests substantially on sovereignty, and a specification owned by a company in one country is not obviously more sovereign, from another country's perspective, than one owned by a company in a third.

The response. This is correct, and it is why Chapter 19 treats custodianship as determining what the work *is* rather than as an administrative arrangement. The X-Road precedent - a custodian founded by two governments jointly - indicates the shape of the answer.

Where this is weak. The answer requires a second government to agree before the proposition is fully credible to a third, and the first participant must therefore accept a proposition that is not yet complete.

25.6 "THE GUARD ARGUMENT WILL NOT BE ACCEPTED IN TIME TO MATTER"

The argument. Substituting verification for content inspection at a domain boundary is a multi-year accreditation conversation with an uncertain outcome, and the architecture's most distinctive claim therefore cannot be relied upon within any relevant planning horizon.

The response. The claim is severable. Every other property - source-controlled disclosure, cryptographic marking, callback, exit - holds within a single domain and requires no change to guard policy. The cross-domain reduction is an additional benefit rather than a dependency, and §11.5 recommends pursuing it as a paper years before it appears in a proposal.

Where this is weak. Nowhere in particular. This is the objection with the best answer.

25.7 "THE UNBUNDLING ANALOGY DOES NOT HOLD"

The argument. Networking unbundled because the protocol layer was genuinely generic and its interface narrow. The federation layer proposed here is neither: it embeds a data model, a marking vocabulary, an identity semantics and a disclosure policy language, all of which are contested and domain-specific. A layer that thick will not commoditise; it will simply relocate the argument about whose definition wins.

The response. The eight normative contracts are deliberately narrower than the capability they support, and everything above them - index, graph, resolution, interface, models - is explicitly non-normative and substitutable. The claim is not that the layer is thin in the sense a protocol is thin, but that it is *thin relative to what it enables*, which is the operative property.

Where this is weak. §8.8 concedes that the Expectation does not settle whose definition prevails, and Chapter 7 (L3) records that published models accrete. If the core grows, the layer thickens, and the analogy degrades in exactly the way the objection predicts. The field budget of §8.7 exists for this reason and it is a crude instrument. This objection is best understood as a permanent maintenance obligation rather than a question that can be closed.

25.8 "CAUTIOUS DEPARTMENTS WILL GET A POOR TOOL AND CONCLUDE THE MODEL DOES NOT WORK"

The argument. Value follows trust. A department disclosing almost nothing obtains almost nothing, concludes the architecture is ineffective, and communicates that conclusion to its peers. The design's most distinctive property is also its most likely source of reputational failure.

The response. This is real and the mitigation is expectation-setting rather than engineering. Zero-disclosure participation should be sold as what it is - a federated lookup with an upgrade path - rather than as the full proposition, and the first deployments should be chosen among participants able to disclose enough for the compounding benefits to appear.

Where this is weak. Expectation-setting is a weaker instrument than either party would like, and the first participant to be disappointed will not be persuaded that they were told.

CHAPTER 26 - CONCLUSION

The prevailing model of intelligence and information-sharing platform delivery treats deep integration as an unavoidable cost, and vendor dependency as its natural consequence. This document has argued that both are choices; that the choices were made for commercial rather than technical reasons; and that a different set produces better security properties at lower cost, with sovereignty as a structural fact rather than a contractual assurance.

The mechanism states in a sentence. **The federation publishes what it expects; the source owner decides what to provide, maps it themselves, marks it, signs it and ships it; and the platform verifies rather than interprets.** Everything else here is consequence, discipline, or honesty about the limits.

Almost none of it is individually novel, and Part II says so at length. Mapping to a published contract is thirty years old. Confidentiality labels and their cryptographic binding are ratified standards. Federated sovereignty-preserving data sharing is the alliance's own endorsed direction. What has not existed is a mechanism by which a source owner declares, in a form their governance can approve and their staff can audit, exactly what may leave their estate; binds that declaration to the data at the moment of release; and retains the ability to refuse, revoke and disconnect thereafter.

The hard problems that remain are named rather than hidden. Aggregation is created by federation and is provably not soluble in general. Cryptography locates accountability without establishing correctness. The question of whose definition prevails is political and is a precondition rather than an output. And the first source owner willing to go first is a relationship rather than an architecture, on which everything else depends.

What is offered is not a cheaper platform. It is a different relationship between the organisations that hold information under a duty and the systems that help them share it: one in which participation is a declaration rather than a surrender, and in which leaving is a capability that was built first.

If the argument of Chapter 2 holds - that dependency is a linear function of privately-held integration knowledge - then its scope is not confined to intelligence platforms. It applies wherever a government buys shared digital infrastructure and permits the supplier to perform the integration: the dependency that results is structural, it deepens with the programme's success, and no contractual instrument acts on its cause. The remedy in every such case is the same one specified here, which is to relocate integration authority to the party that holds the information and to publish the contract between them.

The end state that implies is not a better platform but a different market shape: a federation layer that is public infrastructure, governed as a commons by the organisations that depend on it, with suppliers competing above it on the things suppliers are actually good at. That transition has happened once already, one layer down, and the argument for it was not that the incumbent stacks were bad. It was that nobody should have to buy the network in order to buy what runs on it.

Licence

RACHIS - thesis, reference implementation, source-side connector, diagrams and demo.

This project is released under two permissive licences, one for the written work and one for the software. Both permit commercial use, modification, redistribution and forking.

Scope

Component	Licence
Written work - the thesis <i>Disclosure Without Surrender</i> , all .md documentation, the threat model, the architecture diagram and its accompanying text	CC BY 4.0
Software - rachis-reference/, connector/, the diagram generator, the demo UI, all example configuration and test code	MIT

Where a file could reasonably be read as both (for example a documentation file containing code samples), the code samples are MIT and the prose is CC BY 4.0.

Written work - Creative Commons Attribution 4.0 International (CC BY 4.0)

You are free to **share** (copy and redistribute in any medium or format) and **adapt** (remix, transform and build upon the material) for any purpose, including commercially.

Under the following term:

Attribution - You must give appropriate credit, provide a link to the licence, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.

No additional restrictions - you may not apply legal terms or technological measures that legally restrict others from doing anything the licence permits.

Full text: <https://creativecommons.org/licenses/by/4.0/legalcode>

Suggested attribution:

Based on *Disclosure Without Surrender: integration authority, and federated information exchange as public infrastructure* (Copyright (c) 2026 James Patrick CITP MBCS), licensed CC BY 4.0.

Software - MIT Licence

Copyright (c) 2026 James Patrick CITP MBCS

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

Why permissive

This is not an incidental licensing decision. It is the argument.

RACHIS holds that the federated exchange layer should be **public infrastructure**: a published contract that any party may implement, verify, fork, and take to their own accreditation - rather than a proprietary artefact that binds a customer to one supplier. A restrictive licence on this work would reproduce, at the level of the specification, exactly the dependency the thesis exists to dissolve.

So: implement it. Compete with it. Build a better front end, a stronger connector, the platform side that is not yet written. Take it to a standards body. Use it commercially and keep the proceeds. The intended outcome is suppliers competing *above* the federation layer rather than constituting it.

The only thing asked in return is attribution on the written work, so that the argument can be traced, checked, and - where it is wrong - corrected in public.

What this licence is not

It is not a security accreditation. It is not a warranty of fitness for handling classified, protectively marked, personal or otherwise regulated information. It is not a representation that any deployment built from this work is safe, lawful, or compliant in any jurisdiction.

The reference implementation and connector are built to production engineering standards and are written to be *taken to* accreditation. They have not undergone independent security testing, and security tests written and run by the authors against their own code do not constitute independent assurance. Known limitations - including the software key-store stand-in, classical rather than post-quantum signatures, the unresolved correlation-key mechanism, and unimplemented hardware attestation - are documented in `connector/docs/THREAT_MODEL.md` and in the thesis.

Anyone deploying this work against real regulated information is responsible for their own threat modelling, independent security testing, and accreditation.

All data in the examples, tests and demonstration interface is synthetic.

Authorship

Written and built in a personal capacity, on personal time and equipment. The views expressed in the written work are the author's own and are not those of any employer or organisation with which the author is associated.

Bibliography

Reference list for *Disclosure Without Surrender*, grouped by the part of the argument each source supports.

A note on precision. This is a working bibliography assembled from the research underpinning the thesis. Organisation, title and identifier are given where I hold them with confidence; where a document number, edition or date should be checked against the issuing body before formal citation, it is marked **verify**. Several NATO documents exist in multiple editions with differing distribution status, and the exact edition matters for any claim about what a standard does or does not yet specify. Consistent with the rest of this work, I would rather flag that than present a false precision.

1. NATO standards — labelling, binding, and reference architecture

- NATO Standardization Office. **STANAG 4774 — Confidentiality Metadata Label Syntax.**
- NATO Standardization Office. **STANAG 4778 / ADatP-4778 — Metadata Binding Mechanism.**
- NATO Standardization Office. **ADatP-4778.2 — Profiles for Binding Metadata to Data Objects.** (Contains the conformance model, the XML Signature and CMS cryptographic profiles, the embedded/encapsulated/detached binding approaches, and the statement anticipating a JSON Web Signature subprofile.) Published December 2020. **(verify current edition in future referencing)**
- NATO. **ADatP-5636 — NATO Core Metadata Specification.** **(verify current edition in future referencing)**
- NATO. **Data Centric Reference Architecture for the Alliance (DCRA).** Reference AC/322-D(2025)0056, approved May 2025, NATO Unclassified. **(verify current edition in future referencing)**
- NATO. **Federated Mission Networking (FMN)** — programme documentation and spiral specifications. See the NATO FMN public pages and the FMN Spiral Specification series.

2. Prior federation efforts

- **Multilateral Interoperability Programme (MIP)** — programme documentation and the **Joint Consultation, Command and Control Information Exchange Data Model (JC3IEDM)**, ratified as STANAG 5525. See the MIP public documentation set. **(verify current edition in future referencing)**
- **National Information Exchange Model (NIEM)** — model specification, conformance rules, Information Exchange Package Documentation (IEPD) guidance, and the artefact clearinghouse. <https://www.niem.gov>
- **Distributed Common Ground System – Army (DCGS-A)** — see US Government Accountability Office reports on the programme, and Army lessons-learned reporting on field usability. **(verify assorted reports in future referencing)**
- **DCGS Integration Backbone (DIB) / Defense Intelligence Information Enterprise (DI2E)** — programme documentation describing the modular, community-governed, standards-based data services and metadata catalogue.
- **Palantir Technologies Inc. v. United States**, US Court of Federal Claims / Federal Circuit (2016) — the DCGS-A2 procurement dispute, including the Army's assertions about commercial-item availability and DIB compatibility. **(verify assorted reports in future referencing)**
- **X-Road** — Nordic Institute for Interoperability Solutions (NIIS). Technical documentation, governance model and adoption record. <https://x-road.global> and <https://niis.org>
- **Gaia-X** — Gaia-X European Association for Data and Cloud AISBL: architecture documents, policy rules, and independent commentary on delivery and membership.
- **International Data Spaces (IDS)** — International Data Spaces Association, **IDS Reference Architecture Model (IDS-RAM)**, and the **Eclipse Dataspace Components (EDC)** implementation. <https://internationaldataspaces.org>

3. Statistical disclosure control

- Hundepool, A. et al. **Statistical Disclosure Control**. Wiley, 2012.
- Willenborg, L. and de Waal, T. **Elements of Statistical Disclosure Control**. Springer, 2001.

- Duncan, G., Elliot, M. and Salazar-González, J.-J. **Statistical Confidentiality: Principles and Practice**. Springer, 2011.
- Literature on the computational hardness of **cell suppression** — the problem of selecting a minimum-cost complementary suppression set is established as strongly NP-hard. See Kelly, Golden and Assad, and subsequent operational-research treatments. **(verify assorted reports in future referencing)**
- Systematic review literature on **disclosure risk in web-based public health query systems**, reporting recoverability of suppressed cells via marginal totals and differencing. **(verify assorted reports in future referencing)**
- **Eurostat / ONS statistical disclosure control guidance** — practical treatments of primary and secondary suppression, threshold rules, and disclosure by differencing.

4. Cryptography

- National Institute of Standards and Technology. **FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA)**. 2024.
- National Institute of Standards and Technology. **FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA)**. 2024.
- National Institute of Standards and Technology. **FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM)**. 2024.
- Merkle, R. **A Digital Signature Based on a Conventional Encryption Function**. CRYPTO '87. (The Merkle tree construction underpinning selective disclosure.)
- Laurie, B., Langley, A. and Kasper, E. **RFC 6962 — Certificate Transparency**. IETF, 2013. (Merkle inclusion proofs in production practice.)
- Islam, M. S., Kuzu, M. and Kantarcioglu, M. **Access Pattern Disclosure on Searchable Encryption: Ramification, Attack and Mitigation**. NDSS, 2012.
- Cash, D. et al. **Leakage-Abuse Attacks Against Searchable Encryption**. ACM CCS, 2015.
- Zhang, Y., Katz, J. and Papamanthou, C. **All Your Queries Are Belong to Us: The Power of File-Injection Attacks on Searchable Encryption**. USENIX Security, 2016.
- Literature on **oblivious pseudorandom functions (OPRFs)** and threshold OPRF constructions — the basis for the §16.6 correlation resolution. See the IETF CFRG OPRF draft/RFC and the threshold-OPRF literature. **(verify current RFC status in future referencing)**

- Krawczyk, H. and Eronen, P. **RFC 5869 — HMAC-based Extract-and-Expand Key Derivation Function (HKDF)**. IETF, 2010.

5. Institutional economics and commons governance

- Ostrom, E. **Governing the Commons: The Evolution of Institutions for Collective Action**. Cambridge University Press, 1990. (The eight design principles used as a diagnostic in Part VI.)
- Frischmann, B., Madison, M. and Strandburg, K. (eds). **Governing Knowledge Commons**. Oxford University Press, 2014.
- Frischmann, B. **Infrastructure: The Social Value of Shared Resources**. Oxford University Press, 2012. (The infrastructure argument underpinning the unbundling claim.)
- Ostrom, E. and Hess, C. (eds). **Understanding Knowledge as a Commons**. MIT Press, 2007.
- Farrell, J. and Klemperer, P. **Coordination and Lock-In: Competition with Switching Costs and Network Effects**. Handbook of Industrial Organization, Vol. 3, 2007. (Standard treatment against which the switching-cost model in §§2.7–2.10 should be checked.)

6. Conformance and standards governance

- Cloud Native Computing Foundation. **Certified Kubernetes Conformance Program** — programme terms, test suite, and the separation of change control, test mechanics and programme governance. <https://github.com/cncf/k8s-conformance>
- Internet Engineering Task Force. **RFC 7282 — On Consensus and Humming in the IETF**. 2014. (Rough consensus and running code, cited in the sequencing argument of Chapter 19.)

7. UK context

- Royal United Services Institute (RUSI). Commentary and analysis on the UK **Digital Targeting Web** and associated programmes, including the observation on orchestrating rather than commanding authority. **(verify assorted reports in future referencing)**

- UK Parliamentary Office of Science and Technology (POST). Briefing material on defence digital transformation and technology dependency. **(verify POSTnote number in future referencing)**
- Ministry of Defence. **Defence Data Strategy** and successor digital strategy publications.

Suggested checking order

If you are validating this bibliography before formal use, the four that matter most to the argument are:

1. **ADatP-4778.2** — the claims about what the standard does *not* yet specify (JSON subprofile, post-quantum algorithms, selective disclosure, absence of a reference test facility) are load-bearing for Chapter 10 and Chapter 18. Check the current edition.
2. **DCRA** — the reframing in §6.3 depends on the reference architecture endorsing federated, sovereignty-preserving exchange. Check the reference number and current status.
3. **Cell suppression hardness** — the claim that aggregation is provably intractable in the general case rests on this. Cite the specific result.
4. **Farrell and Klemperer** — the switching-cost model in §§2.7–2.10 should be positioned against the existing lock-in literature before it is presented as novel.

APPENDIX A - A WORKED EXAMPLE

A.1 SCOPE, AND WHY THIS APPENDIX EXISTS

Every claim in Part III about effort, size and comprehensibility is unfalsifiable in prose. This appendix makes them checkable by carrying a single record through the entire mechanism: Expectation, mapping, disclosure policy, validator, binding, transmission, verification, and a callback that is refused.

The domain is maritime domain awareness. A vessel entity was chosen deliberately. It is a real multinational information-sharing problem with genuine sovereignty tension; it exercises identity resolution properly, because vessels change names, flags and ownership routinely and often in order to be difficult to follow; it exercises all four dispositions naturally; and it avoids constructing a worked example around a named individual, which would be an unwise thing to publish regardless of how synthetic the data.

Everything here is fictitious. Identifiers, organisations, hashes and signatures are illustrative and truncated for legibility. The Expectation is deliberately small - sixteen fields where a production contract would carry sixty - because the purpose is to demonstrate mechanism rather than completeness.

Section A.11 records what writing this appendix broke in Part III. Five defects were found. That section is the most useful part of the appendix and should be read even by those who skip the rest.

A.2 THE EXPECTATION

Published by the federation's model authority, retrieved by the source owner's connector over a connection the source initiates.

```
expectation:
  canonical: urn:rachis:expectation:maritime:vessel
  version:   urn:rachis:expectation:maritime:vessel:2:1
  supersedes: urn:rachis:expectation:maritime:vessel:2:0
  deprecation_horizon: 2027-06-30
  marking:
    policy_id: urn:rachis:policy:nato-amoco-demo
    label_syntax: adatp-4774-json:1:0
    field_labels: required
    record_label: required
  core_field_budget: 40      # constitutional; see §8.7
```

entity: vessel

fields:

- name: imoNumber
type: string
pattern: '^IMO[0-9]{7}\$'
obligation: required
note: Persistent hull identifier; survives renaming and reflagging.
- name: currentName
type: string
obligation: required
- name: priorNames
type: array<string>
obligation: optional
- name: flagState
type: enum
enumeration: urn:rachis:codelist:iso3166-alpha3
obligation: required
- name: callSign
type: string
obligation: optional
- name: mmsi
type: string
pattern: '^[0-9]{9}\$'
obligation: optional
- name: grossTonnage
type: integer
obligation: optional
- name: registeredOwner
type: string
obligation: optional
- name: beneficialOwner
type: string
obligation: optional
note: Where established. Frequently the analytically significant field.
- name: lastPortCall.locode
type: string
enumeration: urn:rachis:codelist:unlocode
obligation: optional
- name: lastPortCall.departedAt
type: datetime
obligation: optional
- name: lastKnownPosition.lat

```
type: decimal
obligation: optional
```

- name: lastKnownPosition.lon
type: decimal
obligation: optional
- name: lastKnownPosition.observedAt
type: datetime
obligation: conditional
condition: required if lastKnownPosition.lat present
- name: assessment.summary
type: string
obligation: optional
note: Source's own assessment. Never a federation product.
- name: collectionMeans
type: enum
enumeration: urn:rachis:codelist:collection-means
obligation: optional

temporal:

```
source_supplied: [validFrom, validUntil, observedAt]
platform_derived: [recordedAt]
```

provenance:

```
required: [sourceOrganisation, sourceRecordId, policyHash, mappingHash]
```

Sixteen fields, one page, no tooling required to read it. That is the standard against which Chapter 7 (L3) should be judged.

A.3 THE SOURCE SYSTEM

A national maritime authority operates a vessel register of unremarkable vintage. Relevant columns:

VSL_MASTER		
VSL_PK	NUMBER(12)	-- internal surrogate key
IMO_NO	VARCHAR2(10)	-- '7712345' - no IMO prefix
VSL_NM	VARCHAR2(60)	
VSL_NM_PREV	VARCHAR2(240)	-- pipe-delimited history
FLAG_CD	CHAR(2)	-- ISO 3166 alpha-2
CALL_SGN	VARCHAR2(12)	
MMSI_NO	VARCHAR2(9)	
GT	NUMBER(8)	
OWNR_REG_NM	VARCHAR2(120)	
OWNR_BEN_NM	VARCHAR2(120)	-- populated ~30% of records
OWNR_BEN_SRC	VARCHAR2(20)	-- provenance of beneficial owner
PORT_LAST_CD	VARCHAR2(5)	-- national port code, not UN/LOCODE
PORT_DEP_DTM	DATE	

POS_LAT	NUMBER(9,6)	
POS_LON	NUMBER(9,6)	
POS_DTM	DATE	
ASSESS_TXT	VARCHAR2(4000)	
ASSESS_CLASS	CHAR(1)	-- 'U','O','S' - local scheme
COLL_MEANS_CD	VARCHAR2(8)	
REC_UPD_DTM	DATE	

This is a realistic schema in every respect that causes difficulty: a local port codelist, a two-letter flag code where the Expectation wants three, a pipe-delimited history in a single column, an identifier without its conventional prefix, and a classification column using a national single-character scheme.

A.4 THE MAPPING

Written once by the source owner's data team, held in their repository, under their change control. Reusable by any other authority running the same register product.

```
mapping:
  id: urn:example:mapping:natmaritime-vslmaster:1:4
  expectation: urn:rachis:expectation:maritime:vessel:2:1
  source: VSL_MASTER
  local_key: VSL_PK

fields:
  imoNumber:
    from: IMO_NO
    transform: [prefix('IMO')]

  currentName:
    from: VSL_NM
    transform: [trim, upper]

  priorNames:
    from: VSL_NM_PREV
    transform: [split('|'), trim_each, drop_empty]

  flagState:
    from: FLAG_CD
    transform: [codelist('iso3166-alpha2-to-alpha3')]

  callSign: { from: CALL_SGN, transform: [trim, upper] }
  mmsi: { from: MMSI_NO }
  grossTonnage: { from: GT }
  registeredOwner: { from: OWNR_REG_NM, transform: [trim] }
  beneficialOwner: { from: OWNR_BEN_NM, transform: [trim] }

  lastPortCall.locode:
    from: PORT_LAST_CD
    transform: [codelist('natports-to-unlocode')]
```

```

on_unmapped: omit_field_with_reason

lastPortCall.departedAt:
  from: PORT_DEP_DTM
  transform: [tz_assume('UTC'), iso8601]

lastKnownPosition.lat: { from: POS_LAT }
lastKnownPosition.lon: { from: POS_LON }
lastKnownPosition.observedAt:
  from: POS_DTM
  transform: [tz_assume('UTC'), iso8601]

assessment.summary: { from: ASSESS_TXT }
collectionMeans:
  from: COLL_MEANS_CD
  transform: [codelist('natcoll-to-rachis-means')]

labels:
  field_default:
    from: ASSESS_CLASS
    transform: [codelist('natclass-to-amoco')]
  overrides:
    imoNumber:      { classification: UNMARKED }
    currentName:    { classification: UNMARKED }
    flagState:      { classification: UNMARKED }
    beneficialOwner: { classification: GENERAL }
    assessment.summary: { from: ASSESS_CLASS }
    collectionMeans: { classification: RESTRICTED }

temporal:
  observedAt: { from: REC_UPD_DTM, transform: [tz_assume('UTC'), iso8601] }
  validFrom:  { from: REC_UPD_DTM, transform: [tz_assume('UTC'), iso8601] }

```

Fifty lines. No procedural logic. Every transform is a named function from the published standard library, per §9.2 - `codelist`, `split`, `prefix`, `tz_assume` - so a reviewer needs to understand the declarations, not a program.

Two things a reader should notice. `on_unmapped: omit_field_with_reason` handles the national-to-UN/LOCODE gap explicitly rather than silently, so a port with no international equivalent produces an absent field and a local log entry rather than a wrong one. And the `labels` block derives field labels from the source's own classification column through a published codelist - the source's local scheme is preserved and translated, not replaced.

A.5 THE DISCLOSURE POLICY

A separate artefact, separately signed, approved by the information asset owner rather than the data team (§9.1).

```
disclosure_policy:
  id: urn:example:policy:natmaritime-vessel:3:2
  mapping: urn:example:mapping:natmaritime-vslmaster:1:4
  expectation: urn:rachis:expectation:maritime:vessel:2:1
  approved_by: Information Asset Owner, Maritime Directorate
  approved_at: 2026-03-11
  default: withheld

dispositions:
  imoNumber: clear
  currentName: clear
  priorNames: clear
  flagState: clear
  callSign: clear
  mmsi: clear
  grossTonnage: clear
  lastPortCall.locode: clear
  lastPortCall.departedAt: clear

  registeredOwner: clear
  beneficialOwner: hash-only
  lastKnownPosition.lat: pointer
  lastKnownPosition.lon: pointer
  lastKnownPosition.observedAt: clear

  assessment.summary: pointer
  collectionMeans: withheld

derivations:
  - field: lastKnownPosition.lat
    derivable_from: [lastPortCall.locode, lastPortCall.departedAt]
    granularity: 200km
    accepted: true
    rationale: >
      A port call locates the vessel to the port at the stated time. This is
      a coarse derivation and does not defeat the protection of a precise
      position, which is what the pointer disposition protects.

  - field: collectionMeans
    derivable_from: [lastKnownPosition.observedAt, lastPortCall.departedAt]
    granularity: exact
    accepted: false
    resolution: demote lastKnownPosition.observedAt to pointer
    rationale: >
      Observation timing relative to port departure discloses revisit
      interval, from which collection means is inferable exactly. Since
      collectionMeans is withheld, the deriving set must be constrained.

callback:
  handler: enabled
  requires: [requesterIdentity, organisation, nationality, clearance,
            statedPurpose, devicePosture, requestedDuration]
  max_duration: PT72H
  auto_deny:
    - purpose in [commercial, research]
```

```
- clearance below RESTRICTED
kill_switch: enabled
```

Note the second derivation. It changes a disposition elsewhere in the policy: `lastKnownPosition.observedAt` was declared `clear` and is demoted to `pointer` as the resolution. This is secondary suppression (§9.5) doing exactly what the statistical disclosure control literature says it must - suppressing a value that is not itself sensitive in order to protect one that is.

A.6 THE VALIDATOR RUN

Executed inside the source estate, with no network connection to the platform (§8.4).

```
$ rachis-validate --mapping natmaritime-vslmaster:1:4 \
                  --policy natmaritime-vessel:3:2 \
                  --expectation maritime:vessel:2:1 \
                  --sample ./local_extract.csv --offline
```

MAPPING

```
16/16 Expectation fields addressed
1 conditional satisfied (lastKnownPosition.observedAt)
0 unmapped required fields
2 namespaced extensions declared, carried
```

DERIVATION ANALYSIS

```
schema-derived      3 candidates
library-matched     6 candidates
statistical         2 candidates (sample n=4,812)
-----
11 proposed, 2 retained by asset owner, 9 dismissed
```

RETAINED

```
lastKnownPosition.lat <- {lastPortCall.locode, lastPortCall.departedAt}
  source: library granularity: 200km ACCEPTED (coarse)
collectionMeans       <- {lastKnownPosition.observedAt,
  lastPortCall.departedAt}
  source: statistical (Cramer's V 0.71, n=4,812)
  granularity: exact  RESOLVED by demotion
```

DISMISSED (9) - see `./validate-report.html` for asset owner rationale

POLICY

```
16/16 fields have an explicit or default disposition
default=withheld applied to 0 fields (all explicit)
0 unresolved derivation conflicts
RESULT: policy validates
```

DISCLOSURE PREVIEW - what would leave, per record

```
clear      9 fields
hash-only  1 field
```


pointer 4 fields
withheld 2 fields (collectionMeans, ASSESS_CLASS-derived internal marker)

Never leaves the estate: collectionMeans, and the values behind all 4 pointer fields unless individually authorised.

The collectionMeans derivation was found statistically, not from the schema and not from the library. No human reading the schema would have noticed that observation timing relative to port departure discloses collection method. That is the case for the validator, and it is also the case for the statistical mode being available even though §9.6 rates it the weakest and most false-positive-prone.

A.7 THE DISCLOSURE PACKAGE

Merkle leaves, canonically ordered by qualified field name, extensions sorted after core (see A.11, D5).

leaf[00]	header		7f2a...c19d
leaf[01]	callSign		1b8e...44a0
leaf[02]	collectionMeans	[W]	9d31...07fe
leaf[03]	currentName		3c0f...b271
leaf[04]	flagState		a417...5e88
leaf[05]	grossTonnage		6620...13cc
leaf[06]	imoNumber		e04b...9a35
leaf[07]	lastKnownPosition.lat	[P]	2d79...c604
leaf[08]	lastKnownPosition.lon	[P]	88f1...3ba7
leaf[09]	lastKnownPosition.observedAt	[P]	5a6c...d012
leaf[10]	lastPortCall.departedAt		0e93...7751
leaf[11]	lastPortCall.locode		b25d...ae38
leaf[12]	mmsi		4f80...6c9b
leaf[13]	priorNames		c318...20d4
leaf[14]	registeredOwner		7ab6...f145
leaf[15]	assessment.summary	[P]	91c2...8e60
leaf[16]	beneficialOwner	[H]	dd47...31a9

root = 5e1f7c...a8b3 signed once, ML-DSA-65, source HSM

[W] withheld, [P] pointer, [H] hash-only. Withheld and pointer leaves contribute to the tree and nothing else.

The transmitted package:

```
{
  "binding": {
    "profile": "urn:rachis:4778:profile:json:1:0",
    "alg": "ML-DSA-65",
    "digest": "SHA-384",
    "root": "5e1f7c...a8b3",
  }
}
```

```

    "signature": "MEUCIQ...",
    "createdAt": "2026-07-19T04:12:33Z"
  },
  "header": {
    "expectation": "urn:rachis:expectation:maritime:vessel:2:1",
    "mappingHash": "sha384:9c02...7d1e",
    "policyHash": "sha384:31af...c50b",
    "connectorMeasurement": "sev-snp:ld:4b19...0c7a",
    "sourceIdentity": "urn:example.org:natmaritime-authority",
    "inclusionProof": ["..."]
  },
  "recordLabel": {
    "policyIdentifier": "urn:rachis:policy:nato-amoco-demo",
    "classification": "GENERAL",
    "creationDateTime": "2026-07-19T04:12:33Z"
  },
  "fields": [
    { "name": "imoNumber", "value": "IM07712345",
      "label": { "classification": "UNMARKED" },
      "salt": "9f1c...", "inclusionProof": ["a417...5e88", "..."] },

    { "name": "beneficialOwner",
      "hash": "hmac-sha384:c81d...4e02",
      "correlationKeyId": "urn:rachis:corrkey:maritime:2026H2",
      "label": { "classification": "GENERAL" },
      "salt": null, "inclusionProof": ["..."] },

    { "name": "lastKnownPosition.lat", "pointer": "ptr:8f30...a1",
      "label": { "classification": "RESTRICTED" },
      "inclusionProof": ["..."] }
  ],
  "temporal": {
    "observedAt": "2026-07-18T22:04:00Z",
    "validFrom": "2026-07-18T22:04:00Z"
  }
}

```

Nine fields carry values. Four carry pointers. One carries a keyed correlation digest. One is absent, and its absence is not distinguishable from non-existence.

A.8 INGRESS VERIFICATION

[ingress] pkg 4c81b... src=urn:example.org:natmaritime-authority

1	signature	ML-DSA-65 vs anchor natmaritime-2026	PASS
2	attestation	sev-snp ld:4b19...0c7a in permitted set	PASS
3	expectation version	vessel:2:1 current, horizon 2027-06-30	PASS
4	schema conformance	16/16, 1 conditional satisfied	PASS
5	labels present	all disclosed fields labelled	PASS
6	marking arithmetic	max(field) GENERAL <= record GENERAL	PASS
7	policy hash	31af...c50b recorded	PASS
8	provenance + proofs	14 inclusion proofs verify to root	PASS

9 replay binding ts unseen in 24h window PASS

ADMIT indexed 4c81b... 9 clear, 1 correlation, 4 pointers
NOTE platform holds no key for any pointer value

Nothing in this trace involves judgement. Every check is a comparison, a verification or an arithmetic evaluation, which is the whole argument of §11.3.

A.9 A CALLBACK, REFUSED

Twelve days later, an analyst in a partner nation working a sanctions-evasion case requests the withheld position.

```
{ "request": "callback",
  "record": "4c81b...", "field": "lastKnownPosition.lat",
  "pointer": "ptr:8f30...a1",
  "requester": { "id": "urn:partner:org:pna/analyst-2291",
                  "organisation": "urn:partner:org:pna",
                  "nationality": "PNA", "clearance": "RESTRICTED",
                  "statedPurpose": "sanctions-enforcement",
                  "devicePosture": "managed-attested",
                  "requestedDuration": "PT24H" } }
```

The source's handler evaluates and denies:

```
[callback] req 71e0a... ptr:8f30...a1
auto_deny rules      purpose=sanctions-enforcement not matched
clearance            RESTRICTED >= RESTRICTED      ok
duration             PT24H <= PT72H                 ok
releasability caveat record REL-T0 omits PNA        FAIL

DENY reason=releasability notified=platform, requester
LOGGED locally: requester, purpose, timestamp, decision
```

Three things follow. The platform never held the value and could not have released it. The source learns that a partner is interested in this hull, which is a benefit to the source and a leakage to the requester (§12.2), and is recorded in the local log either way. And the denial reason is a category, not an explanation - the requester learns they are not releasable, not why the caveat exists.

A.10 THE SAME POLICY IN A POLICING VOCABULARY

The generalisation claim of Chapter 3 is that the marking *vocabulary* is pluggable while the *arithmetic* is not. The four dispositions, derivation constraints and callback structure are unchanged; only the label policy differs.

```
disclosure_policy:
  id: urn:example:policy:forcerms-person:1:0
  expectation: urn:rachis:expectation:le:person:1:2
  marking:
    policy_id: urn:rachis:policy:uk-le-handling
    label_syntax: adatp-4774-json:1:0
  default: withheld

dispositions:
  person.familyName:      hash-only
  person.givenName:       hash-only
  person.dateOfBirth:     withheld
  person.approximateAge:  clear
  occurrence.reference:   clear
  occurrence.type:        clear
  intelligence.summary:   pointer
  source.identifier:      withheld

derivations:
- field: person.dateOfBirth
  derivable_from: [person.approximateAge, occurrence.recordedAt]
  granularity: 1y
  accepted: true
  rationale: Year-precision derivation does not defeat DOB protection.

- field: source.identifier
  derivable_from: [intelligence.summary]
  granularity: exact
  accepted: false
  resolution: intelligence.summary remains pointer; callback requires
               purpose assessment against source-protection policy
  rationale: Narrative content may identify the reporting source.

callback:
  requires: [requesterIdentity, force, role, vettingLevel,
            statedPurpose, lawfulBasis, devicePosture]
  auto_deny:
    - lawfulBasis absent
    - purpose not within declared processing purposes
```

Two substantive differences and no structural ones. `lawfulBasis` becomes a mandatory callback attribute, because in this domain purpose is a legal test rather than good practice - the promotion of purpose-based access argued for in Chapter 3. And the source-protection derivation is the same mechanism as the maritime collection-means case, protecting a different thing for a different reason.

The label vocabulary, the codelists and the auto-deny rules are domain-specific. The dispositions, the derivation semantics, the Merkle binding, the header and the verification sequence are identical.

A.11 WHAT THIS APPENDIX BROKE

Five defects in Part III, found by writing the example rather than by reasoning about it. Each is corrected in the text; they are recorded here because the corrections are more instructive than the original claims.

D1 - THE CONNECTOR IS STATEFUL, AND §10.4 IMPLIED OTHERWISE

Per-leaf salts defeat enumeration of low-entropy withheld values. But a salt must accompany its leaf for verification, so salts for *released* fields are public and protect nothing; the protection applies only to withheld and pointer leaves, whose salts stay at source. It follows that the source must retain those salts for the lifetime of the record, because a later callback release must produce a leaf that verifies against the original signed root. The connector therefore holds durable state - a salt store - which must be backed up, protected and included in Exit. Section 10.4 has been corrected and the requirement carried into the Connect contract.

D2 - CORRELATION KEY MANAGEMENT FOR HASH-ONLY DISPOSITION IS UNRESOLVED

Hash-only exists so that two sources hashing the same value produce the same digest and correlation becomes possible. That requires a shared key across the federation, and a shared key means any participant holding it can enumerate the value space offline - precisely the attack per-leaf salting prevents elsewhere. This is resolved in §16.6 by selecting the mechanism per field according to the entropy of its value space: a federation key where enumeration is infeasible, and a threshold oblivious pseudorandom function keyed across the sovereign customer members where it is not. The `correlationKeyId` carried in the package above identifies the epoch under which the digest was computed, and permits rotation without a breaking change.

D3 - TWO RECORD LABELS WERE BEING CONFLATED

The label of the record *as held at source* and the label of the *disclosure package* are different objects. A source record classified SECRET because of one withheld field can yield a disclosure package correctly labelled GENERAL, since the SECRET content is absent from it. Part III used "record label" for both. The package carries the label of what it contains; the source's own record label never leaves. Chapters 10 and 11 have been corrected, and ingress check 6 now reads against the package label explicitly.

D4 - THE FOUR-PART HEADER SHOULD BE FIVE-PART

Expectation version, policy hash, connector measurement and source identity do not answer the question "which mapping produced this value". A mapping defect - an inverted codelist, a timezone assumption, a mis-parsed delimiter - is among the more probable causes of a wrong value reaching a decision, and provenance that cannot identify the mapping cannot support the inquiry it exists for. mappingHash is added; §10.5 now specifies five elements.

D5 - CANONICAL LEAF ORDERING WAS UNDERSPECIFIED ACROSS NAMESPACES

"Ordered canonically by field name" is ambiguous once namespaced extensions exist, and any ambiguity in ordering is a verification failure waiting to happen between independent implementations. The profile now specifies: core fields first, ordered by Unicode code point on the qualified name; then extensions, ordered by namespace URI and then by qualified name; the header always at index zero. Minor, and exactly the class of defect a conformance suite exists to catch.

A REFINEMENT RATHER THAN A DEFECT

The derivation constraint rule in §9.5 was binary - a field is derivable or it is not. The worked example shows immediately that derivation has *precision*: a port call locates a vessel to within a couple of hundred kilometres, which does not defeat a protected precise position, whereas revisit-interval timing discloses collection means exactly. Constraints therefore carry a *granularity* qualifier, and a constraint binds only where the derivation is at or finer than the granularity the disposition protects. This makes the mechanism usable; without it, every coarse correlation would block a policy and asset owners would rapidly learn to dismiss the validator's output wholesale.

APPENDIX B - COMPARATIVE POSITION

Ten dimensions, across the prevailing commercial model and the six most relevant prior efforts. Assessments of prior work are drawn from their published descriptions and from the sources discussed in Part II.

	Commercial platform	DIB / DI2E	MIP / JC3IEDM	NIEM	X-Road	IDS / EDC	Pro h
Who learns the source system	Vendor engineer, over months	Programme integrator	Source nation	Source agency	Source agency	Data provider	Source
Where transformation runs	Vendor infrastructure	Programme services	Source-side	Source-side	Source-side	Provider connector	Source attest
Who decides what is shared	Negotiated, then encoded in pipeline	Programme policy	Bilateral and national	Exchange package scope	Provider, per service	Negotiated usage policy	Source owner field, revoc
Who assigns classification	Platform, administratively	Programme	Source, unbound	Not addressed	Not addressed	Not addressed	Source cryptographically b
Enforcement location	Platform	Platform	Recipient	Recipient	Transport only	Consumer	Source before egress
What the centre holds	Full ingested copy	Catalogue and services	Nothing	Nothing	Nothing	Broker metadata	Index asser index reduc pointe
Inbound access to source estate	Persistent vendor agent	Programme services	None	None	None	None	None source initial only
Agreement structure	Per-customer contract	Programme mandate	Multilateral, ratified	Published + grant mandate	Published + federation agreements	Bilateral negotiation	Public adop unila
Integration cost curve	Linear in sources	Linear, high per	Linear per nation	Linear, package reuse	Linear, low per participant	Quadratic in	Sub-l via m

	Commercial platform	DIB / DI2E	MIP / JC3IEDM	NIEM	X-Road	IDS / EDC	Pro
		source		partial		participants	reuse
Where knowledge accrues	Vendor	Programme integrator	Nation	Agency + clearinghouse	Agency	Provider	Source
Conformance model	None	Gated, programme-controlled	Test reference system	Conformance rules + tooling	Reference implementation	Certification scheme	Self-certification end-user repro
Cost of leaving	Prohibitive	High	Low	Low	Low	Moderate	Spec: teste evidence

Three observations.

The rows that matter most are enforcement location, where knowledge accrues, and cost of leaving. Every other advantage claimed in this document is downstream of those three, and a reader with limited time should assess the proposal on them alone.

The efforts that hold nothing centrally - MIP, NIEM, X-Road - score best on portability and worst on capability, since none of them offers entity resolution, accumulated assertion or cross-source correlation. The proposal here accepts a defined quantity of central state in exchange for those capabilities, and answers Chapter 7 (L7) by making the index reducible and the assertion store portable rather than by holding nothing.

The commercial platform column is not uniformly worse and should not be presented as such. It scores well on capability, on time to first value, and on the organisational coherence discussed in §1.4, which none of the alternatives supplies at all. The argument of this document is confined to the claim that its advantages do not require the integration model that produces its dependency.